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Solving many-task optimization problems via online intertask learning

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ABSTRACT

Evolutionary multitasking optimization (EMTO) is a newly emerging search paradigm for tackling multiple problems concurrently via integrating population-based meta-heuristic method and transfer learning technique, which has attracted considerable interest in computational intelligence community. Notably, effective knowledge transfer across tasks is the main challenge for intertask synergies in EMTO. Particularly, how to select proper auxiliary tasks for each constitutive task, how to adapt the intensity of intertask knowledge transfer and how to narrow the discrepancy between tasks are three key concerns in complicated many-task scenario. However, these issues are rarely studied jointly till date. Bearing this in mind, this article introduces a novel adaptive EMTO solver for many-task transfer optimization. Specifically, we develop an adaptive task selection mechanism to choose sources reasonably by the aid of maximum mean discrepancy. Besides, multi-armed bandit model is employed to control the intensity of knowledge transfer across tasks. In addition, restricted Boltzmann machine is utilized to extract latent features between tasks so that their discrepancy is reduced. Experiments are conducted on a series of numerical benchmarks to investigate the performance of our proposal, the comparison results reveal that our proposal manages to solve EMTO problems competitively when compared to several existing counterparts.

1. Introduction

Evolutionary algorithms (EAs) are a kind of guided random search approaches inspired by biological evolution in the nature, which are capable of tackling complicated search and optimization problems with non-differentiable and even black-box objectives. Owing to their generality and simple implementation, EAs have been widely used in a plethora of practical engineering applications. Despite great success, traditional EAs often launch the search from scratch and optimize only one task at a time. However, in many real world applications, especially those in industrial field, EA solvers are expected to tackle multiple related tasks over a lifetime (Zhou, Gao, Lu, & Yao, 2022). Encouraged by the parallel processing ability of human brain, Evolutionary MultiTasking Optimization (EMTO) paradigm has emerged as a new research topic of studying how to address a number of related tasks concurrently via uncovering intertask latent synergies and exploiting implicit parallelism of population. It works under an assumption that problem-solving knowledge learned from one task may assist in solving another if they are related to each other, thus accelerate the search and minimize makespan. Despite popular in machine learning realm, knowledge transfer has received less attention in EA community.

EMTO, representing a kind of multi-task search framework, can accommodate versatile EA search engines. As the pioneer work of EMTO, Multifactorial EA (MFEA) (Gupta, Ong, & Feng, 2016) enables knowledge sharing across tasks via implicit transfer model, population members are associated with different tasks, each constitutive task acts as a unique factor navigating the evolution of population. Intertask knowledge exchange is implicitly realized via assortative mating of individuals belonging to distinct tasks. Given the situation of nowadays cloud based computing platform needing to handle multiple requests from enormous users at the same time, the potential of EMTO paradigm is little in doubt. Since the invention of MFEA, versatile EMTO solvers have been proposed and succeeded in addressing various problems such as vehicle routing (Feng et al., 2020), expensive numerical simulations (Ding, Yang, Jin, & Chai, 2019), sensor coverage problem (Li, Zhang, & Wang, 2022), multilayer network reconstruction (Wu, Wang, & Liu, 2021), cloud service collaboration (Zhou, Gao, Lu, & Yao, 2023)

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and assembly line balancing in production processes (Tang, Meng, Cheng, & Zhang, 2022).

According to the way of implementing knowledge transfer, EMTO approaches can be broadly categorized into two branches. One is single-population approach, which evolves a single population and encodes population members (solutions) in a unified search space, each member is assigned with a skill factor indicating the index of best-performing task amongst all tasks. Then, solutions are associated with different tasks on the basis of their skill factors, knowledge transfer is conducted via crossover between solutions belonging to distinct tasks. The other is multi-population approach, which maintains multiple populations, each task is assigned an unique population, knowledge transfer is implemented via explicitly constructing solution mapping across tasks or employing cross-task genetic operators within the unified search space.

Recently, a number of efforts have been devoted to single-population EMTO framework. For example, Bali, Ong, Gupta, and Tan (2020) extended MFEA with online estimation of cross-task transfer parameter *rmp*, which builds the population probabilistic model of target task as a mixture model of multiple source tasks, the *rmp* is then adjusted through maximum likelihood estimation of probabilistic model. Considering tasks in EMTO may have different hardness, Gong, Tang, Li, and Zhang (2019) attempted to control the resources invested among tasks in MFEA from the perspective of historical success experience. Zhou, Feng and Tan et al. (2021) studied the efficiency of distinct genetic operators on knowledge transfer in MFEA and designed an adaptive scheme to configure the appropriate genetic operator by learning from previous successful experience. Zheng, Qin, Gong, and Zhou (2020) introduced the concept of ability vector that reflects a solution's ability to resolve each constitutive task, then subpopulation of each task is created by sampling the original population in a probabilistic way according to the ability vector, cross-task knowledge transfer is enabled by the overlap members between subpopulations, the intensity of cross-task information transfer is naturally adapted according to the degree of overlap between the corresponding subpopulations. Ding et al. (2019) suggested an enhanced MFEA with decision variable shuffling and translation, which benefits knowledge exchange on tasks with separate optimums and different dimensions. Besides, matching-based strategy can also be injected into MFEA to improve the knowledge transfer quality from the perspective of domain adaptation, Xue et al. (2022) adopted affine transformation to keep the distribution topology invariant features among population members from distinct tasks, which is helpful to avoid chaotic matching issue.

As for multi-population EMTO structure, knowledge transfer can be conducted by the genetic operators within the unified search space or learning the mapping of solutions across tasks. Feng et al. (2019) put forward an autoencoder enabled solution mapping model, of which knowledge transfer across tasks is implemented by a denoising autoencoder to lower the influence of noise. Analogously, linear autoencoder mapping model (Feng et al., 2020) is utilized to establish the intertask connection of different vehicle routing tasks. Tang, Gong, Wu, Qin and Tan (2021) implemented a new eigencoordinate system which serves as intermediate subspace of function landscapes from different tasks. Rather than linear autoencoder, a kernelized version is devised in Zhou, Feng, Gupta and Ong (2021) to build the nonlinear mapping between tasks in pursuit of transferring problem-solving experience across heterogeneous tasks whose search spaces are nonlinearly correlated. In Liang, Dong, Liu, Liang, and Zhu (2022), principle component analysis is employed to project the original task-specific search space into low dimensional subspace and subspace alignment is adopted to establish the intertask subspace connection, such that robustness of knowledge exchange against noises is enhanced. Likewise, Tang, Gong, Wu, Liu and Xie (2021) suggested to project the task-specific decision spaces into low-dimension subspaces firstly and then learn the alignment matrix between subspaces by minimizing KL divergence so that low-drift intertask knowledge sharing is realized. Zhang, Zhou, Chen, Yao, and Cao (2020) presented an adaptive configure method for multi-source helper task selection, where the appropriate source selection strategy would be triggered under certain conditions, autoencoder is employed as domain adaption method to construct source-to-target mappings. Li, Lin, and Gao (2020) embedded a historical experiences based adaptive mechanism to tune *rmp* in explicit multi-population EMTO. Wu and Tan (2020) pointed out that unified representation is not enough to mitigate negative knowledge transfer induced by source–target domain mismatch, they proposed a multi-population EMTO algorithm considering the bias between tasks with different optima locations. Li et al. (2022) put forward an online resource allocation scheme aided by solutions' improvements and knowledge transfer effect to address competitive EMTO problem. Chen, Zhong, Feng, and Zhang (2020) took advantage of tasks similarity and online feedbacks achieved by knowledge transfer to select helper tasks. Likewise, in the context of surrogate-assisted optimization for computationally expensive tasks, Huang, Zhong, and Yu (2021) integrated rewarding operator and task similarity into a pheromone-based mechanism to help each component task determine auxiliary tasks.

The previous research efforts have made substantial progress in alleviating the threat of negative transfer in EMTO, yet they mostly focus on addressing a portion of the challenges and suffer from some drawbacks. Firstly, the number of tasks that most EMTO solvers can handle concurrently is limited to two or three, this is because they fail to perceive the relation between tasks, the knowledge transfer across tasks in MFEA is somewhat blindness with no feedback to guide the selection of suitable auxiliary tasks. Secondly, the intensity of intertask knowledge transfer is fixed in advance without considering the task correlation, which may impede the effective sharing of knowledge across paired tasks with varying task relatedness. At last, unified representation scheme and autoencoder are frequently employed to address domain mismatch between tasks in existing EMTO algorithms, they may not scale well with the increasing dimensionality of search space.

In order to overcome the first shortcoming, several attempts have been made to develop evolutionary many-task optimization (EMaTO) algorithms where the number of tasks exceeds three, the evolution of biocoenosis through symbiosis (EBS) (Liaw & Ting, 2017) is a classical EMaTO solver which adaptively controls the degree of knowledge exchange between tasks on the basis of rewarding mechanism. Nevertheless, EBS lacks the ability to select helper tasks precisely since it roughly collects the whole rewards from foreign tasks, thus the auxiliary tasks are still selected blindly and negative transfer may occur. Although methods suggested in Chen et al. (2020) and Liang, Xu, Liu, Tu, and Zhu (2021) own the capability to select proper auxiliary tasks for knowledge transfer, challenges still exist in them since the intensity of knowledge transfer in paired tasks is predefined using a fixed parameter, this is exactly the second drawback aforementioned. Besides, the unified search space representation assumes the genetic alleles carried by distinct tasks are inherently aligned, this may not hold in practice. Therefore, extracting the underlying complementary and similarity across heterogeneous task search spaces adaptively is pressing.

To address above issues, this article proposes a novel many-task solver equipped with online intertask learning mechanism, labeled as EMaTO-AMR, where the selection of auxiliary tasks, the control of transfer intensity and the domain adaption are resolved together to enhance the efficiency of intertask knowledge exchange. The main contributions are listed as follows.

- To the best of our knowledge, this work serves as the first attempt to coherently integrate the auxiliary task selection and transfer intensity control of paired tasks together to leverage intertask knowledge exchange in many-task scenarios.

- We proposed a new multi-source task selection method via measuring the divergence of task-specific subspaces. In addition, the multi-armed bandit model is, for the first time, introduced in many-task scene to learn the intensity of knowledge exchange for pairwise tasks.
- Rather than unified search space representation or autoencoder, we exploit the potential of Restricted Boltzmann Machine to extract the useful traits among tasks so as to narrow the intertask domain mismatch for knowledge exchanges.

The rest of this article is structured as follows. Section 2 describes the critical components involved in our adaptive many-task knowledge transfer and gives the detailed implementations of EMaTO-AMR. Section 3 presents the experimental settings. Section 4 analyzes the results of numerical experiments. At last, concluding remarks are drawn in Section 5.

2. The proposed algorithm

Compared with the canonical EAs, the most significant difference of EMTO is that it considers to transform genetic material from one task into another. Therefore, the design of our EMaTO-AMR firstly needs to balance self-evolution and cross-task knowledge transfer (transfer intensity control), which serves as the preprocessing step for the subsequent evolution operations. Once cross-task knowledge transfer is triggered, one should select a helpful source task (from many candidates) for a given target task, forming a task pair for the subsequent transfer of genetic materials. At last, domain adaption is to establish spatial transformation for paired tasks, with the objective of inducing greater overlap between their respective search spaces, paving the way for boosting the positive genetic materials transfer.

Thus, it is of great interest for the design of transfer intensity control, knowledge source selection, and domain adaption for bridging the gap between tasks in a synergistic way to boost the search performance, especially when encountering many tasks. To this end, we implement the proposed EMaTO-AMR as follows. Firstly, the adaptive transfer probability (ATP) technique is developed to adaptively tune the intensity of intertask knowledge transfer. Then, a multi-source task selection (MTS) strategy is designed to adaptively select helper tasks from various sources reasonably. Next, we introduce the theoretical basis of Restricted Boltzmann Machine (RBM) and design a RBM-based domain adaptation (RDA) mechanism, paving the way for non-linear mapping of problem-solving knowledge across heterogeneous tasks. Afterwards, the framework of EMaTO-AMR is outlined. At last, we analyze the complexity of EMaTO-AMR.

2.1. Adaptive transfer probability technique

In many-task scenarios, tasks are not always closely related, intertask knowledge transfer facilitates synergistic search only when tasks are closely related whereas knowledge sharing among unrelated tasks may cause negative transfer. Based on this fact, we employ a multi-armed bandit (MAB) approach to learn the transfer intensity of different task pairs online via utilizing feedback from the search process. MAB is a feedback based decision method which performs a sequence of arms with the goal of enlarging the returned reward based on the feedbacks received from previous plays. Owing to the self-adaptive learning trait, MAB has been utilized to address the dynamic selection of genetic operators in EA solvers (Li, Fialho, Kwong, & Zhang, 2014). In this article, we extend the MAB to control the intensity of knowledge transfer for task pairs in many-task environment, labeled as ATP technique.

Generally, the intensity of intertask knowledge transfer (atp) is a real-valued number within $[0, 1]$ in continuous decision space, standing for the probability of knowledge transfer between paired tasks. However, to simplify the decision making of MAB in discrete attribute spaces, here we discretize atp into four intervals, they are

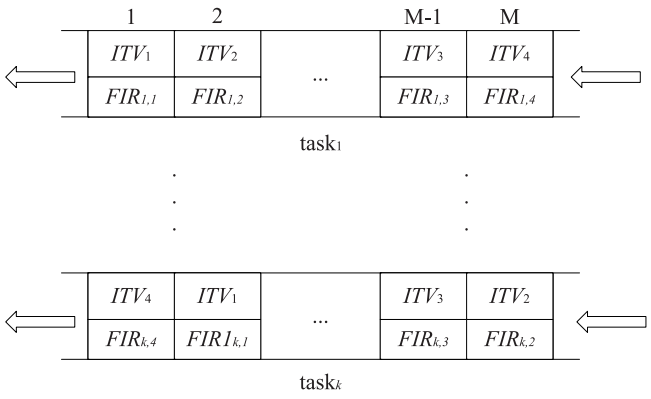


Fig. 1. Illustration of the sliding window.

$ITV_1 = [0.10, 0.25]$, $ITV_2 = [0.25, 0.50]$, $ITV_3 = [0.50, 0.75]$ and $ITV_4 = [0.75, 1.00]$, respectively. Such that, the transfer intensity increases when the interval ranges from ITV_1 to ITV_4. There are two critical issues need to be addressed for bandit-based APT:

(1) how to quantify the impact caused by different configurations of ITV as the search proceeds;

(2) how to choose the suitable ITV according to the above measured impact;

For the first issue, we employ the fitness improvement rate (FIR) of the evolving population over a learning period to quantify the influence of associated ITV. To elaborate, given the cth task F_c, the FIR achieved by population members associated with task F_c under the setting of ITV_t, i.e., $FIR_{c,t}$, is defined as

$$FIR_{c,t} = \frac{pf_{c,t} - cf_{c,t}}{pf_{c,t}} \tag{1}$$

where $pf_{c,t}$ is the best fitness of the one previous generation and $cf_{c,t}$ is the best fitness of the current generation under the influence of ITV_t with respect to task F_c.

For every task F_c, a first-in-first-out sliding window with length of M is employed to collect the used ITVs and associated FIRs dynamically. To elaborate, when the transfer intensity associated with F_c falls into ITV_t, the corresponding ITV_t and evaluated $FIR_{c,t}$ are recorded and put into the sliding window. If the queue is full, the oldest record is removed. Fig. 1 depicts the structure of sliding window.

The main motivation of utilizing sliding window is that it has a good tracking ability to capture the feedbacks associated with ITVs dynamically during the search process so as to guide the selection of most suitable one. As the evolving population is local and dynamic, the best ITV varies for distinct tasks and may change even for different search phases on the same task. Intuitively, sliding window can better reflect the trend of current search since it only records M search experiences being closest to current time point rather than all of them.

As for the second issue, it is desirable to select ITVs with higher rewards as often as possible, however, such doing may lower the chance of earning better rewards for infrequently tried ITVs since currently bad ITVs may perform significantly better in the future search. Aiming at robustness, we take into account of both empirical rewards and the number of tried times for used ITVs to balance the exploitation versus exploration dilemma.

For exploitation aspect, we define the $Reward_{c,t}$ to measure the impact of ITV_t on the search of task F_c, and $Reward_{c,t}$ is the sum of all FIR values associated with ITV_t on the sliding window for task F_c. Herein we define reward-oriented credit $CredR_{c,t}$ that reflects the recent utility of ITV_t with respect to task F_c:

$$CredR_{c,t} = \frac{Reward_{c,t}}{\sum_{t'=1}^{4} Reward_{c,t'}} \tag{2}$$

For exploration aspect, the frequency-oriented credit $CredF_{c,t}$ that reflects the times ITV_t tried before with respect to task F_c is defined

$$CredF_{c,t} = \sqrt{\frac{2 \times ln\sum_{t'=1}^{4} n_{c,t'}}{n_{c,t}}} \tag{3}$$

where $n_{c,t}$ is the number of times that ITV_t has been used with respect to task F_c.

Consequently, credit indicator $Credit_{c,t}$ for deciding which ITV can be used at next time point is obtained by integrating reward and frequency-based credits in a weighted manner

$$Credit_{c,t} = CredR_{c,t} + w \times CredF_{c,t} \tag{4}$$

where w is scaling factor.

Algorithm 1: Pseudocode of ATP

Input: index of current task (c), generation counter (gen), transfer interval set (ITV), scaling factor (w), sliding window (W)
Output: transfer intensity (atp_c), selected interval (t^*)

1 Initialize $Reward_{c,t} \leftarrow 0$ for each $t \in \{1, \cdots, |ITV|\}$;
2 Initialize $n_{c,t} \leftarrow 0$ for each $t \in \{1, \cdots, |ITV|\}$;
3 $nITVs \leftarrow \{1, \cdots, |ITV|\}$; // ITVs that have not been used
4 **while** $nITVs \neq \emptyset$ **do** // There are ITVs not being tried
5 $t' \leftarrow$ randomly select a number from $\{1, \cdots, |ITV|\}$;
6 $nITVs \leftarrow nITVs \backslash \{t'\}$;
7 $atp_c \leftarrow$ randomly generate a number within $ITV_{t'}$;
8 W.SetITV(t');
9 W.SetFIR($FIR_{c,t'}$);
10 **for** $i \leftarrow 1$ **to** $W.length$ **do**
11 $t \leftarrow W$.GetITV(i);
12 $FIR_{c,t} \leftarrow W$.GetFIR(i);
13 $Reward_{c,t} \leftarrow Reward_{c,t} + FIR_{c,t}$;
14 $n_{c,t}++$;
15 **for** $t \leftarrow 1$ **to** $|ITV|$ **do**
16 $CredR_{c,t} \leftarrow$ calculate $CredR_{c,t}$ via Eq. (2);
17 $CredF_{c,t} \leftarrow$ calculate $CredF_{c,t}$ via Eq. (3);
18 $Credit_{c,t} \leftarrow$ calculate $Credit_{c,t}$ via Eq. (4);
19 $t^* \leftarrow \underset{t \in \{1,\cdots,|ITV|\}}{\arg\max} (Credit_{c,t})$;
20 $atp_c \leftarrow$ randomly generate a number within ITV_{t^*};
21 W.SetITV(t^*);
22 W.SetFIR(FIR_{c,t^*});
23 **return** atp_c, t^*

The pseudocode of ATP is given in Algorithm 1. Given the current task F_c, $Reward_{c,t}$ and $n_{c,t}$ are initialized for each transfer interval firstly (lines 1–2). Note that when no ITV has yet been tried at the onset of search, the slots in sliding window W are empty, we randomly select one (e.g., t') from ITV pool and record $FIR_{c,t'}$ and $ITV_{t'}$ in W, this is repeated until all the possible ITVs have been tried (lines 3–9). Otherwise, the credit of each ITV is estimated based on the $FIR_{c,t}$ and ITV_t recorded previously in W (lines 10–18), followed by bandit based selection of ITVs and updating of W (lines 19–22).

2.2. MMD-based task selection

EMaTO involves more than three tasks to be solved simultaneously, each task owns multiple alternative sources for knowledge transfer. Generally, the degree of task relatedness varies among different pairs of component tasks, selecting unrelated sources is prone to incur negative transfer and poses great challenges to the efficiency of EMaTO solver, especially in many-task scenario where a large number of tasks are solved jointly. Therefore, to promote positive knowledge transfer, it is necessary to select highly related tasks for each task rather than random task selection.

Intuitively, the relatedness between tasks can be reflected by the similarity of evolving populations associated with different tasks in decision space. However, canonical distance measure is difficult to measure similarity between the high dimensional datasets. Herein maximum mean discrepancy (MMD) (Gretton, Borgwardt, Rasch, Schoelkopf, & Smola, 2012) is employed to measure distances between populations associated with different tasks. MMD measures the distance between two distributions in the reproducing kernel Hilbert space (RKHS), a kind of Hilbert space with reproducing kernel, which can regenerate the inner product of two functions. MMD in RKHS can reflect the difference between two distributions accurately in high-dimensional space.

Therefore, we employ a MMD-assisted task selection (MTS) strategy to adaptively choose a number of suitable sources for knowledge transfer on each component task as the search proceeds. Specifically, each task is associated with a probabilistic model, which can be learned via the distribution of associated population for related task, multivariate normal distribution is adopted to characterize the population distribution. MMD measures the distance between probabilistic models and thus reflects the relatedness between corresponding tasks. As such, the smaller value the MMD is, the more closely the paired tasks are related. In our proposal, we select l most suitable transfer sources for each component task in terms of MMD.

Assume $\boldsymbol{X} = (x_1, \ldots, x_m)$ and $\boldsymbol{Y} = (y_1, \ldots, y_n)$ are samples from probabilistic models P and Q derived from different tasks, respectively. Let $\mathcal{X}$ be the universal search space and $\mathcal{F}$ be a class of functions $f : \mathcal{X} \to \mathcal{R}$ mapping $\mathcal{X}$ to the set of real values $\mathcal{R}$. Assume $\mathcal{H}$ is the unit ball RKHS, $f(x)$ can be expressed as $f(x) = \langle f, \phi(x) \rangle_{\mathcal{H}}$, where $\phi(x)$ represents the mapping from $\mathcal{X}$ to $\mathcal{H}$, the MMD value between P and Q can be formulated as:

$$\begin{aligned} \mathrm{MMD}[\mathcal{F}, P, Q] &= \sup_{\|f\|_{\mathcal{H}} \leq 1} \boldsymbol{E}_P[f(x)] - \boldsymbol{E}_Q[f(y)] \\ &= \sup_{\|f\|_{\mathcal{H}} \leq 1} \boldsymbol{E}_P[\langle f, \phi(x) \rangle_{\mathcal{H}}] - \boldsymbol{E}_Q[\langle f, \phi(y) \rangle_{\mathcal{H}}] \\ &= \sup_{\|f\|_{\mathcal{H}} \leq 1} \langle f, \mu_P - \mu_Q \rangle_{\mathcal{H}} = \|\mu_P - \mu_Q\|_{\mathcal{H}} \end{aligned} \tag{5}$$

where $\boldsymbol{E}_P[f(x)](\boldsymbol{E}_Q[f(y)])$ represents the expectation of $f(x)(f(y))$ with $x \in P(y \in Q)$. $\mu_P = \boldsymbol{E}_P[\phi(x)]$ and $\mu_Q = \boldsymbol{E}_P[\phi(y)]$ are the expectations of P and Q mapped to $\mathcal{H}$, respectively. Then, we have the square of Eq. (5):

$$\begin{aligned} \mathrm{MMD}^2[\mathcal{F}, P, Q] &= \langle \mu_P - \mu_Q, \mu_P - \mu_Q \rangle_{\mathcal{H}} \\ &= \langle \mu_P, \mu_P \rangle_{\mathcal{H}} + \langle \mu_Q, \mu_Q \rangle_{\mathcal{H}} - 2\langle \mu_P, \mu_Q \rangle_{\mathcal{H}} \\ &= \boldsymbol{E}_P \langle \phi(x), \phi(x') \rangle_{\mathcal{H}} + \boldsymbol{E}_Q \langle \phi(y), \phi(y') \rangle_{\mathcal{H}} \\ &\quad - 2\boldsymbol{E}_{P,Q} \langle \phi(x), \phi(y) \rangle_{\mathcal{H}} \end{aligned} \tag{6}$$

where x' and x are variables satisfying the probability distribution P, variables y' and y satisfy distribution Q. Since the inner product of $\phi(x)$ and $\phi(y)$ can be defined as kernel

$$k(x, y) = \langle \phi(x), \phi(y) \rangle_{\mathcal{H}} \tag{7}$$

Eq. (6) can be rewritten as

$$\begin{aligned} \mathrm{MMD}^2[\mathcal{F}, P, Q] &= \boldsymbol{E}_{x,x'}[k(x, x')] \\ &\quad - 2\boldsymbol{E}_{x,y}[k(x, y)] + \boldsymbol{E}_{y,y'}[k(y, y')] \end{aligned} \tag{8}$$

which can be estimated via

$$\begin{aligned} \mathrm{MMD}^2[\mathcal{F}, P, Q] &= \frac{1}{m^2} \sum_{i,j=1}^{m} k(x_i, x_j) \\ &\quad - \frac{2}{mn} \sum_{i,j=1}^{m,n} k(x_i, y_j) + \frac{1}{n^2} \sum_{i,j=1}^{n} k(y_i, y_j) \end{aligned} \tag{9}$$

In this article, Gaussian kernel (detailed in Eq. (10)), which can map data from search space to infinite dimension space with the least information loss, is utilized to evaluate Eq. (9):

$$k(x, y) = \exp(-\|x - y\|^2 / (2\sigma^2)) \tag{10}$$

where σ is the width parameter of $k(x, y)$.

The pseudocode of MTS strategy is given in Algorithm 2.

Algorithm 2: Pseudocode of MTS

Input: populations for K tasks $\{\boldsymbol{P}_1, ..., \boldsymbol{P}_K\}$, maximal number of transfer sources (l), Gaussian kernel width (σ)
Output: transfer source sets $L = \{L_1, ..., L_K\}$

```
{L_1, L_2, ..., L_K} ← {∅, ∅, ..., ∅};
for c ← 1 to K do
    for k ← 1 to K do
        if k ≠ c then
            MMD_{c,k} ← Calculate MMD between P_k and P_c via Eqs. (5)–(10);
        else
            MMD_{c,k} ← +∞;
    L_c ← indices of tasks with the top l smallest MMD values;
```

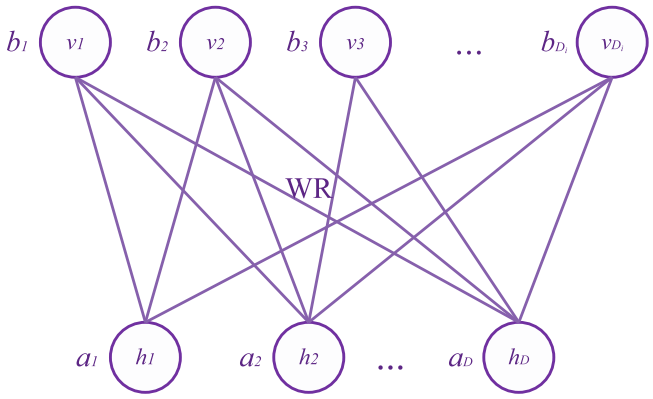


Fig. 2. The structure of RBM.

2.3. RBM-based domain adaptation

In EMTO scene, tasks are usually heterogeneous in terms of function landscapes and decision spaces, in order to transfer knowledge across heterogeneous tasks with diverse properties, we innovatively introduce the RBM to build an intermediate subspace between search spaces of different tasks, which serves the purpose of identifying the underlying complementaries or unveiling the significant invariant features between search spaces, and thus facilitates the exchange of problem solving experience across corresponding tasks.

RBM is essentially an neural network model which can learn the compact representation of input dataset, the learned compact representation of solutions from one task can be mapped back to the decision space of another task, serving as the bridge of knowledge transfer across heterogeneous tasks. The shape of RBM is a bipartite graph with no intra-layer connections (shown in Fig. 2), consisting of visible layer $\boldsymbol{v} = (v_1, v_2, \dots, v_{D_i})^T$ with D_i nodes and a hidden layer $\boldsymbol{h} = (h_1, h_2, \dots, h_D)^T$ with D nodes, $D \ll D_i$, the information of input $\boldsymbol{v}$ is extracted by its hidden layer. This is done by firstly defining an energy function

$$
\begin{aligned}
&E(\boldsymbol{v}, \boldsymbol{h}) = -\sum_{i=1}^{D_i}\sum_{j=1}^{D} WR_{i,j} v_i h_j - \sum_{i=1}^{D_i} b_i v_i - \sum_{j=1}^{D} a_j h_j \\
&\boldsymbol{WR} = \begin{bmatrix} WR_{1,1} & \cdots & WR_{1,D} \\ \vdots & \ddots & \vdots \\ WR_{D_i,1} & \cdots & WR_{D_i,D} \end{bmatrix}, \boldsymbol{a} = [a_1, a_2, \dots, a_D]^T \\
&\boldsymbol{b} = [b_1, b_2, \dots, b_{D_i}]^T
\end{aligned} \tag{11}
$$

with parameters $\theta = (\boldsymbol{WR}, \boldsymbol{a}, \boldsymbol{b})$ and where $WR_{i,j}$ is the connection weight between visible node i and hidden node j, b_i and a_j are the biases of them, respectively. Then, we have the joint distribution of $\boldsymbol{v}$ and $\boldsymbol{h}$ as

$$
P(\boldsymbol{v}, \boldsymbol{h}) = \frac{1}{Z}\exp(-E(\boldsymbol{v}, \boldsymbol{h}; \theta)) \tag{12}
$$

where $Z = \sum_{\boldsymbol{v},\boldsymbol{h}} \exp(-E(\boldsymbol{v}, \boldsymbol{h}; \theta))$ is a normalization constant. Since there is no intra-layer connection in RBM, the node activations in $\boldsymbol{h}$ are mutually independent given the current values of nodes in $\boldsymbol{v}$, and vice versa. We have the conditional distributions of $\boldsymbol{v}$ for $\boldsymbol{h}$ and $\boldsymbol{h}$ for $\boldsymbol{v}$, respectively.

$$
\begin{aligned}
&P(\boldsymbol{v}|\boldsymbol{h}) = \prod_{i=1}^{D_i} P(\boldsymbol{v}_i|\boldsymbol{h}) \\
&P(\boldsymbol{h}|\boldsymbol{v}) = \prod_{j=1}^{D} P(\boldsymbol{h}_j|\boldsymbol{v})
\end{aligned} \tag{13}
$$

Based on Eq. (12) and Bayesian formula, we can rewrite Eq. (13) as:

$$
P(v_i = 1|\boldsymbol{h}) = \sigma(a_i + \sum_{j=1}^{D} WR_{i,j} h_j) \tag{14}
$$

$$
P(h_j = 1|\boldsymbol{v}) = \sigma(b_j + \sum_{i=1}^{D_i} WR_{i,j} v_i) \tag{15}
$$

where $\sigma(\cdot)$ represents sigmoid function.

Given a decision vector $\boldsymbol{x}$ as the input, the value of node h_j is set to 1 with a probability calculated by $P(h_j = 1|\boldsymbol{x}) = \sigma(b_j + \sum_{i=1}^{D_i} WR_{i,j} x_i)$(refer to Eq. (15)). Similarly, the reconstructed value of node x'_i is set to 1 with a probability calculated by $P(x'_i = 1|\boldsymbol{h}) = \sigma(a_i + \sum_{j=1}^{D} WR_{i,j} h_j)$(refer to Eq. (14)). Since RBM is trained to minimize the reconstruction error (solved by the contrastive divergence algorithm), the reconstructed decision vector $\boldsymbol{x}'$ can be reinterpreted as a compact representation of the original decision vector $\boldsymbol{x}$. Following this idea, RBM can be adopted to learn a compact representation from the decision vector of source task, which is then recovered to the decision vector of target task, knowledge transfer across tasks occurs in such transformation process. More specifically, we construct RBM model $\boldsymbol{RBM}_i$ via the corresponding population of task i, $i = 1, \dots, K$. Then, elite individuals from source task i are reduced to compact representations in D-dimensional space through $\boldsymbol{RBM}_i$, which are mapped back into the original decision space of task k through $\boldsymbol{RBM}_k$. The compact representation performs as the bridge for knowledge transfer, problem-solving experiences carried by individuals for task i are transferred to help solve task k.

Although RBM has been utilized to learn the Pareto-optimal subspace in large-scale sparse optimization (Tian, Lu, Zhang, Tan, & Jin, 2021) and multilayer network reconstruction (Wu et al., 2021), it has not been used in many-task scene for EMTO. Herein, we utilize RBM to extract the intrinsic low-dimensional representations of solutions from multiple source tasks, which are then mapped back into the decision space of target task to boost the search. The details are elaborated as follows. Firstly, one RBM for each component task is established. Then, we train these RBMs via corresponding populations of tasks. As such, individuals of component tasks are mapped into hidden layer with recoverability. For ith task, intrinsically, a nonlinear mapping from task i's search space (D_i-dimension) to the hidden common feature subspace (D-dimension) is built. Intuitively, D_i is identical to the dimension of corresponding task decision space, whereas D is determined by the sparsity of top N_{top} individuals (Wu et al., 2021), that is

$$
D = \sum_{d=1}^{D_i} Y_d \tag{16}
$$

where Y_d is a binary variable indicating the connection structure between dth node and other ones in RBM. The probability that $Y_d = 1$ is defined as

$$
p(Y_d = 1) = \frac{1}{N_{\text{top}}} \sum_{r=1}^{N_{\text{top}}} |f(x_d^r)|, \forall d \in \{1, 2..., D_i\} \tag{17}
$$

where x_d^r is the dth variable of the rth individual, $f(x) \in \{0, 1\}$ is a binary indicator variable whose value is 0 when $x < 0.05$ and 1 otherwise.

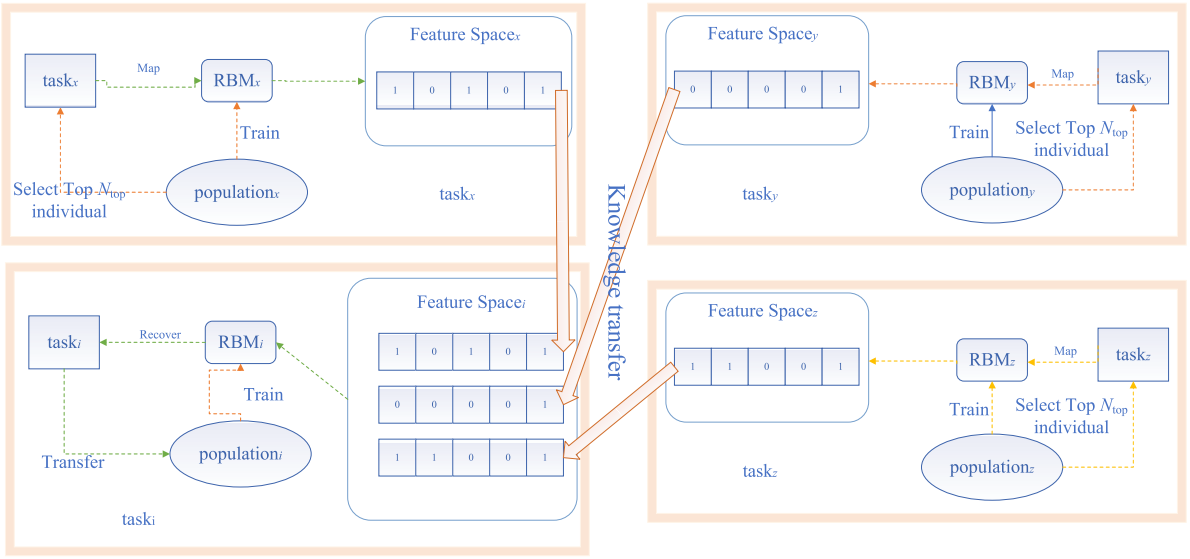


Fig. 3. Intertask knowledge transfer assisted by RBM model.

The procedures of RBM-based domain adaptation are given in Algorithm 3. Given target task c and its transfer sources $L_c = \{c_1, \dots, c_l\}$ identified in Algorithm 2. Firstly, the transferred solution set $\boldsymbol{T}_c$ to target task c is initialized (line 1), then we evaluate the dimension of reduced feature space D and train the RBM model for $\boldsymbol{P}_c$ (lines 2–3). Afterwards, we recursively collect the transferred solutions from all sources $c_j \in L_c$, $j \in \{1, 2, \dots, l\}$ into $\boldsymbol{T}_c$ (lines 4–9). To elaborate, for each $c_j \in L_c$, the best N_{top} individuals from $\boldsymbol{P}_{c_j}$ are selected, notated as $S1$. Then, $S1$ are mapped into the hidden layer of $\boldsymbol{RBM}_{c_j}$, denoted by $S2$. Next, $S2$ are recovered to the search space of task c via $\boldsymbol{RBM}_c$, i.e., transferred solution set $S3$, which are reserved in $\boldsymbol{T}_c$. Herein $\boldsymbol{T}_c$ can be viewed as the carrier of knowledge, which is then implicitly or explicitly transmitted to task c. Fig. 3 illustrates the intertask knowledge transfer via RBM.

Algorithm 3: Pseudocode of RDA

Input: populations of K tasks $\boldsymbol{P} = \{\boldsymbol{P}_1, ..., \boldsymbol{P}_K\}$, index of target task ($c$), knowledge transfer task list $L_c = \{c_1, ..., c_l\}$, number of transferred tasks (l), number of elite solutions from each transfer source (N_{top})
Output: transfer solution set for target task ($\boldsymbol{T}_c$)

1 Set $\boldsymbol{T}_c \leftarrow \emptyset$;
2 $D \leftarrow$ calculate feature space dimension of RBM via Eqs. (16)–(17);
3 $\boldsymbol{RBM}_c \leftarrow$ trainRBM($\boldsymbol{P}_c$, D, contrastive divergence algorithm);
4 **for** $j \leftarrow 1$ **to** l **do**
5 $\quad \boldsymbol{RBM}_{c_j} \leftarrow$ trainRBM($\boldsymbol{P}_{c_j}$, D, contrastive divergence algorithm);
6 $\quad S1 \leftarrow$ select the top N_{top} best individuals from $\boldsymbol{P}_{c_j}$;
7 $\quad S2 \leftarrow$ transform $S1$ to feature space of $\boldsymbol{RBM}_{c_j}$;
8 $\quad S3 \leftarrow$ map $S2$ to search space of cth task by $\boldsymbol{RBM}_c$;
9 $\quad \boldsymbol{T}_c \leftarrow \boldsymbol{T}_c \cup S3$;

2.4. The framework of EMaTO-AMR

The framework of EMaTO-AMR is outlined in Algorithm 4. To facilitate intertask knowledge transfer, solutions for all tasks are encoded in the unified search space $\mathcal{X} \in [0, 1]^{D_{\max}}$, $D_{\max} = \max_{k=1,\dots,K}\{D_k\}$, where D_k is the search space dimension of the kth task. Therefore, in the initialization, solutions of K tasks are randomly generated in $\mathcal{X}$. Suppose that the ith solution of task k is $\boldsymbol{x}_{k,i} = (x_{k,i,1}, \dots, x_{k,i,D})$, when performing fitness evaluation, we decode the solution $\boldsymbol{x}_{k,i}$ from the unified search space $\mathcal{X}$ to task-specific solution $\bar{\boldsymbol{x}}_{k,i}$ in decision space $\mathcal{X}_k$: $\bar{x}_{k,i,d} = l_{k,d} + x_{k,i,d}(u_{k,d} - l_{k,d})$, where $d = 1, 2, \dots, D_k$, $\bar{x}_{k,i,d}$ is the value of the dth dimension of $\bar{\boldsymbol{x}}_{k,i}$, $u_{k,d}$ and $l_{k,d}$ are the upper and lower bounds of the dth dimension task-specific decision space for task k, respectively. In main loop, for each task F_c, the intensity of intertask knowledge transfer (atp_c) is adjusted by ATP, which is elaborated in Algorithm 1. Given a random number ($rand \in [0, 1]$), if $rand \leq atp_c$, knowledge transfer is triggered, MTS is employed to select l source tasks and the transferred solutions ($\boldsymbol{T}_c$) are generated by RBM-based domain adaption mechanism. Next, $\boldsymbol{T}_c$ and $\boldsymbol{P}_c$ are combined to generate offspring $\boldsymbol{O}_c$ via Simulated Binary Crossover (SBX) and Polynomial Mutation (PM) (Deb, Agrawal, et al., 1995). Then, we select half of solution from the union of $\boldsymbol{P}_c$, $\boldsymbol{T}_c$ and $\boldsymbol{O}_c$. When knowledge transfer is not triggered, $\boldsymbol{O}_c$ is generated via classical offspring generation operator and environmental selection. Finally, FIR_{i,t^*} and the best individual $\boldsymbol{X}_c^*$ for task F_c are updated.

Algorithm 4: Framework of EMaTO-AMR

Input: task set $\{F_1, ..., F_K\}$, population size per task (N), N_{top}
Output: the best solutions $\boldsymbol{X}^* \leftarrow \{\boldsymbol{X}_1^*, ..., \boldsymbol{X}_K^*\}$

1 Initialize populations $\boldsymbol{P} \leftarrow \{\boldsymbol{P}_1, ..., \boldsymbol{P}_K\}$;
2 Initialize sliding window W, FIR and atp;
3 Evaluate fitness values of populations on corresponding tasks;
4 **while** *Stopping criterion is not satisfied* **do**
5 $\quad$ **for** $c \leftarrow 1$ **to** K **do**
6 $\quad\quad [atp_c, t^*] \leftarrow$ calculate atp_c and t^* by Algorithm 1;
7 $\quad\quad$ **if** $rand \leq atp_c$ **then**
8 $\quad\quad\quad L_c = \{c_1, ..., c_l\} \leftarrow$ select transfer sources for F_c by Algorithm 2;
9 $\quad\quad\quad \boldsymbol{T}_c \leftarrow$ get the transferred solutions for F_i by Algorithm 3;
10 $\quad\quad\quad \boldsymbol{O}_c \leftarrow$ OffspringGeneration($\boldsymbol{P}_c \cup \boldsymbol{T}_c$);
11 $\quad\quad\quad \boldsymbol{P}_c \leftarrow$ EliteSelection($\boldsymbol{P}_c \cup \boldsymbol{O}_c \cup \boldsymbol{T}_c$);
12 $\quad\quad$ **else**
13 $\quad\quad\quad \boldsymbol{O}_c \leftarrow$ OffspringGeneration($\boldsymbol{P}_c$);
14 $\quad\quad\quad \boldsymbol{P}_c \leftarrow$ EliteSelection($\boldsymbol{P}_c \cup \boldsymbol{O}_c$);
15 $\quad\quad FIR_{c,t^*} \leftarrow$ update FIR_{c,t^*} via Eq. (1);
16 $\quad\quad \boldsymbol{X}_c^* \leftarrow$ the best individual of $\boldsymbol{P}_c$;

2.5. Complexity analysis

In one generation, the computational cost of EMaTO-AMR mainly comes from three parts, transfer intensity assessment ATP, transfer source selection MTS, and intertask domain adaptation RDA. For each

Table 1
Time complexity of peer algorithms.

Algorithm	Time complexity
EMaTO-AMR	$\max\{O(K^2N^2D_{\max}), O(lKNED_{\max}D)\}$
EEMTA (Shang et al., 2019)	$O(KND_{\max}^2)$
AT-MFEA (Xue et al., 2022)	$O(KND_{\max})$
SREMTO (Zheng et al., 2020)	$O(K^2ND_{\max})$
MaTEA (Chen et al., 2020)	$O(K^2ND_{\max}^2)$
MFEA (Gupta et al., 2016)	$O(KND_{\max})$

task, the time complexity of ATP mainly depends on the credit evaluation based on data recorded in sliding window with size of M, which requires $O(M)$ computations. MTS needs $O(KN^2D_{\max})$ computations to evaluate task relatedness, where K is the number of tasks, N is population size per task. RDA consists of two main steps: (1) RBM training, and (2) the evaluation of hidden layer size D. Training RBMs consumes $O(lNED_{\max}D)$ operations, where l is the size of transfer source list, E is number of epochs for training. The evaluation of D requires $O(ND_{\max})$ computations. Complexity of offspring generation is $O(ND_{\max})$. Elite selection requires $O(N)$ comparisons. Therefore, the complexity for one component task is $O(\max\{M, KN^2D_{\max}, lNED_{\max}D, ND_{\max}, N\})$. Since there are totally K tasks, the time complexity of EMaTO-AMR in one generation is $\max\{O(K^2N^2D_{\max}), O(lKNED_{\max}D)\}$. The time complexity of some compared EMTO solvers is summarized in Table 1.

3. Experimental setup

3.1. Benchmark test suite

Two many-task test suites are adopted in experiments. The first is 10-task problem (Chen et al., 2020) composed of seven basic single objective tasks and their variants, they are with diverse properties such as distinct function landscapes, rotations, optimums, dimensions, etc. For clarity, some key features of tasks are outline in Table 2, where the optimums of tasks in decision space can be shifted to varying degrees via shift vectors ($\boldsymbol{O}$), the tasks have different complexity degrees, E-task refers to easy task while C-task means complicated one. Notice that in "ideal assistant task" column, one task is considered as ideal assistant tasks when their optimums are similar in the unified decision space.

The second test suite WCCI2020-SOMATP[1] contains 10 composite 50-task problems (P1–P10), among which the problems involve 7 basic single objective tasks and their variants through translation and rotation, the specific combination of tasks for each problem is listed in Table 3.

3.2. Competing algorithms

As an adaptive many-task solver, our proposal is compared with several state-of-the-art EMaTO solvers and a classical EMTO algorithm. They are EEMTA (Shang et al., 2019), AT-MFEA (Xue et al., 2022), SREMTO (Zheng et al., 2020), MaTEA (Chen et al., 2020) and MFEA (Gupta et al., 2016). Among them, EEMTA is an explicit EMaTO solver whose task selection is assisted by a credit assignment strategy reflecting how good the transferred solutions for target task. AT-MFEA adopts affine transformation to enhance the knowledge transferability across tasks. SREMTO can regulate the intensity of intertask knowledge transfer via a new concept of ability vector. MaTEA is an archive-based EMaTO solver whose task selection is determined by feedback of knowledge transfer and intertask similarity jointly. MFEA is a classical EMTO algorithm.

Population size for each component task is set as 100, the maximal generation ($maxgen$) for all algorithms is 1000. The algorithm-specific parameters for competing algorithms are the same as those in original Refs. Chen et al. (2020), Shang et al. (2019), Xue et al. (2022) and Zheng et al. (2020). Parameters involved in EMaTO-AMR are listed in Table 4. Each algorithm runs 20 times independently on each problem, and their averaged results are recorded.

[1] http://www.bdsc.site/websites/MTO_competition_2020/MTO_Competition_WCCI_2020.html.

4. Results and discussion

In this section, the performance of EMaTO-AMR is examined via comparing with state-of-the-art EMaTO algorithms on 10-task test suite firstly. Then, the effectiveness of each component is verified. Next, we analyze the sensitivity of parameters in our proposal. Finally, the scalability of EMaTO-AMR in solving a series of 50-task problems is investigated.

4.1. Many-task optimization with ten tasks

The means and standard deviations (in brackets) obtained by competing solvers on 10-task test suite over 20 independent runs are reported in Table 5, where the symbols •, § and ∘ indicate that the result obtained by EMaTO-AMR is better than, similar to and worse than its rivals on a specific task, respectively, in terms of Wilcoxon signed-rank test at 95% confidence level (Zhou et al., 2019). In the last line, w, s and l summarize the number of tasks that EMaTO-AMR is significantly better than, similar to and worse than its rivals, respectively. The best results are highlighted in boldfaced font and the second best results are colored in gray background. Moreover, ranks of algorithms obtained by Friedman test (Zhou, Gao and Li, 2021; Zhou et al., 2019) are listed.

In terms of mean objective values, it can be observed from Table 5 that EMaTO-AMR is the most competitive solver achieving the best values on 9 out of 10 tasks, and then it is the MaTEA which achieves the second best results on 7 tasks. When it comes to Friedman test, EMaTO-AMR gives the best total rank, followed by MaTEA, which ranks in the second place. Next, we pay attention to Wilcoxon's rank sum test results, EMaTO-AMR performs significantly better than its rivals on all tasks except the 8th one, where SREMTO gives the best result and MaTEA does not show significant difference with EMaTO-AMR. Notably, 5 out of 9 tasks that EMaTO-AMR wins are C-task (T5, T6, T7, T9, T10) and optimization of them often requires more computational resource.

As can be seen from the above results in three aspects, our proposal has a clear advantage over the competitors. This is because EMaTO-AMR integrates multiple useful components together to leverage the search ability. In particular, ATP adaptively controls the intensity of knowledge transfer on the basis of feedbacks recorded in sliding window, MTS selects appropriate transfer sources, RDA mitigates the barriers of knowledge transfer across heterogeneous task domains. In contrast, there are some shortcomings in competing solvers. For MFEA, the transfer intensity rmp is fixed and the transfer sources are randomly selected, which incurs the risk of negative transfer. AT-MFEA establishes the affine transformation to narrow the task in scenarios of domain mismatch, yet the affine transformation is essentially a linear mapping which struggles to capture the nonlinear correlation between tasks in complicated many-task environment. Both MaTEA and EEMTA select only one task as the transfer source, which is hard to exploit the knowledge from multiple sources. Although intertask knowledge transfer intensity in SREMTO is determined by the overlap degree between distinct tasks, domain adaption mechanism is omitted.

The average convergence traces of all competing solvers on T1–T10 are depicted in Fig. 4, where the x-axis represents the generation number, and the y-axis is the averaged objective value. It is apparent from Fig. 4 that the convergence performance of EMaTO-AMR is superior than its rivals on all tasks except T8. On E-task T2 and T4, EMaTO-AMR is the best on the whole course of search, and it finds the global optimum on T4. For C-task T5, T7, T9 and T10, all the solvers fall into local optimum except EMaTO-AMR. On T6, although EMaTO-AMR

Table 2
Details for 10-task test suite.

Task	Function	Shift vector(O)	Category	Ideal assistant task
T1	Sphere_1	$(o_1, o_2, \ldots o_D) = (0, 0, \ldots, 0)$ $x \in [-100, 100]^D$, $D = 50$	E-task	None
T2	Sphere_2	$(o_1, o_2, \ldots o_D) = (80, 80, \ldots, 80)$ $x \in [-100, 100]^D$, $D = 50$	E-task	None
T3	Sphere_3	$(o_1, o_2, \ldots o_D) = (-80, -80, \ldots, -80)$ $x \in [-100, 100]^D$, $D = 50$	E-task	None
T4	Weierstrass_{25D}	$(o_1, o_2, \ldots o_D) = (-0.4, -0.4, \ldots, -0.4)$ $x \in [-0.5, 0.5]^D$, $D = 25$	E-task	None
T5	Rosenbrock	$(o_1, o_2, \ldots o_D) = (0, 0, \ldots, 0)$ $x \in [-50, 50]^D$, $D = 50$	C-task	T1
T6	Ackley	$(o_1, o_2, \ldots o_D) = (40, 40, \ldots, 40)$ $x \in [-50, 50]^D$, $D = 50$	C-task	T2
T7	Weierstrass_{50D}	$(o_1, o_2, \ldots o_D) = (-0.4, -0.4, \ldots, -0.4)$ $x \in [-0.5, 0.5]^D$, $D = 50$	C-task	T3, T4
T8	Schwefel	$(o_1, o_2, \ldots o_D) = (420.9687, 420.9687, \ldots,$ $420.9687)$, $x \in [-500, 500]^D$, $D = 50$	C-task	None
T9	Griewank	$(o_1, o_2, \ldots o_{D/2}) = (-80, -80, \ldots, -80)$ $(o_{D/2+1}, o_{D/2+2}, \ldots o_D) = (80, 80, \ldots, 80)$ $x \in [-100, 100]^D$, $D = 50$	C-task	T4
T10	Rastrgin	$(o_1, o_2, \ldots o_{D/2}) = (40, 40, \ldots, 40)$ $(o_{D/2+1}, o_{D/2+2}, \ldots o_D) = (-40, -40, \ldots, -40)$ $x \in [-50, 50]^D$, $D = 50$	C-task	None

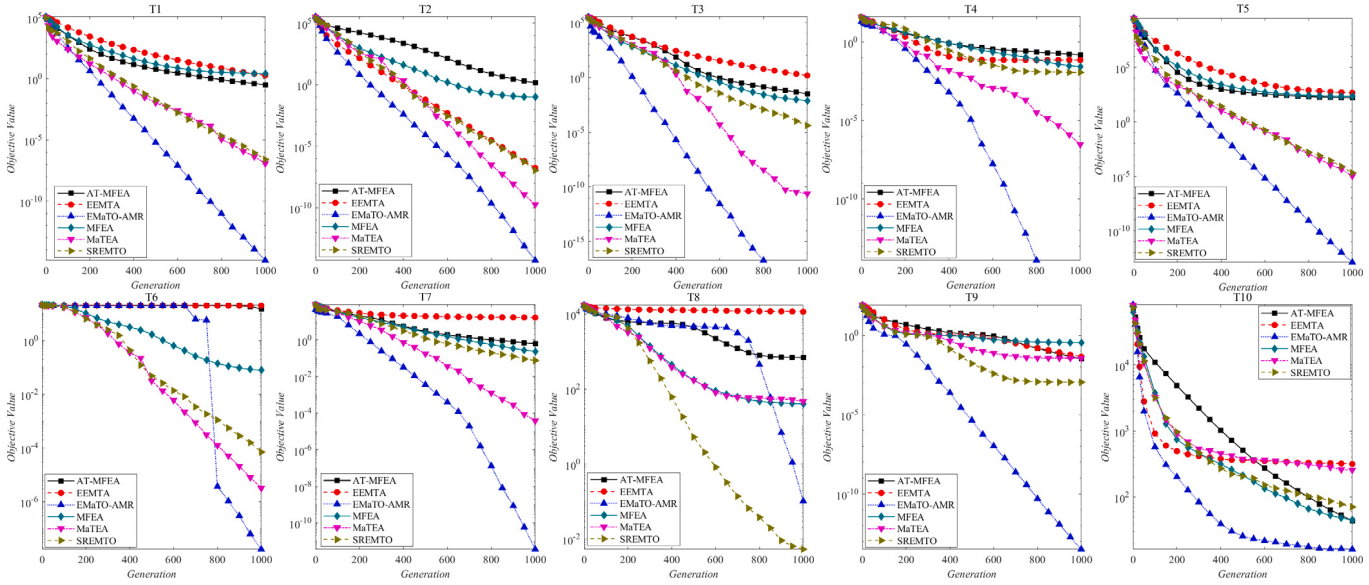


Fig. 4. The average convergence curves of competing algorithms on 10-task test suite.

Table 3
The function combination of WCCI2020-SOMATP.

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
Sphere	✓			✓						
Rosenbrock		✓		✓		✓		✓	✓	
Ackley				✓			✓	✓	✓	✓
Rastrigin			✓		✓		✓	✓	✓	✓
Griewank					✓	✓		✓	✓	✓
Weierstrass					✓		✓	✓	✓	✓
Schwefel						✓			✓	✓

Table 4
Parameter setting for experiments.

Parameter	Value
Distribution index of SBX	2
Distribution index of PM	5
Number of selected tasks for knowledge transfer l	5
Number of transferred individuals for each source task N_{top}	10
Size of sliding window (M)	$0.05 * maxgen$
Value of scaling factor (ω)	1.0

converges slowly in the early stage, it exceeds its rival at the latter stage of search and obtains the best objective value at last, one reason for this occurrence is that efficient knowledge transfer for EMaTO-AMR helps it escape from local optimum. Although SREMTO obtains slightly better results than our proposal on T8, EMaTO-AMR is still satisfied and better than other competitors. Also, EMaTO-AMR surpasses SREMTO on all the other tasks. Based on above analysis, EMaTO-AMR can offer competitive results on 10-task test suite.

Table 5
Mean (Std) of objective values obtained by compared algorithms on 10-task test suite.

	k	EMaTO-AMR	AT-MFEA	EEMTA	MFEA	MaTEA	SREMTO
	1	**1.42E−15(3.2E−16)**	3.28E−1(1.1E−1)•	1.85E+0(4.7E−1)•	2.66E+0(1.3E+0)•	1.19E−7(2.4E−7)•	2.52E−7(7.0E−8)•
	2	**5.38E−15(5.8E−15)**	1.45E+0(6.9E−1)•	1.78E−7(4.6E−8)•	9.83E−2(2.6E−1)•	1.87E−10(2.2E−10)•	1.11E−7(1.3E−7)•
	3	**0.00E+0(0.0E+0)**	3.15E−2(2.4E−2)•	1.44E+0(3.5E−1)•	7.45E−3(8.8E−3)•	2.39E−11(9.5E−11)•	3.96E−5(4.0E−5)•
	4	**0.00E+0(0.0E+0)**	1.49E−1(4.6E−2)•	7.05E−2(2.8E−1)•	2.66E−2(1.2E−2)•	2.94E−7(1.2E−6)•	1.14E−2(5.2E−3)•
	5	**1.27E−13(2.6E−14)**	1.65E+2(8.9E+1)•	4.56E+2(2.4E+2)•	2.18E+2(8.1E+1)•	1.06E−5(1.8E−5)•	2.12E−5(5.9E−6)•
	6	**1.69E−8(1.1E−8)**	1.51E+1(8.7E+0)•	2.00E+1(3.4E−3)•	7.94E−2(9.0E−2)•	3.18E−6(2.2E−6)•	7.05E−5(6.5E−5)•
	7	**3.66E−12(4.4E−12)**	6.14E−1(1.5E−1)•	1.66E+1(4.7E+0)•	2.31E−1(6.3E−2)•	3.78E−5(8.0E−5)•	7.46E−2(4.8E−2)•
	8	1.02E−1(1.2E−1)	6.91E+2(3.3E+2)•	1.14E+4(4.3E+2)•	4.05E+1(1.9E+1)•	4.67E+1(1.1E+2)§	**5.32E−3(5.3E−3)**∘
	9	**3.19E−14(2.5E−14)**	3.30E−2(7.2E−3)•	4.42E−2(1.1E−2)•	3.43E−1(1.2E−1)•	3.58E−2(6.5E−2)•	1.08E−3(4.3E−3)•
	10	**1.62E+1(2.9E+0)**	4.36E+1(1.2E+1)•	3.16E+2(1.5E+1)•	4.47E+1(8.2E+0)•	2.57E+2(1.5E+2)•	7.09E+1(2.9E+1)•
Best/All		9/10	0/10	0/10	0/10	0/10	1/10
Friedman		1.1	4.5	5.5	4.4	2.7	2.8
$w/s/l$		–	10/0/0	10/0/0	10/0/0	9/1/0	9/0/1

Table 6
Experimental results of EMaTO-AMR versus its variants on 10-task test suite.

	k	EMaTO-AMR	EMaTO-noATP	EMaTO-noMTS	EMaTO-noRDA
	1	1.42E−15(3.2E−16)	1.75E−15(4.5E−16)•	9.10E−15(6.7E−15)•	**0.00E+0(0.0E+0)**∘
	2	**5.38E−15(5.8E−15)**	5.68E−15(2.2E−15)§	9.18E−14(8.2E−14)•	3.24E−13(2.1E−13)•
	3	**0.00E+0(0.0E+0)**	**0.00E+0(0.0E+0)**§	**0.00E+0(0.0E+0)**§	1.03E−12(4.1E−13)•
	4	**0.00E+0(0.0E+0)**	**0.00E+0(0.0E+0)**§	**0.00E+0(0.0E+0)**§	**0.00E+0(0.0E+0)**§
	5	1.27E−13(2.6E−14)	1.73E−13(5.5E−14)•	1.08E−12(9.1E−13)•	**0.00E+0(0.0E+0)**∘
	6	**1.69E−8(1.1E−8)**	2.07E−8(4.3E−9)§	7.91E−8(5.2E−8)•	1.56E+1(2.9E−1)•
	7	**3.66E−12(4.4E−12)**	7.64E−12(3.9E−12)•	8.27E−11(3.9E−11)•	5.36E−8(3.8E−8)•
	8	**1.02E−1(1.2E−1)**	3.12E+2(1.7E+2)•	9.72E+2(6.5E+2)•	6.07E+2(2.9E+2)•
	9	3.19E−14(2.5E−14)	**0.00E+0(0.0E+0)**∘	6.02E−15(5.2E−15)∘	1.75E−14(2.4E−14)∘
	10	1.62E+1(2.9E+0)	**1.31E+1(4.8E+0)**§	1.65E+1(5.4E+0)§	1.54E+1(3.8E+0)§
Best/All		6/10	4/10	2/10	3/10
Friedman		1.7	1.8	2.9	2.5
$w/s/l$		–	4/5/1	6/3/1	5/2/3

Table 7
Result obtained by EMaTO-AMR with different configurations of l on 10-task test suite.

	k	$l = 1$	$l = 3$	$l = 5$	$l = 7$	$l = 9$
	1	**4.37E−17(6.9E−17)**	4.26E−16(6.5E−16)•	1.42E−15(3.2E−16)•	4.49E−13(1.3E−12)•	5.53E−6(2.2E−5)•
	2	**3.42E−15(3.6E−15)**	2.49E−14(1.3E−14)•	5.38E−15(5.8E−15)§	1.01E−13(9.1E−14)•	7.99E−11(1.6E−10)•
	3	5.12E−15(3.7E−15)	4.36E−14(2.5E−14)•	**0.00E+0(0.0E+0)**∘	**0.00E+0(0.0E+0)**∘	5.22E−5(1.8E−4)•
	4	**0.00E+0(0.0E+0)**	**0.00E+0(0.0E+0)**§	**0.00E+0(0.0E+0)**§	**0.00E+0(0.0E+0)**§	2.44E−15(6.7E−15)§
	5	**3.90E−15(5.2E−15)**	3.57E−14(5.3E−14)•	1.27E−13(2.6E−14)•	3.65E−11(1.0E−10)•	4.20E−4(1.7E−3)•
	6	2.00E+1(1.6E−6)	2.00E+1(5.5E−3)∘	**1.69E−8(1.1E−8)**∘	8.01E−8(3.6E−8)∘	1.76E−6(1.9E−6)∘
	7	8.53E−10(7.3E−10)	3.22E−5(1.3E−4)•	**3.66E−12(4.4E−12)**∘	1.37E−10(3.0E−10)∘	1.79E−3(5.2E−3)•
	8	1.69E+3(1.2E+3)	4.07E+3(3.3E+2)•	**1.02E−1(1.2E−1)**∘	8.14E+2(1.1E+3)∘	5.18E+2(2.9E+2)∘
	9	1.08E−3(3.0E−3)	**8.06E−16(8.5E−16)**∘	3.19E−14(2.5E−14)∘	3.19E−13(2.5E−13)∘	9.25E−4(2.5E−3)∘
	10	**1.36E+1(4.8E+0)**	1.67E+1(6.6E+0)§	1.62E+1(2.9E+0)•	1.94E+1(4.8E+0)•	1.69E+1(6.8E+0)•
Best/All		5/10	2/10	5/10	2/10	0/10
Friedman		2.4	2.8	1.7	2.9	3.9

4.2. Effectiveness of components

In order to examine the effect of components in EMATO-AMR, three variants of EMaTO-AMR are developed after removing one of three components, the performance comparison between these variants and EMaTO-AMR is investigated. The variants without ATP, MTS and RDA are denoted by EMaTO-noATP, EMaTO-noMTS, EMaTO-noRDA, respectively. In EMaTO-noATP, ATP is replaced by a fixed transfer intensity $atp = 0.1$. EMaTO-noMTS randomly selects l tasks as the transfer sources. In EMaTO-noRDA, RBM-based transfer model is replaced by explicit denoising autoencoder (Feng et al., 2019).

The comparison results on 10-task test suite is reported in Table 6, the average convergence traces of competing solvers are given in Fig. 5. It can be observed from Table 6 that the original EMaTO-AMR ranks the first according to the results from Friedman test. In terms of averaged objective values, EMaTO-AMR wins 6 out of 10 tasks. In addition, the proportion of tasks where EMaTO-AMR significantly outperforms its variants is 4/10, 6/10, 5/10 respectively. Conversely, the proportion of tasks where EMaTO-AMR is significantly inferior to its variants is 1/10, 1/10, 3/10 respectively. The inferiority of EMaTO-noATP is due to that it fails to control the knowledge transfer intensity. For EMaTO-noMTS with randomly selected transfer sources, problem solving knowledge from unrelated tasks may disrupt the search of current task. The superiority EMaTO-AMR over EMaTO-noRDA demonstrates the efficacy of RBM model for domain adaptation. Overall, the joint use of three components leads to the remarkable search ability of EMATO-AMR.

4.3. Sensitivity analysis of parameters

There are three control parameters involved in our proposal, they are the number of selected tasks l, the size of sliding window M, and the scaling factor w. In order to testify how EMaTO-AMR is sensitive to such parameters, experiments are conducted on EMaTO-AMR with different configurations of above parameters. The sensitivity analysis of EMaTO-AMR with different configurations of l, M and w are presented in Tables 7, 8, and 9, respectively, the average convergence traces of EMaTO-AMR with different configurations of parameters are shown in Figs. 6–8, respectively. When analyzing one parameter, the others are set the same as those in original EMaTO-AMR.

Table 8
Result obtained by EMaTO-AMR with different configurations of M on 10-task test suite.

	k	$M = 0.01 * maxgen$	$M = 0.03 * maxgen$	$M = 0.05 * maxgen$	$M = 0.07 * maxgen$	$M = 0.1 * maxgen$
	1	2.23E−15(3.0E−15)	3.32E−15(3.4E−15)§	**1.42E−15(3.2E−16)§**	1.00E−14(1.1E−14)•	5.27E−15(7.5E−15)•
	2	**3.56E−15(2.8E−15)**	1.02E−14(1.5E−14)§	5.38E−15(5.8E−15)§	1.02E−14(8.8E−15)•	2.22E−14(3.2E−14)•
	3	2.03E−14(8.1E−14)	2.34E−14(9.4E−14)§	**0.00E+0(0.0E+0)§**	**0.00E+0(0.0E+0)§**	**0.00E+0(0.0E+0)§**
	4	**0.00E+0(0.0E+0)**	**0.00E+0(0.0E+0)§**	**0.00E+0(0.0E+0)§**	**0.00E+0(0.0E+0)§**	**0.00E+0(0.0E+0)§**
	5	1.90E−13(2.6E−13)	2.55E−13(2.5E−13)§	**1.27E−13(2.6E−14)§**	8.77E−13(9.4E−13)•	4.53E−13(5.8E−13)•
	6	**1.56E−8(6.4E−9)**	2.41E−8(1.6E−8)§	1.69E−8(1.1E−8)§	2.58E−8(1.1E−8)•	3.48E−8(2.4E−8)•
	7	2.81E−10(1.1E−9)	3.22E−9(1.3E−8)§	**3.66E−12(4.4E−12)**∘	6.02E−12(1.0E−11)∘	5.51E−12(5.7E−12)∘
	8	2.00E+2(4.8E+2)	7.50E+1(1.8E+2)§	**1.02E−1(1.2E−1)§**	1.57E+1(4.0E+1)§	1.11E+2(1.6E+2)§
	9	1.12E−14(7.3E−15)	9.25E−4(2.5E−3)§	3.19E−14(2.5E−14)•	1.11E−14(1.0E−14)§	**1.02E−14(5.9E−15)§**
	10	1.63E+1(4.2E+0)	1.75E+1(7.5E+0)§	1.62E+1(2.9E+0)§	1.53E+1(5.5E+0)§	**1.39E+1(4.2E+0)§**
Best/All		3/10	1/10	6/10	2/10	4/10
Friedman		2.5	3.5	1.7	2.8	2.8

Table 9
Result obtained by EMaTO-AMR with different configurations of w on 10-task test suite.

	k	$w = 0$	$w = 0.5$	$w = 1$	$w = 2$	$w = 3$
	1	**9.77E−16(1.2E−15)**	4.19E−15(5.2E−15)•	1.42E−15(3.2E−16)•	7.98E−15(2.1E−14)•	1.92E−14(4.0E−14)•
	2	2.47E−14(5.2E−14)	1.44E−12(5.7E−12)§	**5.38E−15(5.8E−15)§**	9.48E−15(1.1E−14)§	2.06E−14(4.8E−14)§
	3	**0.00E+0(0.0E+0)**	**0.00E+0(0.0E+0)§**	**0.00E+0(0.0E+0)§**	**0.00E+0(0.0E+0)§**	**0.00E+0(0.0E+0)§**
	4	**0.00E+0(0.0E+0)**	**0.00E+0(0.0E+0)§**	**0.00E+0(0.0E+0)§**	**0.00E+0(0.0E+0)§**	**0.00E+0(0.0E+0)§**
	5	**9.81E−14(1.1E−13)**	3.52E−13(4.1E−13)•	1.27E−13(2.6E−14)•	5.89E−13(1.4E−12)•	1.40E−12(2.8E−12)•
	6	3.14E−8(3.0E−8)	1.07E−7(3.3E−7)§	**1.69E−8(1.1E−8)§**	2.38E−8(1.4E−8)§	2.80E−8(3.0E−8)§
	7	6.08E−12(4.6E−12)	3.05E−12(4.0E−12)∘	3.66E−12(4.4E−12)∘	1.46E−11(5.1E−11)•	**2.16E−12(1.9E−12)**∘
	8	2.40E+2(4.5E+2)	7.57E+0(3.0E+1)∘	**1.02E−1(1.2E−1)**∘	8.49E+0(3.0E+1)∘	9.38E+1(2.1E+2)§
	9	**3.48E−16(4.0E−16)**	4.62E−4(1.8E−3)•	3.19E−14(2.5E−14)•	6.16E−4(2.5E−3)•	4.62E−4(1.8E−3)•
	10	1.62E+1(5.5E+0)	**1.54E+1(3.9E+0)§**	1.62E+1(2.9E+0)§	1.69E+1(3.8E+0)§	1.66E+1(5.3E+0)§
Best/All		5/10	3/10	5/10	2/10	3/10
Friedman		2.5	2.7	1.6	3.2	3

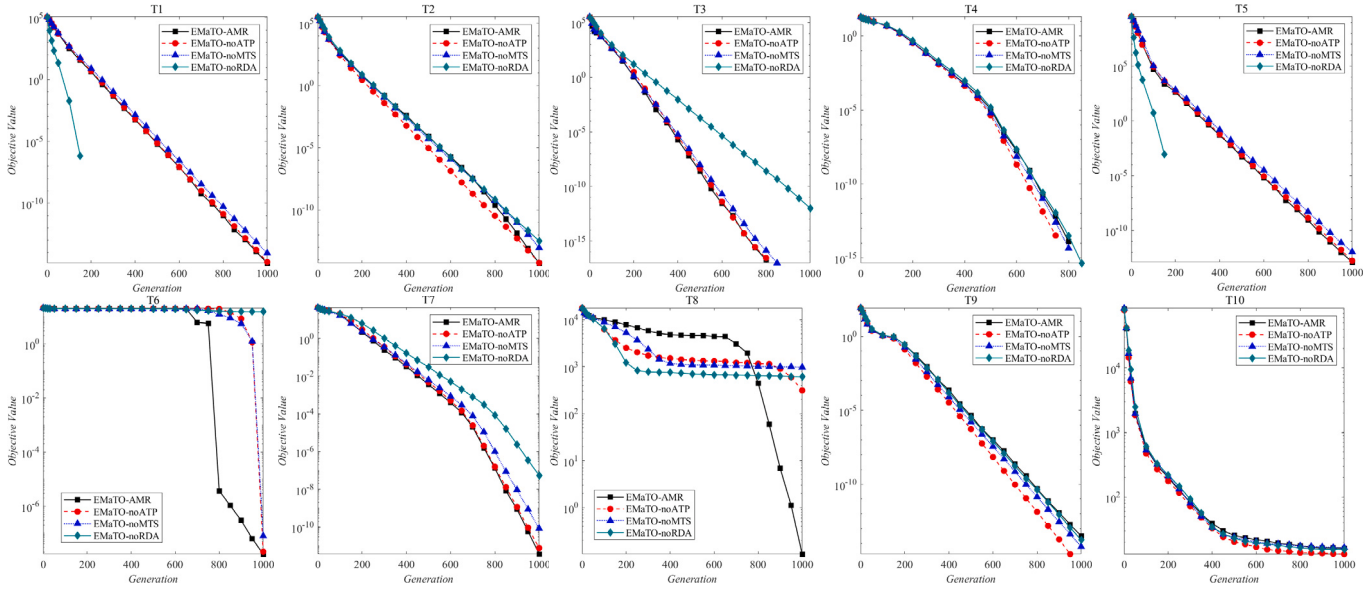


Fig. 5. The average convergence curves of EMaTO-AMR and its variants on 10-task test suite.

As is shown in Table 7, EMaTO-AMR achieves the best performance when $l = 5$, a low value of l may omit some useful transfer sources and incur insufficient knowledge transfer, whereas a large value of l may mislead the selection of related sources and thus results in negative

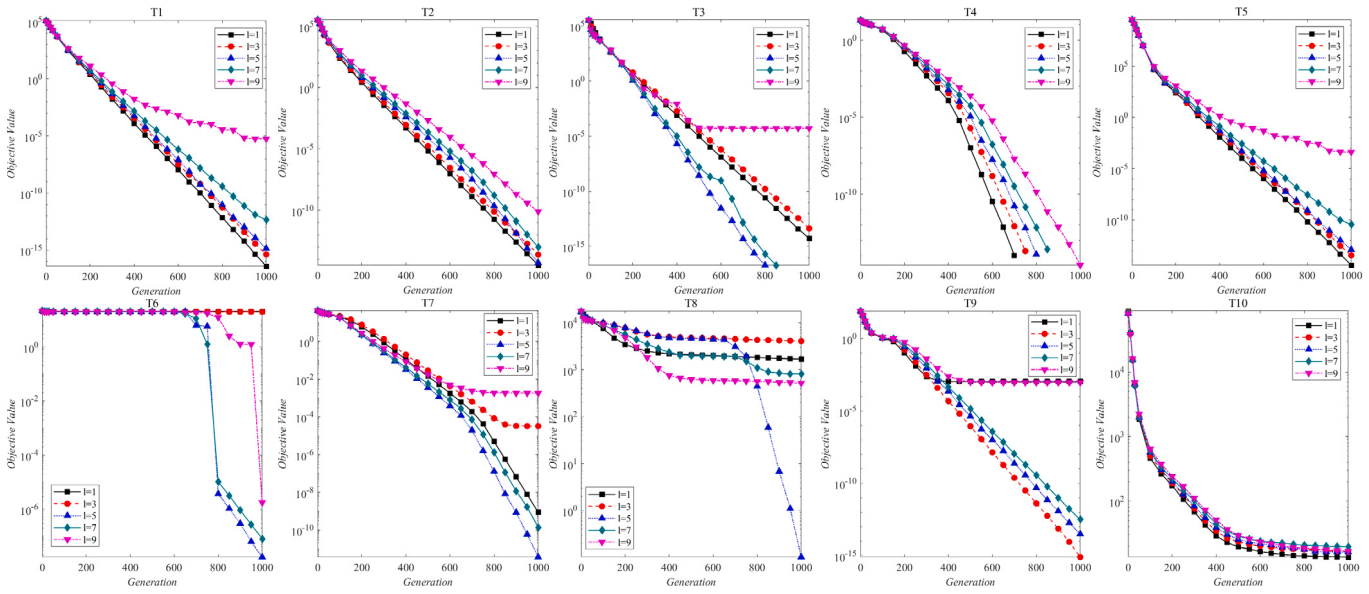


Fig. 6. The average convergence curves of EMaTO-AMR with different configurations of l on 10-task test suite.

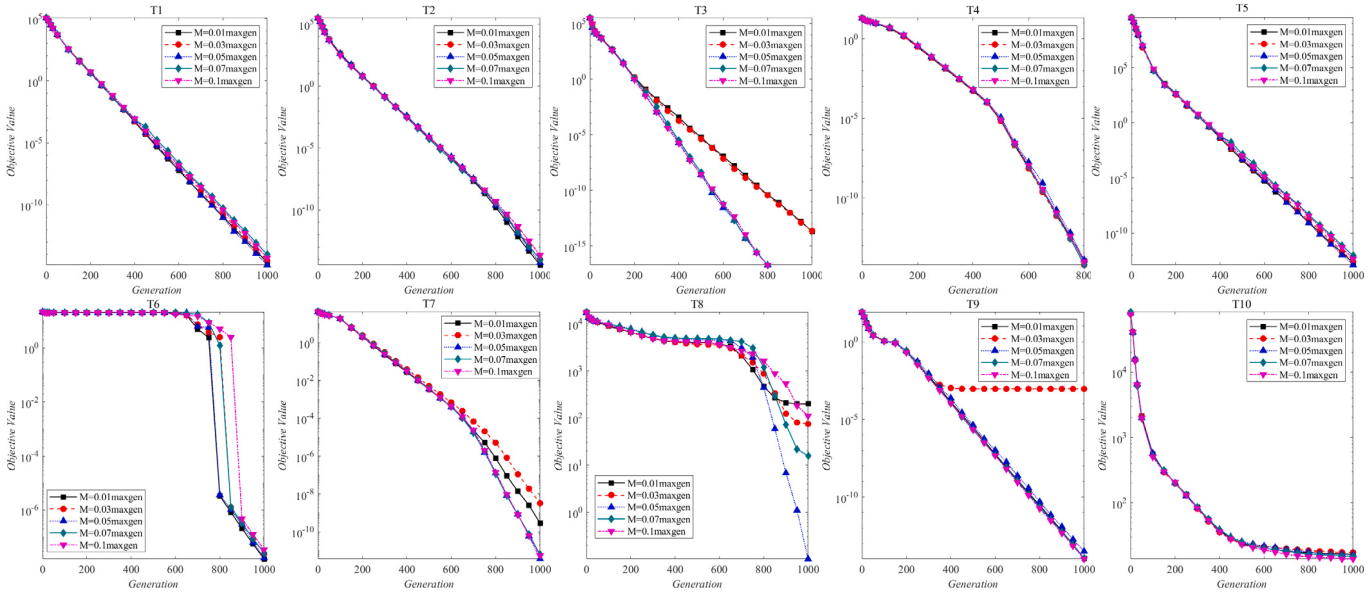


Fig. 7. The average convergence curves of EMaTO-AMR with different configurations of M on 10-task test suite.

transfer. In Table 8, EMaTO-AMR performs the best roughly when $M = 0.05 * maxgen$. The reason is that a small M value is hard to accumulate enough feedbacks for a reliable estimation of credit and may cause the credit evaluation fluctuates violently. On the other hand, when M rises considerably, a large portion of collected feedbacks are outdated and cannot track the dynamics rapidly. In Table 9, it seems that $w = 1$ can well balance the effects of reward-oriented credit and frequently-oriented one in evaluating the credit of distinct ITVs for EMaTO-AMR. Based on above sensitivity analysis, it is plausible to configure $l = 5$, $M = 0.05 * maxgen$ and $w = 1$, respectively.

4.4. Many-task optimization in more complex environment

In this section, we investigate the scalability of EMaTO-AMR in a series of 50-task problems, the statistical results of competing solvers on 50-task test suite are summarized in Table 10, where the values outside brackets represent the number of best results achieved by corresponding solver on each 50-task problem, the values inside brackets are the rank obtained by Friedman test. More details for comparison results of competing solvers on WCCI-SOMATP1 to WCCI-SOMATP10 are listed in Tables 11–20.

As shown in Table 10 for scenarios of 50-task, EMaTO-AMR still performs the best among all the compared solvers. In particular, EMaTO-AMR ranks the first on 8 out of 10 EMaTO problems. According to Table 3, component tasks of WCCI-SOMATP1-3 are composed of the same basic functions with distinct biases and rotation matrices. EMaTO-AMR works quite well on all problems except WCCI-SOMATP2-3 where MaTEA exhibits remarkable search ability. EMaTO-AMR fails to maintain the top position in WCCI-SOMATP2-3, one important reason for this occurrence is that component tasks in these two problems are Rosenbrock and Rastrigin functions with many local optimums, the top N_{top} individuals of each selected source task for carrier of knowledge

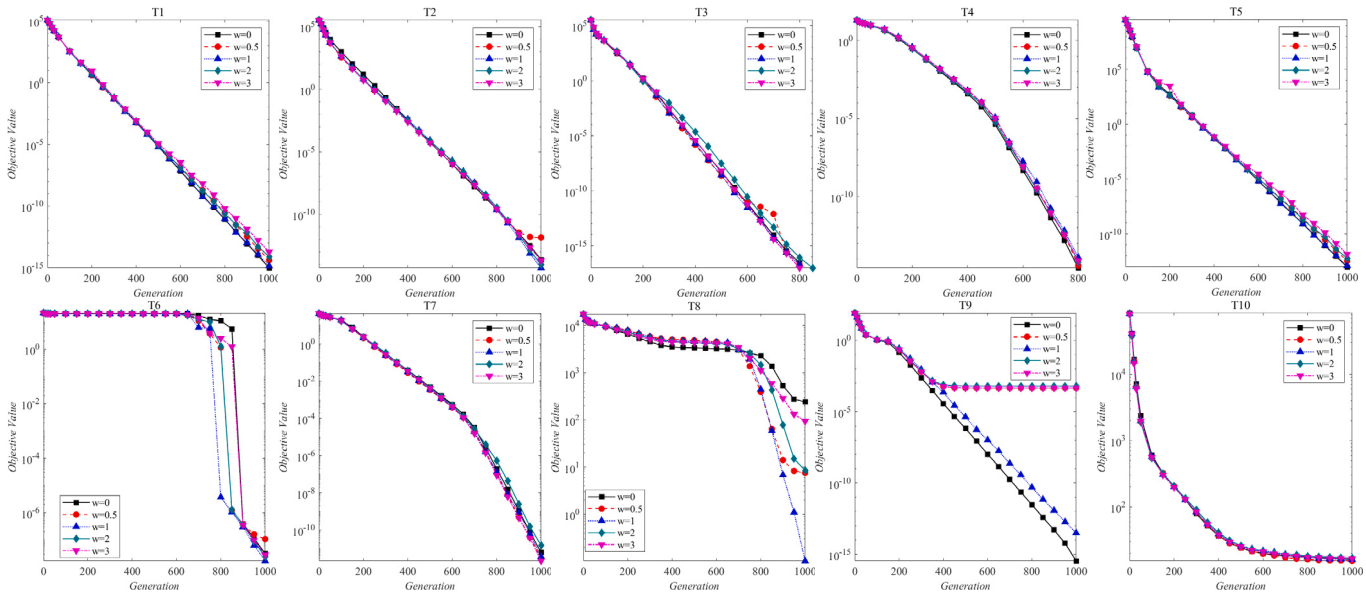


Fig. 8. The average convergence curves of EMaTO-AMR with different configurations of w on 10-task test suite.

Table 10
The number (rank) of best results obtained by competing solvers on each problem of 50-task test suite.

Problem	EMaTO-AMR	AT-MFEA	EEMTA	MaTEA	MFEA	SREMTO
WCCI2020-SOMATP1	**50(1)**	0(5)	0(4)	0(3)	0(6)	0(2)
WCCI2020-SOMATP2	23(2)	0(3)	3(5)	**24(1)**	0(6)	0(4)
WCCI2020-SOMATP3	0(2)	0(3)	0(4)	**50(1)**	0(6)	0(5)
WCCI2020-SOMATP4	**35(1)**	0(5)	4(4)	3(3)	0(6)	8(2)
WCCI2020-SOMATP5	**29(1)**	0(5)	11(2)	10(1)	0(4)	0(3)
WCCI2020-SOMATP6	**40(1)**	0(5)	1(4)	0(3)	0(6)	9(2)
WCCI2020-SOMATP7	**33(1)**	0(6)	9(3)	8(2)	0(4)	0(5)
WCCI2020-SOMATP8	**33(1)**	0(6)	8(3)	6(2)	0(5)	3(4)
WCCI2020-SOMATP9	**29(1)**	0(6)	4(5)	9(2)	3(4)	5(3)
WCCI2020-SOMATP10	**31(1)**	0(6)	10(2)	5(3)	2(4)	2(5)

transfer may decrease the population diversity of target task, thereby posing a big challenge for escaping from local optimum. MaTEA ranks the first in WCCI-SOMATP2-3, the possible reason is that transferred individuals are randomly selected from the source task in MaTEA, giving rise to better diversity maintenance. WCCI-SOMATP4-10 comprise a hybrid of distinct basis functions with great difficulty. Thanks to the joint use of ATP, MTS and RDA components, EMaTO-AMR works quite well on them.

5. Conclusion

Possessing the ability to solve many tasks simultaneously is highly desirable for EAs. Unfortunately, most of the existing EMTO solvers struggle to handle problems with more than three tasks due to a variety of challenges unique to many-task scenarios. In this article, we have devised an efficient EMaTO solver with adaptive knowledge transfer, named EMaTO-AMR. Particularly, ATP exploits the historical search data to adjust the intensity of knowledge transfer, MTS helps each component task choose suitable sources for knowledge transfer, RDA takes advantage of the RBM model to reduce the discrepancy between task-specific search spaces. A series of experiments have been carried out to testify the efficacy of our proposal against several state-of-the-art methods. Besides, the effectiveness of newly introduced components in EMaTO-AMR is empirically verified.

Although EMaTO-AMR has achieved encouraging results, there is still much room to further enhance its search ability in complicated problems. For example, an adaptive way to select the control parameters in EMaTO-AMR is more desirable. The elitist based selection of transferred individuals in RDA may cause premature convergence, how to draw a good balance between refining the search and knowledge transfer is worthy of further study. In addition, it is essential to extend EMaTO-AMR in solving real-world engineering cases.

CRediT authorship contribution statement

Jiajun Zhou: Conceptualization, Methodology, Writing – review & editing. **Shijie Rao:** Writing – original draft, Validation & experiments. **Liang Gao:** Conceptualization, Methodology, Supervision. **Chunjiang Zhang:** Writing – review & editing. **Hongtao Tang:** Methodology, Reviewing & editing. **Yun Li:** Formal analysis. **Felix T.S. Chan:** Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table 11
The comparison results of objective values for EMaTO-AMR, MFEA, EEMTA, AT-MFEA, SREMTO, MaTEA on WCCI2020-SOMATP1.

	k	EMaTO-AMR	AT-MFEA	EEMTA	MaTEA	MFEA	SREMTO
P1	1	**2.20E−13(1.3E−13)**	1.22E+0(2.5E−1)•	1.93E+0(5.0E−1)•	1.05E−2(4.0E−2)•	1.16E+2(1.8E+1)•	3.14E−6(3.7E−6)•
	2	**2.62E−13(2.5E−13)**	1.11E+0(2.7E−1)•	1.04E−6(3.6E−7)•	1.09E−5(2.0E−5)•	1.18E+2(2.1E+1)•	4.34E−6(5.5E−6)•
	3	**2.59E−13(2.5E−13)**	1.18E+0(2.0E−1)•	1.99E+0(4.6E−1)•	5.01E−5(1.4E−4)•	1.13E+2(1.7E+1)•	6.57E−6(1.6E−5)•
	4	**3.25E−13(2.2E−13)**	1.13E+0(2.2E−1)•	8.35E−7(2.8E−7)•	1.85E−5(5.3E−5)•	1.21E+2(2.7E+1)•	6.11E−6(1.5E−5)•
	5	**2.37E−13(1.9E−13)**	1.13E+0(2.0E−1)•	1.92E+0(4.2E−1)•	9.45E−5(3.7E−4)•	1.13E+2(1.2E+1)•	3.78E−6(7.4E−6)•
	6	**2.25E−13(2.3E−13)**	1.21E+0(2.0E−1)•	8.35E−7(3.0E−7)•	7.06E−6(1.0E−5)•	1.20E+2(2.2E+1)•	7.38E−6(1.7E−5)•
	7	**2.14E−13(1.3E−13)**	1.07E+0(2.8E−1)•	2.10E+0(6.1E−1)•	3.53E−4(1.1E−3)•	1.17E+2(1.7E+1)•	5.43E−6(1.0E−5)•
	8	**2.43E−13(1.9E−13)**	1.13E+0(2.6E−1)•	1.01E−6(3.7E−7)•	7.84E−5(1.9E−4)•	1.21E+2(2.9E+1)•	4.46E−6(6.1E−6)•
	9	**2.84E−13(2.1E−13)**	1.17E+0(3.4E−1)•	2.03E+0(5.0E−1)•	3.80E−5(7.8E−5)•	1.19E+2(1.5E+1)•	3.78E−6(4.5E−6)•
	10	**3.39E−13(2.3E−13)**	1.31E+0(2.7E−1)•	9.42E−7(3.8E−7)•	2.20E−5(5.1E−5)•	1.29E+2(1.3E+1)•	4.90E−6(9.0E−6)•
	11	**2.93E−13(2.7E−13)**	1.10E+0(1.9E−1)•	1.90E+0(4.6E−1)•	5.04E−4(1.8E−3)•	1.17E+2(1.9E+1)•	3.07E−6(5.5E−6)•
	12	**2.12E−13(1.3E−13)**	1.08E+0(2.2E−1)•	8.19E−7(3.9E−7)•	3.58E−4(1.4E−3)•	1.09E+2(1.8E+1)•	2.77E−6(4.0E−6)•
	13	**3.69E−13(5.5E−13)**	1.38E+0(2.4E−1)•	1.96E+0(5.2E−1)•	1.48E−3(5.6E−3)•	1.17E+2(1.8E+1)•	5.98E−6(1.1E−5)•
	14	**4.20E−13(3.0E−13)**	1.20E+0(2.7E−1)•	1.02E−6(4.9E−7)•	6.59E−5(1.7E−4)•	1.15E+2(2.0E+1)•	3.84E−6(5.8E−6)•
	15	**3.18E−13(3.2E−13)**	1.09E+0(2.1E−1)•	1.73E+0(3.6E−1)•	1.84E−5(3.7E−5)•	1.13E+2(1.8E+1)•	3.86E−6(5.8E−6)•
	16	**3.70E−13(2.8E−13)**	1.17E+0(2.5E−1)•	8.98E−7(4.1E−7)•	6.69E−4(2.6E−3)•	1.16E+2(2.2E+1)•	4.50E−6(7.4E−6)•
	17	**1.79E−13(1.0E−13)**	1.26E+0(3.7E−1)•	2.00E+0(5.3E−1)•	1.14E−3(4.4E−3)•	1.14E+2(2.3E+1)•	3.48E−6(4.7E−6)•
	18	**2.46E−13(1.6E−13)**	1.15E+0(2.9E−1)•	1.11E−6(4.1E−7)•	9.49E−4(3.6E−3)•	1.10E+2(1.7E+1)•	5.07E−6(7.2E−6)•
	19	**2.30E−13(1.3E−13)**	1.18E+0(3.5E−1)•	1.95E+0(3.6E−1)•	7.47E−4(3.0E−3)•	1.12E+2(1.8E+1)•	1.14E−5(2.8E−5)•
	20	**4.17E−13(4.7E−13)**	1.13E+0(2.4E−1)•	9.53E−7(2.2E−7)•	3.56E−6(4.1E−6)•	1.07E+2(1.9E+1)•	4.15E−6(5.8E−6)•
	21	**3.11E−13(2.4E−13)**	1.01E+0(2.4E−1)•	2.06E+0(5.2E−1)•	8.19E−5(2.7E−4)•	1.04E+2(1.9E+1)•	5.38E−6(9.6E−6)•
	22	**3.33E−13(2.8E−13)**	1.12E+0(2.1E−1)•	9.17E−7(3.6E−7)•	6.80E−5(2.2E−4)•	1.11E+2(1.3E+1)•	4.43E−6(7.7E−6)•
	23	**2.52E−13(1.3E−13)**	1.30E+0(3.7E−1)•	1.91E+0(6.5E−1)•	4.93E−5(1.2E−4)•	1.26E+2(2.2E+1)•	5.95E−6(9.9E−6)•
	24	**2.68E−13(1.7E−13)**	1.14E+0(2.7E−1)•	9.78E−7(4.0E−7)•	1.53E−4(4.0E−4)•	1.16E+2(2.6E+1)•	5.20E−6(9.6E−6)•
	25	**2.61E−13(1.6E−13)**	1.14E+0(3.0E−1)•	2.05E+0(5.8E−1)•	2.98E−5(6.2E−5)•	1.11E+2(2.2E+1)•	3.68E−6(6.7E−6)•
	26	**2.73E−13(2.5E−13)**	1.20E+0(2.8E−1)•	1.07E−6(4.0E−7)•	1.75E−4(4.5E−4)•	1.09E+2(1.6E+1)•	5.51E−6(8.6E−6)•
	27	**2.36E−13(1.8E−13)**	1.23E+0(3.0E−1)•	1.89E+0(5.6E−1)•	1.36E−5(2.2E−5)•	1.17E+2(1.4E+1)•	4.89E−6(1.0E−5)•
	28	**1.98E−13(1.5E−13)**	1.22E+0(2.9E−1)•	1.11E−6(3.7E−7)•	4.12E−5(1.5E−4)•	1.20E+2(2.1E+1)•	4.52E−6(7.6E−6)•
	29	**2.88E−13(2.1E−13)**	1.10E+0(1.6E−1)•	2.08E+0(4.8E−1)•	1.44E−5(2.4E−5)•	1.15E+2(1.8E+1)•	6.27E−6(1.3E−5)•
	30	**2.51E−13(2.6E−13)**	1.32E+0(2.6E−1)•	8.87E−7(4.4E−7)•	8.67E−6(2.0E−5)•	1.10E+2(1.5E+1)•	5.45E−6(1.2E−5)•
	31	**2.89E−13(1.7E−13)**	1.06E+0(2.1E−1)•	1.83E+0(5.6E−1)•	9.59E−4(2.9E−3)•	1.21E+2(3.0E+1)•	3.83E−6(5.4E−6)•
	32	**3.29E−13(3.0E−13)**	1.23E+0(3.1E−1)•	9.50E−7(3.3E−7)•	2.99E−4(1.0E−3)•	1.17E+2(2.1E+1)•	4.19E−6(5.9E−6)•
	33	**2.60E−13(1.9E−13)**	1.05E+0(2.7E−1)•	2.11E+0(4.5E−1)•	3.63E−4(1.1E−3)•	1.11E+2(1.4E+1)•	3.98E−6(6.9E−6)•
	34	**2.63E−13(2.3E−13)**	1.12E+0(2.0E−1)•	8.86E−7(3.4E−7)•	2.33E−5(5.4E−5)•	1.14E+2(1.0E+1)•	5.80E−6(1.1E−5)•
	35	**2.64E−13(2.4E−13)**	1.11E+0(1.9E−1)•	2.03E+0(4.9E−1)•	1.16E−5(3.2E−5)•	1.07E+2(1.5E+1)•	7.95E−6(1.9E−5)•
	36	**3.23E−13(3.6E−13)**	1.20E+0(2.0E−1)•	1.01E−6(4.2E−7)•	2.40E−5(6.3E−5)•	1.25E+2(2.8E+1)•	4.80E−6(7.8E−6)•
	37	**2.91E−13(1.9E−13)**	1.15E+0(2.7E−1)•	1.90E+0(6.0E−1)•	8.42E−5(2.6E−4)•	1.09E+2(1.6E+1)•	6.43E−6(1.7E−5)•
	38	**1.87E−13(1.1E−13)**	1.22E+0(3.2E−1)•	1.12E−6(4.2E−7)•	1.96E−5(4.4E−5)•	1.06E+2(1.7E+1)•	5.56E−6(1.0E−5)•
	39	**2.72E−13(2.2E−13)**	1.22E+0(2.2E−1)•	1.92E+0(4.7E−1)•	5.32E−4(1.5E−3)•	1.08E+2(1.7E+1)•	4.98E−6(8.3E−6)•
	40	**2.83E−13(2.6E−13)**	1.11E+0(3.1E−1)•	9.72E−7(2.5E−7)•	4.72E−6(1.1E−5)•	1.16E+2(1.9E+1)•	2.92E−6(3.8E−6)•
	41	**3.40E−13(4.5E−13)**	1.12E+0(2.0E−1)•	1.85E+0(5.4E−1)•	3.83E−6(4.9E−6)•	1.14E+2(2.1E+1)•	3.74E−6(6.9E−6)•
	42	**3.26E−13(3.3E−13)**	1.22E+0(2.7E−1)•	9.19E−7(3.1E−7)•	2.34E−5(8.4E−5)•	1.20E+2(1.7E+1)•	3.93E−6(5.7E−6)•
	43	**2.37E−13(2.0E−13)**	1.17E+0(2.5E−1)•	2.10E+0(5.2E−1)•	5.92E−5(2.1E−4)•	1.14E+2(1.7E+1)•	4.24E−6(8.8E−6)•
	44	**2.44E−13(1.6E−13)**	1.25E+0(2.5E−1)•	9.43E−7(3.7E−7)•	3.34E−5(8.3E−5)•	1.06E+2(2.2E+1)•	3.44E−6(6.3E−6)•
	45	**3.79E−13(3.8E−13)**	1.25E+0(2.7E−1)•	1.93E+0(4.1E−1)•	7.40E−6(1.3E−5)•	1.12E+2(1.5E+1)•	5.62E−6(8.7E−6)•
	46	**2.54E−13(1.6E−13)**	1.19E+0(2.9E−1)•	1.13E−6(6.1E−7)•	9.89E−6(2.0E−5)•	1.11E+2(2.0E+1)•	4.96E−6(9.7E−6)•
	47	**1.77E−13(1.1E−13)**	1.23E+0(2.5E−1)•	1.93E+0(3.8E−1)•	4.24E−5(7.5E−5)•	1.22E+2(1.8E+1)•	5.51E−6(8.1E−6)•
	48	**1.83E−13(1.3E−13)**	1.07E+0(2.3E−1)•	9.15E−7(4.4E−7)•	2.11E−2(8.4E−2)•	1.11E+2(1.7E+1)•	7.39E−6(1.4E−5)•
	49	**3.82E−13(3.5E−13)**	1.14E+0(2.5E−1)•	1.73E+0(4.3E−1)•	5.11E−5(1.3E−4)•	1.19E+2(1.2E+1)•	4.56E−6(8.2E−6)•
	50	**2.97E−13(2.9E−13)**	1.23E+0(2.8E−1)•	1.03E−6(5.2E−7)•	6.04E−5(1.1E−4)•	1.12E+2(1.4E+1)•	3.08E−6(5.3E−6)•
Best/All		50/50	0/50	0/50	0/50	0/50	0/50
Friedman		1	4.5	3.5	3.46	6	2.54
$w/s/l$		–	50/0/0	50/0/0	50/0/0	50/0/0	50/0/0

Table 12
The comparison results of objective values for EMaTO-AMR, MFEA, EEMTA, AT-MFEA, SREMTO, MaTEA on WCCI2020-SOMATP2.

	k	EMaTO-AMR	AT-MFEA	EEMTA	MaTEA	MFEA	SREMTO
P2	1	**9.65E+1(1.0E+2)**	4.82E+2(4.2E+2)•	1.44E+4(3.5E+4)•	1.44E+2(9.6E+1)•	1.84E+4(9.6E+3)•	3.92E+2(4.4E+2)•
	2	3.49E+2(5.9E+2)	4.50E+2(2.2E+2)•	**1.04E+2(9.7E+1)**∘	3.57E+2(2.9E+2)§	1.20E+4(3.9E+3)•	8.18E+2(1.4E+3)§
	3	3.25E+2(7.5E+2)	4.15E+2(2.1E+2)•	1.29E+4(2.5E+4)•	**1.33E+2(9.5E+1)§**	1.48E+4(1.1E+4)•	7.31E+2(9.3E+2)§
	4	**4.80E+1(1.1E+0)**	6.42E+2(4.7E+2)•	2.58E+2(4.2E+2)•	1.44E+2(6.6E+1)•	1.48E+4(6.6E+3)•	9.91E+2(1.1E+3)•
	5	**1.66E+2(1.9E+2)**	6.37E+2(4.3E+2)•	1.06E+4(2.7E+4)•	2.58E+2(2.1E+2)•	1.44E+4(7.6E+3)•	5.78E+2(7.1E+2)•
	6	1.49E+2(1.9E+2)	6.13E+2(5.1E+2)•	5.29E+2(1.8E+3)§	**1.35E+2(1.6E+2)§**	1.49E+4(8.0E+3)•	6.64E+2(9.1E+2)•
	7	3.06E+2(7.3E+2)	4.73E+2(2.2E+2)•	2.17E+4(5.3E+4)•	**1.56E+2(1.3E+2)§**	1.41E+4(6.4E+3)•	2.22E+2(2.1E+2)§
	8	1.81E+2(3.9E+2)	3.40E+2(1.6E+2)•	7.15E+2(1.2E+3)•	**8.87E+1(4.0E+1)§**	1.51E+4(7.2E+3)•	2.25E+2(2.3E+2)•
	9	7.34E+2(1.2E+3)	7.65E+2(7.1E+2)•	1.04E+4(2.3E+4)•	**3.44E+2(4.1E+2)§**	1.24E+4(4.5E+3)•	8.37E+2(9.9E+2)§
	10	3.46E+2(1.0E+3)	4.91E+2(2.9E+2)•	6.34E+2(1.5E+3)•	**1.23E+2(1.1E+2)**∘	1.67E+4(7.1E+3)•	7.38E+2(1.1E+3)•
	11	2.00E+2(4.7E+2)	4.49E+2(4.8E+2)•	9.94E+3(3.3E+4)•	**1.08E+2(5.0E+1)**∘	1.81E+4(1.2E+4)•	8.55E+2(9.4E+2)•
	12	2.41E+2(5.8E+2)	5.48E+2(4.3E+2)•	1.99E+2(3.8E+2)§	**1.23E+2(5.5E+1)**∘	1.59E+4(1.1E+4)•	7.13E+2(1.0E+3)§
	13	**2.05E+2(4.9E+2)**	4.46E+2(2.8E+2)•	9.68E+3(2.7E+4)•	2.70E+2(3.3E+2)•	1.55E+4(5.8E+3)•	7.99E+2(9.9E+2)§
	14	2.34E+2(5.7E+2)	4.30E+2(3.8E+2)•	2.98E+2(5.1E+2)§	**1.91E+2(2.8E+2)**∘	1.64E+4(9.6E+3)•	6.68E+2(9.8E+2)•
	15	**1.34E+2(2.2E+2)**	7.47E+2(6.5E+2)•	2.20E+4(5.9E+4)•	3.97E+2(4.7E+2)•	1.22E+4(4.9E+3)•	1.21E+3(1.4E+3)•
	16	4.08E+2(8.4E+2)	4.81E+2(4.6E+2)•	6.03E+2(1.2E+3)§	**1.17E+2(1.4E+2)§**	1.56E+4(7.6E+3)•	6.52E+2(1.1E+3)§
	17	1.52E+2(2.2E+2)	4.60E+2(3.2E+2)•	6.57E+4(1.4E+5)•	**1.45E+2(5.7E+1)**∘	1.50E+4(8.9E+3)•	3.17E+2(5.1E+2)§
	18	3.46E+2(6.5E+2)	6.78E+2(5.5E+2)•	6.48E+2(1.1E+3)§	**3.00E+2(2.9E+2)§**	1.37E+4(5.1E+3)•	1.10E+3(1.3E+3)•
	19	**9.10E+1(7.3E+1)**	3.20E+2(3.8E+2)•	2.72E+3(3.4E+3)•	2.13E+2(3.1E+2)•	1.83E+4(8.8E+3)•	7.64E+2(9.1E+2)§
	20	**1.00E+2(1.2E+2)**	4.02E+2(1.8E+2)•	3.42E+2(7.4E+2)§	1.29E+2(4.6E+1)•	1.53E+4(1.1E+4)•	1.52E+2(8.6E+1)•
	21	2.82E+2(7.2E+2)	5.04E+2(3.2E+2)•	1.48E+4(1.8E+4)•	**1.54E+2(1.2E+2)§**	1.59E+4(7.5E+3)•	4.79E+2(1.0E+3)§
	22	2.83E+2(6.5E+2)	6.19E+2(5.7E+2)•	2.96E+2(8.8E+2)§	**1.01E+2(4.8E+1)**∘	1.77E+4(1.1E+4)•	7.59E+2(8.7E+2)•
	23	**2.28E+2(3.6E+2)**	4.39E+2(2.7E+2)•	1.64E+4(3.4E+4)•	3.30E+2(3.5E+2)§	1.44E+4(7.3E+3)•	8.56E+2(1.1E+3)§
	24	2.79E+2(7.7E+2)	5.05E+2(3.9E+2)•	1.12E+3(1.9E+3)§	**1.22E+2(9.0E+1)**∘	1.63E+4(7.3E+3)•	6.56E+2(1.0E+3)§
	25	**1.24E+2(2.8E+2)**	6.24E+2(5.4E+2)•	9.73E+3(2.7E+4)•	1.25E+2(6.0E+1)•	1.51E+4(7.2E+3)•	7.12E+2(1.2E+3)•
	26	1.37E+2(1.9E+2)	3.72E+2(1.9E+2)•	5.01E+2(8.6E+2)§	**1.08E+2(5.0E+1)§**	1.58E+4(9.1E+3)•	1.86E+2(1.7E+2)§
	27	**1.52E+2(2.4E+2)**	4.24E+2(1.9E+2)•	2.37E+4(7.8E+4)•	2.03E+2(1.8E+2)•	1.48E+4(6.8E+3)•	4.24E+2(6.2E+2)§
	28	2.55E+2(6.0E+2)	6.39E+2(7.0E+2)•	5.43E+2(1.7E+3)§	**1.43E+2(1.3E+2)**∘	1.72E+4(1.1E+4)•	8.73E+2(1.1E+3)•
	29	3.41E+2(9.5E+2)	4.54E+2(2.2E+2)•	3.15E+4(5.6E+4)•	**1.57E+2(1.9E+2)§**	1.49E+4(5.5E+3)•	3.53E+2(4.7E+2)§
	30	3.76E+2(6.9E+2)	7.41E+2(8.1E+2)•	**3.71E+2(9.9E+2)§**	5.12E+2(4.6E+2)•	1.25E+4(2.9E+3)•	1.26E+3(1.5E+3)•
	31	**1.48E+2(2.2E+2)**	6.60E+2(6.8E+2)•	1.30E+4(3.1E+4)•	2.04E+2(2.3E+2)•	1.25E+4(3.4E+3)•	7.61E+2(8.8E+2)•
	32	**1.63E+2(1.5E+2)**	3.58E+2(1.4E+2)•	5.74E+2(1.1E+3)§	1.93E+2(9.8E+1)§	1.32E+4(3.7E+3)•	2.92E+2(2.4E+2)§
	33	**7.32E+1(6.6E+1)**	3.66E+2(1.8E+2)•	5.71E+3(1.1E+4)•	1.51E+2(1.1E+2)•	1.74E+4(1.1E+4)•	1.97E+2(2.1E+2)•
	34	1.25E+2(2.0E+2)	5.65E+2(5.3E+2)•	6.58E+2(1.3E+3)§	**9.60E+1(5.1E+1)**∘	1.58E+4(7.9E+3)•	8.28E+2(8.9E+2)•
	35	**8.54E+1(1.1E+2)**	3.95E+2(2.3E+2)•	1.47E+4(3.9E+4)•	1.45E+2(1.0E+2)•	1.82E+4(1.4E+4)•	4.58E+2(5.7E+2)•
	36	**1.14E+2(1.6E+2)**	5.27E+2(5.5E+2)•	7.36E+2(1.3E+3)•	2.06E+2(1.3E+2)•	1.34E+4(3.9E+3)•	4.25E+2(5.9E+2)•
	37	**9.86E+1(1.2E+2)**	4.72E+2(2.8E+2)•	1.34E+4(3.0E+4)•	1.34E+2(7.9E+1)•	1.34E+4(7.4E+3)•	3.26E+2(3.1E+2)§
	38	**1.10E+2(1.4E+2)**	5.22E+2(3.6E+2)•	8.52E+2(2.1E+3)§	2.97E+2(4.3E+2)•	1.34E+4(3.1E+3)•	1.89E+2(1.7E+2)•
	39	5.72E+2(6.9E+2)	9.54E+2(8.1E+2)•	3.59E+4(9.7E+4)•	**3.76E+2(2.7E+2)§**	1.44E+4(6.5E+3)•	1.14E+3(1.1E+3)§
	40	**1.15E+2(1.4E+2)**	5.34E+2(4.5E+2)•	2.55E+2(6.3E+2)§	1.79E+2(2.3E+2)•	1.34E+4(6.9E+3)•	6.44E+2(8.5E+2)§
	41	**1.39E+2(1.1E+2)**	5.37E+2(2.1E+2)•	1.73E+4(4.3E+4)•	2.68E+2(1.1E+2)•	1.37E+4(5.5E+3)•	2.64E+2(2.3E+2)§
	42	4.47E+2(1.1E+3)	6.48E+2(6.3E+2)•	**1.79E+2(2.5E+2)§**	1.82E+2(2.7E+2)∘	1.51E+4(5.3E+3)•	8.94E+2(1.1E+3)•
	43	**9.07E+1(1.1E+2)**	4.40E+2(2.8E+2)•	3.12E+4(8.5E+4)•	1.27E+2(1.5E+2)•	1.66E+4(1.2E+4)•	2.45E+2(3.1E+2)§
	44	3.71E+2(1.0E+3)	4.94E+2(4.5E+2)•	2.00E+2(3.5E+2)§	**1.64E+2(1.2E+2)**∘	1.64E+4(9.8E+3)•	7.54E+2(9.2E+2)§
	45	4.19E+2(1.1E+3)	6.18E+2(6.6E+2)•	2.97E+4(1.0E+5)•	**1.62E+2(1.1E+2)§**	1.38E+4(4.1E+3)•	4.48E+2(5.8E+2)§
	46	**1.22E+2(1.4E+2)**	7.21E+2(6.2E+2)•	5.79E+2(1.2E+3)§	1.55E+2(1.0E+2)•	1.59E+4(8.7E+3)•	9.13E+2(1.1E+3)•
	47	3.29E+2(8.4E+2)	4.94E+2(4.0E+2)•	7.96E+3(1.3E+4)•	**2.31E+2(1.4E+2)**∘	1.34E+4(4.6E+3)•	7.48E+2(8.7E+2)§
	48	4.30E+2(6.7E+2)	8.79E+2(6.3E+2)•	3.73E+2(5.4E+2)§	**2.66E+2(2.5E+2)§**	1.50E+4(5.4E+3)•	9.89E+2(1.2E+3)§
	49	**6.61E+1(3.9E+1)**	6.29E+2(5.1E+2)•	8.68E+3(2.3E+4)•	2.13E+2(3.7E+2)•	1.40E+4(5.4E+3)•	5.83E+2(9.4E+2)•
	50	**7.93E+1(9.1E+1)**	4.02E+2(1.9E+2)•	1.52E+2(2.0E+2)§	1.76E+2(1.4E+2)•	1.65E+4(1.1E+4)•	7.29E+2(8.0E+2)•
Best/All		23/50	0/50	3/50	24/50	0/50	0/50
Fried		1.64	3.7	4.34	1.62	5.78	3.92
R		2	3	5	1	6	4
$w/s/l$		–	50/0/0	29/20/1	22/16/12	50/0/0	25/25/0

Table 13
The comparison results of objective values for EMaTO-AMR, MFEA, EEMTA, AT-MFEA, SREMTO, MaTEA on WCCI2020-SOMATP3.

	k	EMaTO-AMR	AT-MFEA	EEMTA	MaTEA	MFEA	SREMTO
P3	1	3.04E+2(8.8E+1)	3.46E+2(3.8E+1)§	3.14E+2(4.8E+1)§	**1.87E+2(3.6E+1)**∘	4.93E+2(2.4E+1)•	4.51E+2(7.6E+1)•
	2	2.88E+2(9.8E+1)	3.11E+2(4.1E+1)§	4.07E+2(1.5E+1)•	**1.71E+2(4.4E+1)**∘	4.94E+2(2.2E+1)•	4.53E+2(7.5E+1)•
	3	3.22E+2(9.3E+1)	3.07E+2(4.3E+1)∘	3.18E+2(5.0E+1)§	**1.64E+2(3.2E+1)**∘	4.84E+2(3.1E+1)•	3.95E+2(7.7E+1)•
	4	2.57E+2(1.2E+2)	3.31E+2(4.0E+1)§	3.90E+2(1.9E+1)•	**1.89E+2(6.1E+1)§**	4.91E+2(1.8E+1)•	4.25E+2(6.2E+1)•
	5	2.58E+2(1.3E+2)	3.03E+2(5.3E+1)§	3.46E+2(7.0E+1)§	**1.95E+2(4.6E+1)§**	4.94E+2(2.0E+1)•	3.86E+2(8.4E+1)•
	6	3.14E+2(9.4E+1)	3.46E+2(5.9E+1)§	3.98E+2(1.4E+1)•	**1.85E+2(4.2E+1)**∘	4.89E+2(3.2E+1)•	4.19E+2(8.1E+1)•
	7	2.78E+2(1.0E+2)	3.95E+2(4.6E+1)•	3.33E+2(6.8E+1)§	**1.89E+2(4.5E+1)**∘	5.01E+2(2.5E+1)•	4.83E+2(7.8E+1)•
	8	2.83E+2(9.0E+1)	3.69E+2(5.7E+1)•	4.07E+2(1.3E+1)•	**2.01E+2(4.9E+1)**∘	5.00E+2(2.1E+1)•	4.64E+2(6.5E+1)•
	9	3.20E+2(7.0E+1)	3.69E+2(6.7E+1)§	3.19E+2(4.3E+1)§	**1.70E+2(3.9E+1)**∘	4.91E+2(2.4E+1)•	4.28E+2(8.2E+1)•
	10	3.23E+2(8.8E+1)	3.39E+2(7.5E+1)§	4.02E+2(1.4E+1)•	**2.00E+2(5.1E+1)**∘	4.98E+2(2.2E+1)•	4.38E+2(1.0E+2)•
	11	2.86E+2(1.1E+2)	3.53E+2(7.2E+1)§	3.21E+2(7.3E+1)§	**1.83E+2(5.7E+1)**∘	4.88E+2(3.2E+1)•	4.30E+2(6.1E+1)•
	12	3.25E+2(7.6E+1)	3.34E+2(6.6E+1)§	4.03E+2(1.6E+1)•	**2.04E+2(5.1E+1)**∘	4.86E+2(2.9E+1)•	5.05E+2(1.5E+2)•
	13	2.92E+2(1.1E+2)	3.47E+2(4.9E+1)§	3.24E+2(6.8E+1)§	**1.84E+2(4.5E+1)**∘	4.90E+2(2.3E+1)•	4.38E+2(7.8E+1)•
	14	3.03E+2(8.3E+1)	3.60E+2(5.3E+1)•	4.04E+2(1.6E+1)•	**1.84E+2(3.7E+1)**∘	4.88E+2(1.6E+1)•	4.31E+2(1.2E+2)•
	15	3.36E+2(4.9E+1)	3.51E+2(5.8E+1)§	3.29E+2(6.8E+1)§	**1.79E+2(3.2E+1)**∘	5.05E+2(1.7E+1)•	4.55E+2(7.1E+1)•
	16	3.22E+2(8.9E+1)	3.66E+2(6.8E+1)§	4.04E+2(1.0E+1)•	**1.70E+2(3.5E+1)**∘	4.91E+2(1.3E+1)•	3.95E+2(5.6E+1)•
	17	2.91E+2(1.1E+2)	3.55E+2(4.2E+1)§	3.08E+2(5.4E+1)§	**1.69E+2(3.1E+1)**∘	4.90E+2(3.3E+1)•	4.11E+2(7.4E+1)•
	18	3.06E+2(9.0E+1)	3.28E+2(5.1E+1)§	4.01E+2(1.3E+1)•	**1.74E+2(3.9E+1)**∘	4.96E+2(2.3E+1)•	4.49E+2(8.4E+1)•
	19	2.96E+2(1.2E+2)	3.16E+2(6.5E+1)§	3.22E+2(4.0E+1)§	**1.98E+2(4.8E+1)**∘	4.81E+2(3.1E+1)•	4.48E+2(8.6E+1)•
	20	2.89E+2(1.1E+2)	4.00E+2(6.2E+1)•	4.03E+2(1.2E+1)•	**2.06E+2(3.2E+1)**∘	4.99E+2(1.7E+1)•	4.79E+2(8.2E+1)•
	21	2.75E+2(1.2E+2)	3.23E+2(6.4E+1)§	3.28E+2(7.5E+1)§	**1.76E+2(3.8E+1)§**	4.96E+2(2.8E+1)•	4.01E+2(7.2E+1)•
	22	2.73E+2(1.1E+2)	3.46E+2(6.3E+1)§	4.01E+2(1.4E+1)•	**1.91E+2(3.9E+1)**∘	5.00E+2(1.9E+1)•	4.85E+2(7.5E+1)•
	23	3.21E+2(1.0E+2)	3.28E+2(7.0E+1)§	3.38E+2(8.5E+1)§	**1.89E+2(4.1E+1)**∘	4.89E+2(2.6E+1)•	4.42E+2(1.0E+2)•
	24	2.91E+2(9.0E+1)	2.96E+2(5.1E+1)§	4.04E+2(1.6E+1)•	**1.72E+2(3.3E+1)**∘	4.83E+2(2.4E+1)•	4.21E+2(1.1E+2)•
	25	3.27E+2(7.4E+1)	3.19E+2(5.5E+1)§	3.57E+2(7.2E+1)§	**1.74E+2(3.6E+1)**∘	4.97E+2(1.8E+1)•	3.49E+2(7.8E+1)§
	26	2.72E+2(1.1E+2)	3.61E+2(4.4E+1)•	4.06E+2(2.0E+1)•	**1.87E+2(4.0E+1)§**	4.92E+2(2.8E+1)•	4.81E+2(9.8E+1)•
	27	2.83E+2(1.0E+2)	3.74E+2(4.9E+1)•	3.11E+2(6.2E+1)§	**1.95E+2(2.9E+1)**∘	4.83E+2(2.7E+1)•	5.18E+2(1.5E+2)•
	28	3.23E+2(7.1E+1)	3.29E+2(6.7E+1)§	4.03E+2(1.4E+1)•	**1.96E+2(4.8E+1)**∘	4.93E+2(2.4E+1)•	4.59E+2(1.3E+2)§
	29	3.09E+2(8.9E+1)	3.23E+2(5.2E+1)§	3.41E+2(4.6E+1)§	**1.78E+2(4.6E+1)**∘	4.99E+2(2.5E+1)•	4.65E+2(1.1E+2)•
	30	3.25E+2(6.7E+1)	3.12E+2(3.4E+1)§	4.07E+2(1.5E+1)•	**1.74E+2(3.1E+1)**∘	4.85E+2(2.2E+1)•	4.78E+2(8.1E+1)•
	31	3.17E+2(9.0E+1)	3.13E+2(4.0E+1)§	3.25E+2(4.4E+1)§	**1.77E+2(3.1E+1)**∘	4.94E+2(2.3E+1)•	4.19E+2(4.6E+1)•
	32	2.69E+2(1.1E+2)	3.20E+2(4.6E+1)§	3.99E+2(2.0E+1)•	**1.67E+2(4.1E+1)**∘	4.85E+2(3.1E+1)•	4.11E+2(9.5E+1)•
	33	3.07E+2(9.3E+1)	3.38E+2(5.4E+1)§	3.00E+2(4.4E+1)§	**1.83E+2(2.5E+1)**∘	4.89E+2(1.8E+1)•	4.43E+2(7.0E+1)•
	34	3.19E+2(8.3E+1)	3.29E+2(5.4E+1)§	4.03E+2(1.7E+1)•	**1.72E+2(3.3E+1)**∘	4.92E+2(2.5E+1)•	4.50E+2(1.2E+2)•
	35	3.23E+2(7.8E+1)	3.40E+2(5.1E+1)§	3.35E+2(6.1E+1)§	**1.82E+2(3.2E+1)**∘	4.89E+2(1.6E+1)•	4.53E+2(8.4E+1)•
	36	3.11E+2(9.2E+1)	3.37E+2(7.8E+1)§	4.09E+2(1.3E+1)•	**2.02E+2(4.8E+1)**∘	4.94E+2(3.1E+1)•	4.47E+2(7.4E+1)•
	37	2.88E+2(1.0E+2)	3.70E+2(6.9E+1)•	3.39E+2(4.2E+1)§	**1.73E+2(1.9E+1)**∘	4.98E+2(2.6E+1)•	4.38E+2(7.6E+1)•
	38	2.85E+2(1.1E+2)	3.56E+2(4.0E+1)§	4.04E+2(1.3E+1)•	**1.81E+2(5.3E+1)**∘	4.91E+2(2.1E+1)•	4.21E+2(5.5E+1)•
	39	3.17E+2(7.5E+1)	3.42E+2(5.2E+1)§	2.77E+2(4.5E+1)§	**1.81E+2(3.4E+1)**∘	4.94E+2(1.7E+1)•	4.42E+2(8.1E+1)•
	40	3.14E+2(8.3E+1)	3.23E+2(4.6E+1)§	4.00E+2(1.3E+1)•	**1.68E+2(3.2E+1)**∘	4.92E+2(1.6E+1)•	4.38E+2(7.4E+1)•
	41	3.07E+2(8.5E+1)	3.00E+2(5.2E+1)§	3.16E+2(4.7E+1)§	**1.83E+2(5.0E+1)**∘	4.98E+2(3.0E+1)•	4.20E+2(8.7E+1)•
	42	2.44E+2(1.3E+2)	3.22E+2(5.4E+1)§	4.07E+2(1.8E+1)•	**1.68E+2(5.3E+1)§**	4.92E+2(1.5E+1)•	4.53E+2(6.3E+1)•
	43	2.62E+2(1.2E+2)	3.81E+2(6.4E+1)•	3.39E+2(7.6E+1)§	**2.17E+2(7.0E+1)§**	5.04E+2(2.6E+1)•	4.41E+2(1.2E+2)•
	44	3.22E+2(7.2E+1)	3.24E+2(4.1E+1)§	4.02E+2(1.5E+1)•	**1.73E+2(3.2E+1)**∘	4.86E+2(2.3E+1)•	3.92E+2(8.0E+1)•
	45	3.00E+2(1.1E+2)	2.95E+2(4.7E+1)§	3.21E+2(6.7E+1)§	**1.75E+2(4.2E+1)**∘	4.88E+2(3.5E+1)•	4.01E+2(9.1E+1)•
	46	3.28E+2(7.5E+1)	3.71E+2(6.8E+1)§	3.95E+2(1.8E+1)•	**1.91E+2(5.5E+1)**∘	4.95E+2(1.7E+1)•	5.01E+2(1.3E+2)•
	47	3.04E+2(9.0E+1)	3.14E+2(5.2E+1)§	3.04E+2(6.6E+1)§	**1.78E+2(6.4E+1)**∘	4.84E+2(1.3E+1)•	4.14E+2(6.2E+1)•
	48	3.12E+2(9.7E+1)	3.48E+2(3.5E+1)§	4.04E+2(1.7E+1)•	**1.87E+2(3.9E+1)**∘	4.94E+2(1.9E+1)•	4.74E+2(9.6E+1)•
	49	3.09E+2(8.2E+1)	3.39E+2(6.0E+1)§	3.28E+2(6.3E+1)§	**1.78E+2(5.3E+1)**∘	4.88E+2(2.1E+1)•	4.44E+2(1.0E+2)•
	50	3.21E+2(8.8E+1)	3.34E+2(4.3E+1)§	3.99E+2(2.1E+1)•	**1.82E+2(3.9E+1)**∘	4.76E+2(1.8E+1)•	5.04E+2(8.9E+1)•
Best/All		0/50	0/50	0/50	50/50	0/50	0/50
Fried		2.22	3.18	3.66	1	5.92	5.02
R		2	3	4	1	6	5
$w/s/l$		–	8/41/1	25/25/0	0/6/44	50/0/0	48/2/0

Table 14
The comparison results of objective values for EMaTO-AMR, MFEA, EEMTA, AT-MFEA, SREMTO, MaTEA on WCCI2020-SOMATP4.

	k	EMaTO-AMR	AT-MFEA	EEMTA	MaTEA	MFEA	SREMTO
P4	1	**5.51E−13(3.9E−13)**	1.82E+0(4.1E−1)•	1.91E+0(5.7E−1)•	1.23E−1(3.3E−1)•	1.45E+2(2.4E+1)•	1.80E−5(2.9E−5)•
	2	1.11E+2(1.2E+2)	5.25E+2(5.6E+2)•	8.49E+1(1.1E+2)§	1.51E+2(8.9E+1)•	3.05E+4(2.1E+4)•	**8.36E+1(8.2E+1)§**
	3	**5.64E−7(2.2E−7)**	3.66E+0(1.0E+0)•	3.39E+0(5.9E−1)•	4.96E+0(1.1E+0)•	4.98E+0(2.5E−1)•	3.58E+0(9.7E−1)•
	4	**5.00E−13(3.7E−13)**	1.84E+0(4.5E−1)•	1.00E−6(3.5E−7)•	1.28E−4(4.1E−4)•	1.54E+2(2.1E+1)•	1.23E−5(1.1E−5)•
	5	2.40E+2(5.5E+2)	4.88E+2(3.7E+2)•	1.97E+4(5.6E+4)•	2.09E+2(1.8E+2)○	3.87E+4(2.9E+4)•	**1.67E+2(1.3E+2)§**
	6	**2.19E−5(8.5E−5)**	3.57E+0(7.7E−1)•	6.63E+0(9.2E+0)•	4.70E+0(1.0E+0)•	4.98E+0(2.5E−1)•	4.15E+0(3.4E+0)•
	7	**7.08E−13(5.9E−13)**	1.70E+0(4.1E−1)•	1.95E+0(5.8E−1)•	7.84E−5(2.3E−4)•	1.43E+2(1.9E+1)•	2.51E−5(3.4E−5)•
	8	2.76E+2(5.6E+2)	6.32E+2(6.5E+2)•	**1.55E+2(2.2E+2)§**	3.61E+2(4.4E+2)•	3.75E+4(3.0E+4)•	1.79E+2(3.9E+2)§
	9	**1.04E−6(1.6E−6)**	3.41E+0(6.4E−1)•	3.47E+0(6.0E−1)•	4.35E+0(1.2E+0)•	4.94E+0(2.7E−1)•	3.39E+0(6.0E−1)•
	10	**9.37E−13(8.4E−13)**	1.97E+0(5.3E−1)•	9.00E−7(4.3E−7)•	1.68E−4(3.4E−4)•	1.52E+2(2.3E+1)•	2.31E−5(2.6E−5)•
	11	2.23E+2(4.4E+2)	6.58E+2(5.5E+2)•	1.94E+4(4.2E+4)•	3.82E+2(6.8E+2)•	3.72E+4(4.8E+4)•	**1.29E+2(1.1E+2)§**
	12	**7.04E−7(4.4E−7)**	3.35E+0(7.7E−1)•	8.29E+0(9.4E+0)•	4.84E+0(1.0E+0)•	4.84E+0(3.0E−1)•	5.11E+0(2.5E+0)•
	13	**5.05E−13(2.3E−13)**	1.67E+0(4.3E−1)•	1.76E+0(5.4E−1)•	6.74E−5(1.1E−4)•	1.46E+2(2.5E+1)•	3.14E−5(4.1E−5)•
	14	3.83E+2(5.8E+2)	4.56E+2(2.8E+2)•	**1.99E+2(2.3E+2)§**	3.27E+2(2.0E+2)§	1.95E+4(6.2E+3)•	2.91E+2(3.1E+2)§
	15	**5.81E−7(2.1E−7)**	3.24E+0(4.6E−1)•	3.19E+0(4.2E−1)•	4.38E+0(6.5E−1)•	4.84E+0(1.8E−1)•	3.56E+0(9.2E−1)•
	16	**5.97E−13(2.7E−13)**	1.87E+0(4.2E−1)•	9.40E−7(3.8E−7)•	1.85E−4(5.2E−4)•	1.43E+2(2.8E+1)•	3.59E−5(4.4E−5)•
	17	3.16E+2(4.9E+2)	7.86E+2(6.3E+2)•	5.20E+4(1.3E+5)•	**2.46E+2(1.7E+2)**○	3.00E+4(2.0E+4)•	3.24E+2(5.0E+2)§
	18	**7.50E−7(4.7E−7)**	3.60E+0(6.9E−1)•	1.35E+1(9.3E+0)•	4.71E+0(1.1E+0)•	4.90E+0(3.1E−1)•	3.44E+0(1.3E+0)•
	19	**7.07E−13(4.5E−13)**	1.83E+0(3.9E−1)•	1.81E+0(5.1E−1)•	3.00E−2(1.2E−1)•	1.55E+2(2.1E+1)•	1.73E−5(1.7E−5)•
	20	3.00E+2(5.5E+2)	6.65E+2(4.2E+2)•	4.81E+2(1.5E+3)§	2.92E+2(3.6E+2)○	3.57E+4(3.2E+4)•	**6.19E+1(2.7E+1)§**
	21	**7.43E−7(3.7E−7)**	4.47E+0(4.2E+0)•	3.98E+0(1.7E+0)•	4.76E+0(9.3E−1)•	4.92E+0(1.9E−1)•	5.38E+0(4.8E+0)•
	22	**5.52E−13(5.9E−13)**	1.97E+0(4.8E−1)•	9.73E−7(3.7E−7)•	1.76E−4(3.2E−4)•	1.52E+2(2.6E+1)•	1.24E−5(1.7E−5)•
	23	5.00E+2(1.4E+3)	6.98E+2(4.4E+2)•	4.63E+3(6.8E+3)•	**2.37E+2(1.4E+2)§**	4.44E+4(6.3E+4)•	3.23E+2(2.6E+2)§
	24	**6.47E−7(2.5E−7)**	3.49E+0(7.5E−1)•	8.60E+0(9.4E+0)•	4.51E+0(1.4E+0)•	4.90E+0(2.3E−1)•	3.81E+0(1.0E+0)•
	25	**6.54E−13(4.5E−13)**	1.82E+0(4.6E−1)•	1.75E+0(3.4E−1)•	1.77E−4(5.9E−4)•	1.47E+2(2.9E+1)•	1.83E−5(1.9E−5)•
	26	1.61E+2(2.0E+2)	3.85E+2(2.0E+2)•	1.57E+2(2.0E+2)§	2.60E+2(2.3E+2)•	3.69E+4(2.6E+4)•	**1.07E+2(7.4E+1)§**
	27	**6.11E−7(2.8E−7)**	3.46E+0(5.3E−1)•	4.17E+0(4.3E+0)•	5.11E+0(6.9E−1)•	4.91E+0(2.5E−1)•	3.95E+0(2.4E+0)•
	28	**6.81E−13(8.3E−13)**	1.82E+0(3.0E−1)•	8.75E−7(2.6E−7)•	1.32E−5(3.2E−5)•	1.45E+2(2.5E+1)•	2.11E−5(1.9E−5)•
	29	5.25E+2(7.7E+2)	6.97E+2(3.4E+2)•	7.58E+3(2.1E+4)•	**2.41E+2(1.5E+2)§**	2.19E+4(1.1E+4)•	3.53E+2(3.6E+2)§
	30	**6.82E−7(2.7E−7)**	3.26E+0(6.3E−1)•	9.25E+0(9.6E+0)•	4.69E+0(9.5E−1)•	4.96E+0(1.8E−1)•	7.99E+0(6.3E+0)•
	31	**5.73E−13(4.9E−13)**	1.54E+0(3.4E−1)•	2.15E+0(4.0E−1)•	4.51E−4(1.7E−3)•	1.51E+2(2.7E+1)•	1.78E−5(1.8E−5)•
	32	**7.43E+2(1.0E+3)**	1.45E+3(1.2E+3)•	1.04E+3(1.5E+3)§	8.32E+2(5.9E+2)§	2.40E+4(8.1E+3)•	8.95E+2(8.4E+2)§
	33	**6.07E−7(3.7E−7)**	3.53E+0(7.2E−1)•	3.28E+0(3.2E−1)•	4.61E+0(1.0E+0)•	4.97E+0(2.9E−1)•	5.47E+0(4.9E+0)•
	34	**5.38E−13(4.0E−13)**	1.87E+0(4.8E−1)•	9.58E−7(3.6E−7)•	9.89E−5(1.5E−4)•	1.61E+2(2.6E+1)•	2.74E−5(3.1E−5)•
	35	1.15E+3(1.2E+3)	1.00E+3(7.9E+2)§	1.39E+4(3.0E+4)§	1.01E+3(7.2E+2)§	2.36E+4(1.2E+4)•	**6.99E+2(8.1E+2)§**
	36	**7.22E−7(4.0E−7)**	4.52E+0(4.2E+0)•	1.37E+1(9.1E+0)•	5.12E+0(1.0E+0)•	4.81E+0(1.6E−1)•	3.56E+0(9.7E−1)•
	37	**4.10E−13(2.8E−13)**	1.91E+0(5.5E−1)•	1.92E+0(6.2E−1)•	1.61E−3(6.2E−3)•	1.56E+2(2.0E+1)•	9.69E−6(1.3E−5)•
	38	6.65E+2(6.6E+2)	1.45E+3(1.2E+3)•	**5.96E+2(1.3E+3)§**	1.18E+3(1.0E+3)§	3.86E+4(3.8E+4)•	1.31E+3(1.2E+3)§
	39	**7.00E−7(4.5E−7)**	3.52E+0(4.8E−1)•	3.24E+0(5.5E−1)•	4.49E+0(1.0E+0)•	4.92E+0(1.7E−1)•	4.46E+0(4.2E+0)•
	40	**6.78E−13(5.0E−13)**	2.00E+0(5.4E−1)•	9.77E−7(6.8E−7)•	1.92E−4(3.2E−4)•	1.52E+2(1.6E+1)•	1.21E−5(9.0E−6)•
	41	**1.29E+2(1.2E+2)**	7.68E+2(6.1E+2)•	8.20E+4(1.5E+5)•	3.07E+2(3.7E+2)•	2.92E+4(2.4E+4)•	3.22E+2(7.3E+2)§
	42	**6.36E−7(3.2E−7)**	3.59E+0(6.3E−1)•	1.31E+1(9.7E+0)•	4.38E+0(1.0E+0)•	4.89E+0(2.5E−1)•	3.74E+0(7.8E−1)•
	43	**6.62E−13(4.9E−13)**	1.82E+0(5.5E−1)•	1.87E+0(2.8E−1)•	2.08E−3(8.0E−3)•	1.54E+2(2.3E+1)•	9.52E−6(5.9E−6)•
	44	2.31E+2(4.7E+2)	7.51E+2(6.6E+2)•	**1.44E+2(1.4E+2)§**	3.63E+2(4.4E+2)•	2.73E+4(1.8E+4)•	2.51E+2(4.0E+2)§
	45	**1.42E−6(2.8E−6)**	3.49E+0(5.3E−1)•	3.15E+0(5.0E−1)•	4.66E+0(1.3E+0)•	5.05E+0(2.4E−1)•	5.64E+0(5.6E+0)•
	46	**4.22E−13(1.9E−13)**	1.78E+0(4.5E−1)•	7.90E−7(2.8E−7)•	7.53E−2(3.0E−1)•	1.38E+2(2.4E+1)•	1.65E−5(1.4E−5)•
	47	1.75E+2(3.8E+2)	8.10E+2(5.0E+2)•	5.03E+4(1.0E+5)•	2.54E+2(2.7E+2)•	4.88E+4(4.8E+4)•	**1.57E+2(3.3E+2)§**
	48	**8.41E−7(5.9E−7)**	3.63E+0(8.6E−1)•	1.02E+1(9.2E+0)•	4.53E+0(1.1E+0)•	4.96E+0(2.2E−1)•	3.36E+0(7.6E−1)•
	49	**4.55E−13(2.8E−13)**	1.62E+0(3.0E−1)•	2.17E+0(4.1E−1)•	8.95E−4(2.6E−3)•	1.57E+2(1.6E+1)•	1.03E−5(6.0E−6)•
	50	1.05E+3(1.0E+3)	7.42E+2(6.6E+2)§	8.19E+2(1.8E+3)○	6.56E+2(6.4E+2)§	1.94E+4(6.8E+3)•	**6.41E+2(6.6E+2)§**
Best/All		35/50	0/50	4/50	3/50	0/50	8/50
Fried		1.58	3.88	3.8	3.42	5.64	2.68
R		1	5	4	3	6	2
$w/s/l$		–	48/2/0	40/9/1	40/7/3	50/0/0	33/17/0

Table 15
The comparison results of objective values for EMaTO-AMR, MFEA, EEMTA, AT-MFEA, SREMTO, MaTEA on WCCI2020-SOMATP5.

	k	EMaTO-AMR	AT-MFEA	EEMTA	MaTEA	MFEA	SREMTO
P5	1	**2.93E+2(1.0E+2)**	6.03E+2(1.0E+2)•	3.02E+2(5.7E+1)§	3.74E+2(7.3E+1)§	5.17E+2(1.9E+1)•	5.16E+2(1.0E+2)•
	2	**4.66E−4(1.8E−3)**	1.58E−1(4.0E−2)•	1.18E−3(6.8E−4)•	1.17E−2(2.0E−2)•	1.03E+0(1.3E−2)•	1.10E−2(1.2E−2)•
	3	3.44E+1(5.6E+0)	3.24E+1(7.2E+0)○	3.78E+1(3.3E+0)§	**2.12E+1(2.6E+0)**○	2.91E+1(5.5E+0)○	3.31E+1(4.6E+0)§
	4	**3.26E+2(8.2E+1)**	5.81E+2(1.4E+2)•	4.06E+2(1.4E+1)•	3.61E+2(8.3E+1)§	5.13E+2(2.8E+1)•	5.30E+2(1.9E+2)•
	5	**5.33E−6(3.8E−6)**	1.53E−1(2.6E−2)•	1.73E−1(2.6E−2)•	1.60E−2(1.2E−2)•	1.03E+0(1.1E−2)•	3.11E−3(4.2E−3)•
	6	3.66E+1(2.9E+0)	3.29E+1(7.1E+0)○	**3.96E+0(2.9E+0)**○	2.02E+1(2.9E+0)○	3.13E+1(5.7E+0)○	3.03E+1(3.7E+0)○
	7	**2.96E+2(9.5E+1)**	5.71E+2(1.0E+2)•	3.04E+2(6.2E+1)§	3.80E+2(8.2E+1)•	5.14E+2(2.1E+1)•	5.63E+2(1.8E+2)•
	8	**6.51E−6(8.4E−6)**	1.66E−1(2.2E−2)•	1.16E−3(4.6E−4)•	1.36E−2(1.4E−2)•	1.03E+0(1.5E−2)•	9.11E−3(9.1E−3)•
	9	3.71E+1(1.6E+0)	3.48E+1(7.5E+0)○	3.82E+1(5.2E+0)§	**2.20E+1(2.3E+0)**○	3.13E+1(3.2E+0)○	3.33E+1(2.8E+0)○
	10	**2.89E+2(1.1E+2)**	6.29E+2(1.3E+2)•	4.07E+2(1.7E+1)•	3.11E+2(5.8E+1)§	5.15E+2(1.4E+1)•	5.65E+2(2.3E+2)•
	11	**5.34E−6(4.5E−6)**	1.56E−1(3.6E−2)•	1.78E−1(4.5E−2)•	1.09E−2(1.4E−2)•	1.03E+0(1.1E−2)•	1.08E−2(9.9E−3)•
	12	3.73E+1(6.9E−1)	3.41E+1(7.9E+0)○	**3.91E+0(3.0E+0)**○	2.09E+1(2.4E+0)○	2.93E+1(6.5E+0)○	3.22E+1(4.7E+0)○
	13	**3.39E+2(6.4E+1)**	5.86E+2(1.4E+2)•	3.52E+2(6.7E+1)§	3.62E+2(7.4E+1)§	5.05E+2(2.9E+1)•	4.61E+2(6.5E+1)•
	14	**4.66E−4(1.8E−3)**	1.54E−1(2.8E−2)•	1.11E−3(2.9E−4)•	1.18E−2(9.6E−3)•	1.03E+0(1.5E−2)•	9.45E−3(9.8E−3)•
	15	3.51E+1(4.6E+0)	3.16E+1(6.5E+0)○	4.17E+1(5.3E+0)•	**2.15E+1(2.1E+0)**○	3.12E+1(5.5E+0)○	3.26E+1(4.5E+0)○
	16	**2.93E+2(1.1E+2)**	5.82E+2(1.2E+2)•	3.96E+2(1.7E+1)•	3.67E+2(1.0E+2)§	5.03E+2(2.1E+1)•	4.83E+2(1.1E+2)•
	17	**4.11E−6(2.7E−6)**	1.60E−1(2.7E−2)•	1.65E−1(3.8E−2)•	2.15E−2(3.2E−2)•	1.03E+0(2.6E−2)•	6.65E−3(6.9E−3)•
	18	3.53E+1(4.8E+0)	3.42E+1(8.1E+0)§	**3.11E+0(2.3E+0)**○	2.17E+1(3.2E+0)○	3.15E+1(6.4E+0)○	3.08E+1(3.9E+0)○
	19	3.16E+2(9.1E+1)	5.65E+2(1.2E+2)•	**3.08E+2(6.1E+1)§**	3.52E+2(7.4E+1)§	5.02E+2(2.4E+1)•	5.10E+2(2.1E+2)•
	20	**4.08E−6(3.7E−6)**	1.68E−1(3.2E−2)•	1.26E−3(3.2E−4)•	1.30E−2(1.4E−2)•	1.03E+0(9.5E−3)•	6.04E−3(8.6E−3)•
	21	3.67E+1(2.7E+0)	3.03E+1(5.8E+0)○	3.87E+1(3.6E+0)§	**2.19E+1(2.7E+0)**○	2.77E+1(5.6E+0)○	2.85E+1(2.5E+0)○
	22	3.53E+2(4.5E+1)	5.96E+2(1.5E+2)•	4.04E+2(1.5E+1)•	**3.46E+2(7.6E+1)§**	5.07E+2(2.4E+1)•	5.11E+2(1.6E+2)•
	23	**4.95E−6(4.0E−6)**	1.73E−1(3.8E−2)•	1.55E−1(4.0E−2)•	8.59E−3(1.0E−2)•	1.03E+0(1.4E−2)•	4.66E−3(5.0E−3)•
	24	3.54E+1(4.8E+0)	3.53E+1(8.8E+0)§	**2.38E+0(8.4E−1)**○	2.16E+1(2.2E+0)○	3.11E+1(5.2E+0)○	3.29E+1(2.6E+0)○
	25	**2.82E+2(1.1E+2)**	5.72E+2(9.9E+1)•	3.13E+2(5.8E+1)§	3.66E+2(8.0E+1)§	5.02E+2(2.1E+1)•	5.98E+2(2.3E+2)•
	26	**3.44E−6(1.6E−6)**	1.65E−1(3.1E−2)•	1.10E−3(4.9E−4)•	2.10E−2(2.9E−2)•	1.04E+0(1.1E−2)•	4.52E−3(6.2E−3)•
	27	3.63E+1(2.3E+0)	3.93E+1(1.0E+1)§	3.82E+1(5.2E+0)§	**2.37E+1(2.3E+0)**○	3.29E+1(3.5E+0)○	3.36E+1(2.1E+0)○
	28	**2.65E+2(1.2E+2)**	5.72E+2(1.1E+2)•	4.04E+2(1.8E+1)•	3.38E+2(9.3E+1)§	5.13E+2(1.9E+1)•	5.13E+2(1.1E+2)•
	29	**3.27E−6(1.6E−6)**	1.54E−1(3.8E−2)•	1.69E−1(4.6E−2)•	8.40E−3(8.8E−3)•	1.03E+0(1.1E−2)•	1.14E−2(8.9E−3)•
	30	3.58E+1(4.1E+0)	3.75E+1(8.5E+0)§	**3.58E+0(2.8E+0)**○	2.25E+1(2.7E+0)○	3.20E+1(5.5E+0)○	3.40E+1(3.6E+0)§
	31	**2.90E+2(1.1E+2)**	5.94E+2(9.6E+1)•	3.21E+2(5.6E+1)§	3.98E+2(9.0E+1)§	5.03E+2(2.7E+1)•	5.60E+2(2.7E+2)•
	32	**6.20E−4(2.5E−3)**	1.48E−1(3.5E−2)•	1.12E−3(4.6E−4)•	8.79E−3(8.1E−3)•	1.04E+0(1.3E−2)•	3.88E−3(7.4E−3)•
	33	3.63E+1(3.7E+0)	3.31E+1(7.1E+0)○	3.92E+1(4.8E+0)•	**2.13E+1(2.3E+0)**○	3.11E+1(4.9E+0)○	3.34E+1(3.2E+0)○
	34	**2.73E+2(1.2E+2)**	5.49E+2(1.5E+2)•	4.09E+2(1.4E+1)•	3.53E+2(8.4E+1)§	5.04E+2(2.7E+1)•	5.52E+2(1.4E+2)•
	35	**4.36E−6(4.0E−6)**	1.62E−1(2.9E−2)•	1.67E−1(2.8E−2)•	7.32E−3(8.1E−3)•	1.03E+0(9.8E−3)•	7.27E−3(6.5E−3)•
	36	3.64E+1(3.3E+0)	3.15E+1(7.4E+0)○	**4.51E+0(4.4E+0)**○	2.02E+1(2.5E+0)○	3.03E+1(6.3E+0)○	3.15E+1(4.2E+0)○
	37	**2.93E+2(9.4E+1)**	5.52E+2(1.2E+2)•	3.25E+2(5.8E+1)§	3.48E+2(7.5E+1)§	5.21E+2(2.2E+1)•	5.55E+2(1.3E+2)•
	38	**4.59E−6(4.5E−6)**	1.46E−1(3.1E−2)•	1.14E−3(5.1E−4)•	8.84E−3(9.1E−3)•	1.03E+0(9.2E−3)•	4.05E−3(6.8E−3)•
	39	3.49E+1(6.0E+0)	3.73E+1(8.0E+0)§	3.79E+1(3.4E+0)§	**2.34E+1(2.0E+0)**○	3.29E+1(3.6E+0)○	3.35E+1(3.3E+0)§
	40	3.29E+2(8.6E+1)	5.11E+2(1.1E+2)•	4.00E+2(1.5E+1)•	**3.17E+2(7.3E+1)§**	5.11E+2(2.4E+1)•	5.54E+2(1.5E+2)•
	41	**2.61E−6(1.6E−6)**	1.56E−1(3.0E−2)•	1.73E−1(2.9E−2)•	1.12E−2(1.1E−2)•	1.03E+0(1.1E−2)•	8.82E−3(7.0E−3)•
	42	3.72E+1(7.7E−1)	3.03E+1(5.6E+0)○	**2.96E+0(2.1E+0)**○	2.14E+1(3.5E+0)○	3.04E+1(6.4E+0)○	3.05E+1(2.8E+0)○
	43	3.16E+2(8.9E+1)	6.16E+2(1.2E+2)•	**3.11E+2(7.8E+1)§**	3.53E+2(7.3E+1)§	5.01E+2(2.0E+1)•	6.04E+2(2.8E+2)•
	44	**3.70E−6(2.9E−6)**	1.49E−1(3.0E−2)•	1.02E−3(3.6E−4)•	1.82E−2(2.0E−2)•	1.03E+0(1.8E−2)•	4.66E−3(7.6E−3)•
	45	3.55E+1(4.3E+0)	3.16E+1(3.6E+0)○	3.95E+1(4.6E+0)§	**2.07E+1(2.4E+0)**○	3.36E+1(4.4E+0)§	3.33E+1(3.6E+0)○
	46	**3.19E+2(7.5E+1)**	5.76E+2(1.0E+2)•	4.01E+2(1.6E+1)•	3.48E+2(8.9E+1)§	5.01E+2(3.8E+1)•	4.14E+2(1.2E+2)•
	47	**4.29E−6(3.2E−6)**	1.58E−1(3.0E−2)•	1.71E−1(2.7E−2)•	9.94E−3(8.6E−3)•	1.03E+0(1.0E−2)•	6.21E−3(1.0E−2)•
	48	3.66E+1(3.3E+0)	3.00E+1(4.5E+0)○	**3.83E+0(2.7E+0)**○	2.22E+1(3.1E+0)○	2.97E+1(7.0E+0)○	3.38E+1(3.7E+0)○
	49	3.29E+2(6.3E+1)	5.37E+2(1.2E+2)•	**3.24E+2(6.6E+1)§**	3.72E+2(7.4E+1)§	5.11E+2(3.2E+1)•	5.28E+2(1.2E+2)•
	50	**4.85E−6(3.1E−6)**	1.66E−1(3.0E−2)•	1.12E−3(5.0E−4)•	1.74E−2(8.9E−3)•	1.03E+0(1.9E−2)•	7.74E−3(7.8E−3)•
Best/All		29/50	0/50	11/50	10/50	0/50	0/50
Fried		2.48	4.86	3.02	2.48	4.38	3.78
R		1	5	2	1	4	3
$w/s/l$		–	34/5/11	27/15/8	18/16/16	34/1/15	34/3/13

Table 16
The comparison results of objective values for EMaTO-AMR, MFEA, EEMTA, AT-MFEA, SREMTO, MaTEA on WCCI2020-SOMATP6.

	k	EMaTO-AMR	AT-MFEA	EEMTA	MaTEA	MFEA	SREMTO
P6	1	3.14E+2(4.8E+2)	4.66E+2(4.9E+2)•	3.41E+3(5.0E+3)•	5.22E+2(7.1E+2)•	2.58E+4(1.4E+4)•	**1.76E+2(1.3E+2)§**
	2	**3.25E−6(4.0E−6)**	1.44E−1(2.9E−2)•	9.11E−4(2.9E−4)•	9.56E−3(7.1E−3)•	1.04E+0(7.3E−3)•	6.80E−3(1.5E−2)•
	3	**6.30E+3(1.6E+3)**	7.96E+3(6.9E+2)•	6.56E+3(9.0E+2)§	7.69E+3(1.4E+3)•	6.94E+3(5.3E+2)•	7.04E+3(6.4E+2)•
	4	**6.45E+1(4.7E+1)**	8.67E+2(9.3E+2)•	3.61E+2(1.2E+3)§	3.28E+2(5.7E+2)•	4.71E+4(7.3E+4)•	1.06E+2(1.3E+2)§
	5	**3.63E−6(2.5E−6)**	1.67E−1(3.7E−2)•	1.78E−1(3.9E−2)•	1.14E−2(1.2E−2)•	1.03E+0(2.8E−2)•	9.73E−3(7.2E−3)•
	6	**5.17E+3(8.7E+2)**	8.16E+3(8.9E+2)•	1.29E+4(4.4E+2)•	8.23E+3(1.8E+3)•	7.16E+3(7.8E+2)•	7.52E+3(9.7E+2)•
	7	**7.22E+1(7.8E+1)**	5.47E+2(7.4E+2)•	1.91E+4(4.3E+4)•	1.65E+2(1.0E+2)•	3.27E+4(4.2E+4)•	7.68E+1(3.3E+1)•
	8	**5.67E−6(5.2E−6)**	1.51E−1(3.4E−2)•	1.10E−3(3.6E−4)•	2.53E−2(5.8E−2)•	1.03E+0(1.2E−2)•	1.02E−2(9.2E−3)•
	9	**5.73E+3(1.4E+3)**	8.15E+3(9.1E+2)•	6.59E+3(9.0E+2)§	8.33E+3(2.1E+3)•	7.41E+3(6.8E+2)•	6.72E+3(1.2E+3)§
	10	4.09E+2(8.7E+2)	1.04E+3(1.0E+3)•	3.56E+2(7.4E+2)§	4.09E+2(4.5E+2)○	2.08E+4(5.2E+3)•	**2.55E+2(5.4E+2)§**
	11	**3.59E−6(2.1E−6)**	1.61E−1(3.2E−2)•	1.75E−1(4.0E−2)•	1.11E−2(7.4E−3)•	1.03E+0(1.5E−2)•	1.08E−2(1.3E−2)•
	12	**6.58E+3(2.6E+3)**	7.76E+3(6.4E+2)§	1.27E+4(5.0E+2)•	7.62E+3(1.2E+3)§	6.93E+3(8.4E+2)§	7.33E+3(1.1E+3)§
	13	2.08E+2(4.8E+2)	7.22E+2(6.4E+2)•	8.28E+3(1.7E+4)•	1.15E+2(5.3E+1)○	3.32E+4(3.6E+4)•	**6.76E+1(3.5E+1)§**
	14	**3.54E−6(3.0E−6)**	1.65E−1(4.6E−2)•	9.75E−4(3.5E−4)•	6.75E−3(7.2E−3)•	1.03E+0(8.7E−3)•	5.74E−3(6.6E−3)•
	15	**4.57E+3(1.7E+3)**	7.90E+3(1.1E+3)•	6.67E+3(6.7E+2)•	7.58E+3(2.5E+3)•	7.34E+3(6.6E+2)•	6.79E+3(1.2E+3)•
	16	**2.60E+2(3.8E+2)**	9.78E+2(9.4E+2)•	9.37E+2(1.9E+3)§	8.67E+2(6.9E+2)•	3.95E+4(3.5E+4)•	5.19E+2(4.7E+2)§
	17	**5.50E−6(6.6E−6)**	1.67E−1(3.4E−2)•	1.63E−1(2.5E−2)•	9.44E−3(8.2E−3)•	1.03E+0(8.3E−3)•	1.05E−2(9.5E−3)•
	18	**5.07E+3(9.0E+2)**	8.01E+3(1.1E+3)•	1.29E+4(5.3E+2)•	7.51E+3(1.8E+3)•	7.30E+3(6.3E+2)•	7.10E+3(1.2E+3)•
	19	**7.52E+1(1.1E+2)**	1.17E+3(9.4E+2)•	3.29E+4(6.7E+4)•	1.28E+2(7.9E+1)•	3.40E+4(1.3E+4)•	1.03E+2(1.7E+2)§
	20	**2.97E−6(2.0E−6)**	1.83E−1(3.4E−2)•	1.16E−3(4.2E−4)•	1.51E−2(2.5E−2)•	1.04E+0(1.1E−2)•	5.28E−3(5.9E−3)•
	21	7.25E+3(1.5E+3)	8.02E+3(7.0E+2)§	7.12E+3(1.0E+3)§	7.92E+3(1.2E+3)§	6.94E+3(5.3E+2)§	**6.93E+3(1.2E+3)§**
	22	2.34E+2(3.9E+2)	1.49E+3(8.9E+2)•	**2.34E+2(4.4E+2)§**	5.48E+2(5.6E+2)•	2.10E+4(7.3E+3)•	3.38E+2(4.9E+2)§
	23	**4.66E−4(1.8E−3)**	1.48E−1(2.1E−2)•	1.57E−1(3.1E−2)•	9.37E−3(1.4E−2)•	1.03E+0(9.9E−3)•	5.43E−3(6.6E−3)•
	24	**5.58E+3(9.6E+2)**	8.08E+3(8.8E+2)•	1.29E+4(4.3E+2)•	6.86E+3(1.6E+3)•	7.34E+3(4.4E+2)•	6.75E+3(1.3E+3)•
	25	**1.73E+2(4.7E+2)**	7.23E+2(6.1E+2)•	1.50E+4(3.6E+4)•	3.14E+2(4.6E+2)•	3.40E+4(2.1E+4)•	5.00E+2(8.2E+2)•
	26	**4.68E−4(1.8E−3)**	1.59E−1(4.0E−2)•	9.65E−4(4.0E−4)•	1.22E−2(1.3E−2)•	1.03E+0(1.0E−2)•	3.59E−3(6.7E−3)•
	27	**4.27E+3(1.5E+3)**	7.69E+3(5.9E+2)•	6.99E+3(9.9E+2)•	8.39E+3(1.6E+3)•	6.74E+3(6.1E+2)•	7.20E+3(1.5E+3)•
	28	1.42E+2(1.6E+2)	5.35E+2(5.3E+2)•	4.11E+2(8.4E+2)§	2.08E+2(9.7E+1)•	3.32E+4(3.7E+4)•	**1.36E+2(1.4E+2)§**
	29	**4.66E−4(1.8E−3)**	1.48E−1(4.1E−2)•	1.63E−1(3.2E−2)•	8.01E−3(7.2E−3)•	1.04E+0(8.8E−3)•	7.27E−3(1.2E−2)•
	30	**5.41E+3(1.7E+3)**	7.91E+3(7.8E+2)•	1.30E+4(3.0E+2)•	6.97E+3(1.3E+3)•	7.04E+3(7.7E+2)•	7.56E+3(1.1E+3)•
	31	2.51E+2(5.1E+2)	6.86E+2(8.2E+2)•	2.64E+3(2.2E+3)•	3.06E+2(3.2E+2)•	2.51E+4(1.0E+4)•	**1.31E+2(1.3E+2)§**
	32	**4.77E−6(4.7E−6)**	1.54E−1(4.9E−2)•	9.84E−4(4.5E−4)•	1.59E−2(1.7E−2)•	1.03E+0(8.7E−3)•	3.90E−3(6.1E−3)•
	33	**5.60E+3(9.8E+2)**	7.96E+3(5.9E+2)•	7.00E+3(7.6E+2)•	8.36E+3(2.0E+3)•	7.17E+3(7.9E+2)•	6.56E+3(1.6E+3)§
	34	**1.06E+2(1.1E+2)**	4.86E+2(3.2E+2)•	4.30E+2(1.2E+3)§	3.11E+2(3.0E+2)•	3.55E+4(3.7E+4)•	2.47E+2(4.7E+2)§
	35	**4.64E−6(2.7E−6)**	1.61E−1(3.1E−2)•	1.61E−1(2.8E−2)•	2.00E−2(2.6E−2)•	1.03E+0(1.4E−2)•	1.09E−2(1.4E−2)•
	36	**5.56E+3(9.1E+2)**	8.01E+3(8.1E+2)•	1.27E+4(5.3E+2)•	7.76E+3(1.5E+3)•	7.28E+3(9.1E+2)•	7.48E+3(1.4E+3)•
	37	**7.55E+1(4.5E+1)**	4.81E+2(2.8E+2)•	3.40E+4(7.5E+4)•	1.59E+2(5.7E+1)•	3.42E+4(2.6E+4)•	1.01E+2(5.8E+1)§
	38	**6.57E−6(1.0E−5)**	1.57E−1(2.9E−2)•	1.17E−3(5.5E−4)•	1.37E−2(9.5E−3)•	1.04E+0(1.0E−2)•	5.13E−3(8.6E−3)•
	39	**6.11E+3(1.6E+3)**	7.91E+3(7.6E+2)•	6.87E+3(8.3E+2)•	7.21E+3(1.8E+3)•	7.52E+3(7.5E+2)•	7.25E+3(1.2E+3)•
	40	3.96E+2(9.6E+2)	1.16E+3(1.1E+3)•	1.20E+3(1.6E+3)§	7.31E+2(7.5E+2)•	2.89E+4(1.2E+4)•	**2.99E+2(4.4E+2)§**
	41	**3.79E−6(3.8E−6)**	1.66E−1(2.5E−2)•	1.61E−1(2.7E−2)•	1.65E−2(2.0E−2)•	1.03E+0(1.5E−2)•	8.35E−3(9.2E−3)•
	42	**5.37E+3(9.1E+2)**	7.49E+3(8.0E+2)•	1.29E+4(4.2E+2)•	7.69E+3(2.0E+3)•	7.21E+3(7.5E+2)•	7.22E+3(8.8E+2)•
	43	5.86E+2(7.8E+2)	1.11E+3(1.1E+3)•	1.08E+4(2.3E+4)•	7.39E+2(6.9E+2)§	2.29E+4(6.5E+3)•	**3.59E+2(4.4E+2)§**
	44	**4.82E−6(3.1E−6)**	1.61E−1(3.3E−2)•	9.81E−4(3.4E−4)•	1.31E−2(1.5E−2)•	1.03E+0(1.6E−2)•	4.08E−3(5.8E−3)•
	45	6.88E+3(1.8E+3)	7.93E+3(1.0E+3)§	6.66E+3(9.9E+2)§	7.64E+3(1.5E+3)§	6.92E+3(7.8E+2)§	**6.41E+3(1.4E+3)§**
	46	**7.72E+2(8.4E+2)**	1.96E+3(1.4E+3)•	1.07E+3(2.1E+3)§	1.90E+3(9.2E+2)•	2.36E+4(5.6E+3)•	9.84E+2(8.4E+2)§
	47	**3.94E−6(2.6E−6)**	1.47E−1(3.0E−2)•	1.65E−1(2.8E−2)•	9.60E−3(1.0E−2)•	1.03E+0(1.5E−2)•	5.90E−3(6.4E−3)•
	48	**5.98E+3(1.4E+3)**	8.05E+3(4.8E+2)•	1.29E+4(4.8E+2)•	7.21E+3(1.1E+3)•	7.25E+3(8.7E+2)•	7.24E+3(1.3E+3)•
	49	**1.51E+2(1.3E+2)**	5.21E+2(3.9E+2)•	8.84E+3(2.5E+4)•	5.12E+2(5.5E+2)•	4.27E+4(5.1E+4)•	2.02E+2(2.2E+2)§
	50	**4.66E−4(1.8E−3)**	1.62E−1(4.0E−2)•	1.08E−3(2.6E−4)•	2.63E−2(6.5E−2)•	1.03E+0(1.4E−2)•	7.30E−3(6.3E−3)•
Best/All		40/50	0/50	1/50	0/50	0/50	9/50
Fried		1.32	4.72	3.9	3.62	5.08	2.36
R		1	5	4	3	6	2
$w/s/l$		–	47/3/0	38/12/0	44/4/2	47/3/0	30/20/0

Table 17
The comparison results of objective values for EMaTO-AMR, MFEA, EEMTA, AT-MFEA, SREMTO, MaTEA on WCCI2020-SOMATP7.

	k	EMaTO-AMR	AT-MFEA	EEMTA	MaTEA	MFEA	SREMTO
P7	1	**2.05E−6(2.2E−6)**	3.45E+0(7.0E−1)•	3.81E+0(2.0E+0)•	4.51E+0(1.1E+0)•	4.90E+0(2.6E−1)•	3.95E+0(3.1E+0)•
	2	**3.27E+2(8.9E+1)**	5.70E+2(1.2E+2)•	4.01E+2(2.2E+1)•	3.35E+2(6.9E+1)§	5.09E+2(2.1E+1)•	5.93E+2(2.5E+2)•
	3	3.69E+1(3.0E+0)	3.82E+1(1.1E+1)§	3.89E+1(4.7E+0)§	**2.31E+1(2.4E+0)**○	3.28E+1(5.8E+0)○	3.50E+1(2.9E+0)○
	4	**9.14E−7(5.8E−7)**	3.31E+0(5.3E−1)•	1.38E+1(8.3E+0)•	5.01E+0(1.1E+0)•	5.03E+0(2.1E−1)•	4.10E+0(3.6E+0)•
	5	**3.02E+2(1.0E+2)**	5.22E+2(8.8E+1)•	3.37E+2(5.5E+1)§	3.89E+2(7.1E+1)•	5.04E+2(2.0E+1)•	5.60E+2(1.3E+2)•
	6	3.72E+1(7.0E−1)	3.95E+1(1.2E+1)§	**3.06E+0(2.0E+0)**○	2.09E+1(1.7E+0)○	3.13E+1(3.0E+0)○	3.12E+1(2.2E+0)○
	7	**8.15E−7(3.1E−7)**	3.54E+0(9.3E−1)•	3.42E+0(7.9E−1)•	4.58E+0(9.4E−1)•	5.00E+0(2.7E−1)•	6.96E+0(5.8E+0)•
	8	**2.96E+2(1.0E+2)**	6.05E+2(1.3E+2)•	4.01E+2(1.9E+1)•	3.69E+2(7.9E+1)•	5.10E+2(2.3E+1)•	5.47E+2(2.4E+2)•
	9	3.74E+1(7.5E−1)	3.62E+1(8.1E+0)§	3.98E+1(4.8E+0)•	**2.14E+1(2.4E+0)**○	3.24E+1(2.5E+0)○	3.24E+1(3.4E+0)○
	10	**1.32E−6(1.7E−6)**	3.40E+0(6.4E−1)•	1.47E+1(8.5E+0)•	4.40E+0(9.8E−1)•	4.98E+0(2.6E−1)•	3.48E+0(8.4E−1)•
	11	**2.97E+2(8.6E+1)**	6.49E+2(1.7E+2)•	3.05E+2(6.6E+1)§	4.05E+2(1.5E+2)•	5.03E+2(3.2E+1)•	5.09E+2(1.4E+2)•
	12	3.74E+1(6.2E−1)	3.42E+1(1.0E+1)○	**2.16E+0(9.0E−1)**○	2.19E+1(2.4E+0)○	3.22E+1(3.2E+0)○	3.25E+1(2.6E+0)○
	13	**1.95E−6(4.2E−6)**	3.74E+0(9.0E−1)•	3.27E+0(7.8E−1)•	4.13E+0(9.0E−1)•	4.96E+0(2.8E−1)•	4.58E+0(3.4E+0)•
	14	**3.29E+2(8.1E+1)**	6.02E+2(1.4E+2)•	3.97E+2(1.7E+1)•	3.50E+2(8.7E+1)§	5.17E+2(2.3E+1)•	5.73E+2(1.9E+2)•
	15	3.62E+1(3.3E+0)	3.54E+1(7.5E+0)○	3.93E+1(3.7E+0)•	**2.24E+1(2.1E+0)**○	3.13E+1(5.0E+0)○	3.39E+1(3.6E+0)○
	16	**2.34E−6(6.2E−6)**	3.74E+0(6.7E−1)•	1.48E+1(9.7E+0)•	5.17E+0(9.9E−1)•	4.97E+0(2.6E−1)•	3.48E+0(9.3E−1)•
	17	**2.96E+2(9.7E+1)**	5.90E+2(1.5E+2)•	3.44E+2(6.0E+1)§	3.49E+2(5.9E+1)§	5.12E+2(2.5E+1)•	5.02E+2(2.2E+2)•
	18	3.70E+1(1.6E+0)	3.71E+1(8.1E+0)§	**2.35E+0(9.2E−1)**○	2.21E+1(2.8E+0)○	2.98E+1(4.6E+0)○	3.38E+1(3.2E+0)○
	19	**1.10E−6(7.0E−7)**	3.72E+0(8.5E−1)•	3.39E+0(4.7E−1)•	5.02E+0(8.7E−1)•	4.90E+0(2.7E−1)•	3.50E+0(1.2E+0)•
	20	**2.96E+2(1.1E+2)**	5.69E+2(1.3E+2)•	4.06E+2(9.6E+0)•	3.41E+2(5.4E+1)§	5.03E+2(2.1E+1)•	5.59E+2(1.2E+2)•
	21	3.60E+1(3.9E+0)	2.92E+1(5.0E+0)○	3.71E+1(3.5E+0)§	**2.15E+1(3.1E+0)**○	3.42E+1(4.4E+0)§	3.07E+1(3.0E+0)○
	22	**8.31E−7(5.4E−7)**	3.54E+0(6.4E−1)•	1.11E+1(9.8E+0)•	4.93E+0(1.1E+0)•	4.99E+0(1.6E−1)•	5.32E+0(4.5E+0)•
	23	**3.14E+2(9.8E+1)**	5.83E+2(1.6E+2)•	3.26E+2(6.1E+1)§	3.38E+2(7.0E+1)§	5.09E+2(2.0E+1)•	5.67E+2(4.8E+2)•
	24	3.60E+1(4.7E+0)	3.39E+1(7.9E+0)○	**3.10E+0(2.1E+0)**○	2.15E+1(2.6E+0)○	3.08E+1(5.4E+0)○	3.52E+1(4.6E+0)§
	25	**8.57E−7(5.1E−7)**	3.54E+0(5.1E−1)•	3.36E+0(6.0E−1)•	4.70E+0(1.4E+0)•	5.00E+0(1.9E−1)•	3.63E+0(7.4E−1)•
	26	**2.55E+2(1.2E+2)**	5.43E+2(1.0E+2)•	4.03E+2(1.7E+1)•	4.02E+2(9.0E+1)•	5.16E+2(2.0E+1)•	6.01E+2(2.3E+2)•
	27	3.55E+1(3.3E+0)	3.44E+1(6.4E+0)○	3.91E+1(4.1E+0)§	**2.14E+1(2.2E+0)**○	3.09E+1(3.9E+0)○	3.14E+1(4.8E+0)○
	28	**1.34E−6(1.5E−6)**	3.20E+0(5.8E−1)•	1.17E+1(9.4E+0)•	4.86E+0(9.5E−1)•	4.94E+0(2.7E−1)•	4.30E+0(3.4E+0)•
	29	3.12E+2(8.5E+1)	5.66E+2(9.2E+1)•	**3.11E+2(7.1E+1)**§	3.72E+2(6.2E+1)§	5.12E+2(2.1E+1)•	5.56E+2(1.4E+2)•
	30	3.62E+1(2.9E+0)	3.19E+1(3.4E+0)○	**2.81E+0(2.4E+0)**○	2.20E+1(1.9E+0)○	3.04E+1(3.7E+0)○	3.21E+1(3.8E+0)○
	31	**8.88E−7(5.7E−7)**	3.59E+0(8.5E−1)•	3.33E+0(5.9E−1)•	4.92E+0(1.1E+0)•	4.90E+0(2.6E−1)•	7.70E+0(6.2E+0)•
	32	**3.18E+2(9.4E+1)**	5.55E+2(1.1E+2)•	4.04E+2(1.7E+1)•	3.87E+2(8.0E+1)§	5.09E+2(2.3E+1)•	4.87E+2(1.4E+2)•
	33	3.72E+1(9.7E−1)	3.23E+1(6.0E+0)○	3.98E+1(4.3E+0)•	**2.20E+1(1.8E+0)**○	3.13E+1(3.3E+0)○	3.18E+1(2.8E+0)○
	34	**8.02E−7(4.8E−7)**	3.62E+0(7.6E−1)•	1.17E+1(1.0E+1)•	5.20E+0(1.2E+0)•	5.00E+0(2.5E−1)•	5.98E+0(5.3E+0)•
	35	**2.56E+2(1.3E+2)**	5.82E+2(1.5E+2)•	3.41E+2(5.1E+1)§	3.34E+2(7.6E+1)§	5.08E+2(3.4E+1)•	5.81E+2(2.2E+2)•
	36	3.64E+1(2.7E+0)	3.19E+1(6.6E+0)○	**2.21E+0(7.5E−1)**○	2.10E+1(1.9E+0)○	2.88E+1(4.4E+0)○	3.37E+1(3.4E+0)○
	37	**1.30E−6(1.2E−6)**	3.44E+0(4.9E−1)•	3.84E+0(2.9E+0)•	5.31E+0(1.3E+0)•	4.97E+0(3.1E−1)•	3.62E+0(9.5E−1)•
	38	**3.16E+2(7.5E+1)**	6.18E+2(1.6E+2)•	3.96E+2(2.3E+1)•	3.69E+2(7.1E+1)§	5.05E+2(2.2E+1)•	5.54E+2(1.6E+2)•
	39	3.74E+1(8.7E−1)	3.19E+1(7.0E+0)○	3.88E+1(4.6E+0)§	**2.20E+1(2.7E+0)**○	3.05E+1(6.0E+0)○	2.97E+1(4.2E+0)○
	40	**8.99E−7(8.0E−7)**	3.72E+0(5.3E−1)•	1.01E+1(9.2E+0)•	5.12E+0(1.2E+0)•	4.99E+0(2.6E−1)•	4.38E+0(3.4E+0)•
	41	**3.20E+2(8.5E+1)**	5.46E+2(1.0E+2)•	3.44E+2(6.5E+1)§	3.79E+2(5.8E+1)§	5.01E+2(2.9E+1)•	5.06E+2(1.8E+2)•
	42	3.57E+1(4.7E+0)	3.32E+1(6.0E+0)○	**2.41E+0(1.5E+0)**○	2.16E+1(2.0E+0)○	3.12E+1(4.9E+0)○	3.35E+1(2.4E+0)○
	43	**8.52E−7(7.4E−7)**	3.42E+0(6.7E−1)•	3.21E+0(5.0E−1)•	4.28E+0(9.2E−1)•	4.94E+0(2.1E−1)•	4.18E+0(2.7E+0)•
	44	**3.43E+2(6.7E+1)**	6.20E+2(1.1E+2)•	3.99E+2(2.4E+1)•	3.58E+2(6.7E+1)§	4.97E+2(2.6E+1)•	5.32E+2(2.1E+2)•
	45	3.60E+1(4.5E+0)	3.27E+1(5.2E+0)○	3.83E+1(4.1E+0)§	**2.19E+1(2.6E+0)**○	3.10E+1(4.2E+0)○	3.07E+1(5.3E+0)○
	46	**2.54E−6(6.1E−6)**	3.33E+0(5.3E−1)•	1.22E+1(9.5E+0)•	4.59E+0(8.0E−1)•	5.13E+0(1.8E−1)•	3.10E+0(5.7E−1)•
	47	**3.03E+2(9.1E+1)**	5.30E+2(1.0E+2)•	3.25E+2(5.6E+1)§	3.50E+2(7.1E+1)§	5.05E+2(2.8E+1)•	5.71E+2(2.6E+2)•
	48	3.74E+1(8.1E−1)	3.46E+1(6.6E+0)○	**3.20E+0(3.0E+0)**○	2.18E+1(2.8E+0)○	3.25E+1(3.4E+0)○	3.45E+1(3.7E+0)○
	49	**8.49E−7(4.7E−7)**	4.62E+0(4.2E+0)•	3.21E+0(5.2E−1)•	4.37E+0(8.4E−1)•	4.97E+0(2.4E−1)•	5.92E+0(4.8E+0)•
	50	**3.06E+2(8.2E+1)**	5.41E+2(8.3E+1)•	4.04E+2(1.8E+1)•	3.50E+2(7.9E+1)§	5.20E+2(1.8E+1)•	5.13E+2(1.3E+2)•
Best/All		33/50	0/50	9/50	8/50	0/50	0/50
Fried		2.4	4.24	3.36	2.82	4	4.18
R		1	6	3	2	4	5
$w/s/l$		–	34/4/12	29/13/8	21/13/16	34/1/15	34/1/15

Table 18
The comparison results of objective values for EMaTO-AMR, MFEA, EEMTA, AT-MFEA, SREMTO, MaTEA on WCCI2020-SOMATP8.

	k	EMaTO-AMR	AT-MFEA	EEMTA	MaTEA	MFEA	SREMTO
P8	1	1.65E+2(3.0E+2)	1.02E+3(1.1E+3)•	2.81E+4(6.5E+4)•	4.61E+2(5.5E+2)•	3.64E+4(2.4E+4)•	**8.29E+1(8.2E+1)§**
	2	**1.64E−4(6.4E−4)**	1.99E+1(2.2E−1)•	1.72E+1(8.4E+0)•	5.47E+0(2.2E+0)•	5.20E+0(2.5E−1)•	1.64E+1(6.2E+0)•
	3	3.16E+2(7.7E+1)	6.54E+2(1.8E+2)•	**3.13E+2(5.2E+1)§**	4.89E+2(1.3E+2)•	5.17E+2(2.8E+1)•	4.66E+2(1.5E+2)•
	4	**4.49E−6(2.4E−6)**	1.98E−1(3.4E−2)•	1.18E−3(4.4E−4)•	1.71E−2(2.0E−2)•	1.04E+0(1.2E−2)•	8.53E−3(1.3E−2)•
	5	3.67E+1(2.4E+0)	5.25E+1(5.4E+0)•	4.03E+1(3.0E+0)•	**3.03E+1(2.0E+0)**○	3.40E+1(5.3E+0)§	4.08E+1(4.5E+0)•
	6	**8.48E+1(1.2E+2)**	5.65E+3(8.3E+3)•	4.74E+2(1.5E+3)§	2.96E+2(2.1E+2)•	7.09E+4(9.3E+4)•	2.66E+2(5.1E+2)§
	7	**2.78E−6(3.5E−6)**	1.53E+1(7.2E+0)•	3.54E+0(1.2E+0)•	5.15E+0(1.5E+0)•	5.11E+0(1.7E−1)•	7.81E+0(6.9E+0)•
	8	**3.36E+2(6.4E+1)**	6.25E+2(1.3E+2)•	4.06E+2(9.0E+0)•	4.28E+2(1.4E+2)•	5.19E+2(1.9E+1)•	5.66E+2(2.6E+2)•
	9	**3.03E−6(1.9E−6)**	2.16E−1(5.0E−2)•	1.58E−1(3.5E−2)•	2.27E−2(3.0E−2)•	1.03E+0(1.8E−2)•	3.93E−3(6.6E−3)•
	10	3.64E+1(3.7E+0)	5.46E+1(3.3E+0)•	**2.51E+0(1.5E+0)**○	3.10E+1(2.3E+0)○	3.54E+1(4.2E+0)§	4.41E+1(4.2E+0)•
	11	**5.31E+1(2.1E+1)**	3.18E+3(3.9E+3)•	1.17E+4(3.2E+4)•	3.45E+2(5.8E+2)•	9.68E+4(1.5E+5)•	8.22E+1(3.8E+1)•
	12	**8.08E−5(3.0E−4)**	1.90E+1(1.0E+0)•	1.84E+1(6.4E+0)•	5.41E+0(2.0E+0)•	5.24E+0(3.0E−1)•	1.39E+1(7.2E+0)•
	13	**2.77E+2(1.1E+2)**	6.12E+2(1.6E+2)•	3.28E+2(6.3E+1)§	4.52E+2(1.4E+2)•	5.05E+2(2.3E+1)•	5.71E+2(1.7E+2)•
	14	**4.08E−6(1.5E−6)**	1.89E−1(3.4E−2)•	8.85E−4(3.9E−4)•	2.41E−2(4.1E−2)•	1.04E+0(5.8E−3)•	6.99E−3(6.7E−3)•
	15	3.71E+1(8.4E−1)	5.44E+1(7.2E+0)•	3.94E+1(4.2E+0)•	**3.22E+1(3.2E+0)**○	3.32E+1(4.4E+0)○	4.51E+1(3.7E+0)•
	16	9.09E+2(1.8E+3)	9.26E+2(8.4E+2)•	8.17E+2(1.7E+3)§	**3.12E+2(1.2E+2)§**	1.03E+5(1.1E+5)•	8.14E+2(1.3E+3)§
	17	**8.87E−6(2.7E−5)**	1.95E+1(7.7E−1)•	6.35E+0(6.9E+0)•	5.44E+0(1.5E+0)•	5.20E+0(1.9E−1)•	1.41E+1(7.3E+0)•
	18	**2.80E+2(1.1E+2)**	7.02E+2(1.3E+2)•	4.08E+2(1.5E+1)•	4.24E+2(1.5E+2)•	5.03E+2(2.5E+1)•	5.75E+2(1.8E+2)•
	19	**3.17E−6(2.3E−6)**	2.13E−1(4.6E−2)•	1.52E−1(3.2E−2)•	1.50E−2(9.2E−3)•	1.04E+0(9.9E−3)•	7.45E−3(7.9E−3)•
	20	3.73E+1(2.3E+0)	5.57E+1(3.5E+0)•	**5.49E+0(4.7E+0)**○	3.18E+1(2.6E+0)○	3.23E+1(4.1E+0)○	4.56E+1(4.5E+0)•
	21	**4.75E+1(1.3E+0)**	1.60E+4(1.5E+4)•	7.12E+4(1.1E+5)•	1.50E+4(2.3E+4)•	1.54E+5(2.1E+5)•	5.77E+1(2.1E+1)•
	22	**2.65E+0(7.2E+0)**	1.90E+1(3.9E+0)•	1.86E+1(7.0E+0)•	5.84E+0(2.1E+0)•	5.27E+0(2.6E−1)•	1.92E+1(2.7E−1)•
	23	**3.27E+2(8.4E+1)**	7.22E+2(1.8E+2)•	3.47E+2(7.4E+1)§	4.80E+2(1.1E+2)•	5.19E+2(2.3E+1)•	6.07E+2(1.9E+2)•
	24	**3.60E−6(2.0E−6)**	1.91E−1(4.0E−2)•	1.07E−3(3.2E−4)•	1.03E−2(1.0E−2)•	1.03E+0(1.3E−2)•	6.83E−3(8.1E−3)•
	25	3.72E+1(7.2E−1)	5.24E+1(4.8E+0)•	4.08E+1(3.5E+0)•	**3.17E+1(3.6E+0)**○	3.31E+1(4.4E+0)○	4.46E+1(3.1E+0)•
	26	**1.87E+2(3.3E+2)**	1.86E+3(1.5E+3)•	4.58E+2(8.9E+2)§	6.47E+2(7.3E+2)•	6.64E+4(9.5E+4)•	3.04E+2(4.4E+2)§
	27	**4.39E−6(6.2E−6)**	1.88E+1(4.2E+0)•	6.98E+0(6.9E+0)•	5.93E+0(2.2E+0)•	5.20E+0(3.0E−1)•	1.22E+1(7.6E+0)•
	28	**3.28E+2(8.7E+1)**	6.54E+2(2.0E+2)•	4.04E+2(1.6E+1)•	4.55E+2(9.0E+1)•	5.11E+2(2.2E+1)•	6.21E+2(2.0E+2)•
	29	**3.04E−6(1.6E−6)**	2.00E−1(5.8E−2)•	1.67E−1(4.3E−2)•	1.30E−2(1.2E−2)•	1.04E+0(8.9E−3)•	8.88E−3(1.1E−2)•
	30	3.65E+1(3.4E+0)	5.34E+1(4.0E+0)•	**4.31E+0(3.8E+0)**○	3.19E+1(3.0E+0)○	3.18E+1(4.0E+0)○	4.37E+1(5.2E+0)•
	31	**6.10E+2(1.5E+3)**	2.16E+3(2.2E+3)•	2.80E+3(3.1E+3)•	1.04E+3(9.3E+2)•	4.01E+4(2.4E+4)•	6.61E+2(5.0E+2)•
	32	**1.34E+0(5.3E+0)**	1.99E+1(3.5E−1)•	1.82E+1(6.7E+0)•	5.26E+0(1.4E+0)•	5.26E+0(2.4E−1)•	1.72E+1(5.5E+0)•
	33	**3.22E+2(8.6E+1)**	6.30E+2(1.4E+2)•	3.41E+2(7.9E+1)§	4.84E+2(1.7E+2)•	5.07E+2(1.7E+1)•	5.66E+2(2.0E+2)•
	34	**4.66E−4(1.8E−3)**	2.32E−1(4.6E−2)•	8.79E−4(3.1E−4)•	1.14E−2(1.4E−2)•	1.04E+0(8.1E−3)•	6.42E−3(5.3E−3)•
	35	3.61E+1(4.0E+0)	5.57E+1(4.1E+0)•	3.84E+1(4.1E+0)•	**3.17E+1(4.2E+0)**○	3.36E+1(4.1E+0)§	4.41E+1(6.3E+0)•
	36	2.99E+2(4.6E+2)	1.50E+3(1.2E+3)•	3.48E+2(1.1E+3)§	5.71E+2(5.0E+2)•	4.64E+4(4.3E+4)•	**2.47E+2(4.7E+2)§**
	37	**1.32E+0(5.3E+0)**	1.99E+1(3.0E−1)•	7.18E+0(6.6E+0)•	5.95E+0(1.4E+0)•	5.30E+0(2.6E−1)•	1.32E+1(7.7E+0)•
	38	**3.15E+2(8.8E+1)**	6.29E+2(1.5E+2)•	4.06E+2(1.4E+1)•	4.69E+2(1.3E+2)•	5.09E+2(2.2E+1)•	5.02E+2(1.1E+2)•
	39	**4.46E−6(3.6E−6)**	2.11E−1(5.4E−2)•	1.70E−1(3.1E−2)•	1.20E−2(9.2E−3)•	1.04E+0(8.6E−3)•	6.54E−3(6.0E−3)•
	40	3.64E+1(4.3E+0)	5.33E+1(5.8E+0)•	**3.13E+0(2.1E+0)**○	3.02E+1(3.3E+0)○	3.36E+1(2.8E+0)○	4.50E+1(6.4E+0)•
	41	7.85E+2(1.3E+3)	1.77E+3(1.5E+3)•	6.61E+3(1.2E+4)•	5.87E+2(6.1E+2)§	6.05E+4(7.2E+4)•	**4.68E+2(1.1E+3)§**
	42	**1.13E−4(4.0E−4)**	1.99E+1(2.9E−1)•	1.77E+1(7.7E+0)•	5.84E+0(1.7E+0)•	5.25E+0(2.3E−1)•	1.70E+1(5.6E+0)•
	43	3.26E+2(7.0E+1)	6.20E+2(1.2E+2)•	**3.06E+2(5.5E+1)**○	4.53E+2(1.0E+2)•	5.09E+2(2.4E+1)•	5.66E+2(1.5E+2)•
	44	**3.86E−6(3.6E−6)**	2.00E−1(4.2E−2)•	1.03E−3(4.2E−4)•	7.99E−3(7.1E−3)•	1.04E+0(5.4E−3)•	8.54E−3(7.9E−3)•
	45	3.55E+1(5.1E+0)	5.30E+1(5.0E+0)•	3.99E+1(3.4E+0)•	**3.00E+1(2.5E+0)**○	3.39E+1(2.2E+0)○	4.14E+1(4.1E+0)•
	46	2.44E+2(5.0E+2)	1.24E+3(1.6E+3)•	**1.23E+2(1.7E+2)§**	3.07E+2(1.3E+2)•	9.63E+4(1.1E+5)•	1.35E+2(1.3E+2)§
	47	**7.41E−5(1.9E−4)**	1.98E+1(4.2E−1)•	7.25E+0(7.0E+0)•	5.67E+0(1.2E+0)•	5.28E+0(2.1E−1)•	1.70E+1(5.3E+0)•
	48	**2.60E+2(1.2E+2)**	6.20E+2(1.1E+2)•	4.00E+2(1.8E+1)•	4.57E+2(9.9E+1)•	5.17E+2(2.8E+1)•	4.69E+2(6.9E+1)•
	49	**4.66E−4(1.8E−3)**	2.13E−1(4.8E−2)•	1.74E−1(4.1E−2)•	2.24E−2(3.2E−2)•	1.04E+0(1.0E−2)•	8.67E−3(1.5E−2)•
	50	3.49E+1(6.7E+0)	5.14E+1(5.6E+0)•	**3.52E+0(4.2E+0)**○	2.97E+1(3.2E+0)○	3.35E+1(4.0E+0)○	4.30E+1(3.8E+0)•
Best/All		33/50	0/50	8/50	6/50	0/50	3/50
Fried		1.72	5.48	3.08	2.84	4.16	3.72
R		1	6	3	2	5	4
$w/s/l$		–	50/0/0	35/9/6	38/2/10	40/3/7	43/7/0

Table 19
The comparison results of objective values for EMaTO-AMR, MFEA, EEMTA, AT-MFEA, SREMTO, MaTEA on WCCI2020-SOMATP9.

	k	EMaTO-AMR	AT-MFEA	EEMTA	MaTEA	MFEA	SREMTO
P9	1	6.82E+2(1.2E+3)	1.17E+3(1.0E+3)•	1.23E+4(2.4E+4)•	**5.91E+2(5.7E+2)§**	8.03E+4(1.5E+5)•	6.26E+2(5.9E+2)§
	2	**1.32E+0(5.3E+0)**	1.98E+1(3.4E−1)•	1.84E+1(6.6E+0)•	5.49E+0(1.7E+0)•	5.21E+0(2.3E−1)•	1.31E+1(7.5E+0)•
	3	**3.00E+2(1.1E+2)**	7.11E+2(1.6E+2)•	3.17E+2(6.4E+1)§	4.61E+2(1.4E+2)•	5.17E+2(2.7E+1)•	6.59E+2(2.8E+2)•
	4	**4.67E−4(1.8E−3)**	1.90E−1(3.9E−2)•	1.07E−3(3.8E−4)•	1.74E−2(2.3E−2)•	1.04E+0(8.7E−3)•	1.41E−2(1.8E−2)•
	5	3.70E+1(9.2E−1)	5.77E+1(4.4E+0)•	3.92E+1(4.4E+0)§	**3.23E+1(3.6E+0)**∘	3.24E+1(5.2E+0)∘	4.85E+1(4.9E+0)•
	6	6.67E+3(3.4E+3)	6.08E+3(8.9E+2)§	1.18E+4(6.0E+2)•	6.81E+3(1.6E+3)§	**5.57E+3(7.0E+2)§**	5.61E+3(1.4E+3)§
	7	**5.15E+1(1.5E+1)**	4.68E+3(5.7E+3)•	4.40E+4(6.8E+4)•	2.95E+3(4.9E+3)•	2.14E+5(2.0E+5)•	8.40E+1(4.4E+1)•
	8	**1.01E−4(3.2E−4)**	1.85E+1(4.2E+0)•	1.60E+1(9.2E+0)•	5.99E+0(2.4E+0)•	5.28E+0(2.2E−1)•	1.57E+1(5.8E+0)•
	9	3.18E+2(8.8E+1)	6.56E+2(1.2E+2)•	**3.07E+2(5.3E+1)§**	4.82E+2(1.7E+2)•	5.12E+2(2.7E+1)•	6.06E+2(2.1E+2)•
	10	**3.57E−6(1.8E−6)**	2.06E−1(3.6E−2)•	1.03E−3(2.9E−4)•	1.27E−2(8.4E−3)•	1.04E+0(1.5E−2)•	3.47E−3(6.4E−3)•
	11	3.75E+1(6.4E−1)	5.37E+1(5.1E+0)•	4.19E+1(3.8E+0)•	**3.27E+1(3.1E+0)**∘	3.33E+1(4.2E+0)∘	4.41E+1(5.0E+0)•
	12	8.59E+3(3.2E+3)	6.49E+3(1.3E+3)§	1.16E+4(5.0E+2)•	6.19E+3(1.8E+3)§	5.48E+3(9.2E+2)∘	**4.73E+3(1.4E+3)**∘
	13	**1.07E+2(1.8E+2)**	5.21E+3(8.7E+3)•	1.92E+4(3.4E+4)•	1.95E+3(5.5E+3)•	7.04E+4(8.6E+4)•	5.22E+2(1.2E+3)§
	14	**2.56E−6(3.3E−6)**	1.96E+1(6.6E−1)•	1.46E+1(9.3E+0)•	4.83E+0(1.2E+0)•	5.26E+0(2.5E−1)•	1.09E+1(7.6E+0)•
	15	**3.21E+2(7.6E+1)**	6.49E+2(1.4E+2)•	3.49E+2(6.8E+1)§	5.29E+2(1.6E+2)•	5.14E+2(2.3E+1)•	6.20E+2(1.7E+2)•
	16	**4.56E−6(2.8E−6)**	1.83E−1(3.9E−2)•	9.11E−4(3.5E−4)•	9.18E−3(1.0E−2)•	1.04E+0(8.1E−3)•	6.10E−3(6.4E−3)•
	17	3.56E+1(4.8E+0)	5.36E+1(4.6E+0)•	4.24E+1(4.3E+0)•	**3.27E+1(2.8E+0)**∘	3.38E+1(4.4E+0)§	4.84E+1(3.5E+0)•
	18	**4.78E+3(3.2E+3)**	5.91E+3(1.1E+3)•	1.20E+4(5.2E+2)•	5.47E+3(1.6E+3)§	5.63E+3(8.0E+2)•	5.44E+3(1.3E+3)•
	19	**1.12E+2(1.1E+2)**	1.77E+3(2.0E+3)•	2.93E+4(8.5E+4)•	3.18E+2(2.6E+2)•	6.73E+4(6.4E+4)•	2.53E+2(3.9E+2)§
	20	**2.64E+0(7.2E+0)**	1.88E+1(4.2E+0)•	1.66E+1(8.2E+0)•	5.86E+0(1.9E+0)•	5.17E+0(2.5E−1)•	1.93E+1(2.8E−1)•
	21	**2.96E+2(1.2E+2)**	7.00E+2(1.1E+2)•	3.19E+2(3.9E+1)§	4.96E+2(2.4E+2)•	5.04E+2(2.1E+1)•	6.49E+2(1.8E+2)•
	22	**4.65E−4(1.8E−3)**	1.92E−1(3.6E−2)•	1.22E−3(4.5E−4)•	1.77E−2(1.5E−2)•	1.03E+0(1.9E−2)•	9.94E−3(1.0E−2)•
	23	3.38E+1(8.0E+0)	5.58E+1(4.2E+0)•	4.14E+1(4.0E+0)•	**3.17E+1(3.0E+0)**∘	3.35E+1(4.1E+0)§	4.41E+1(4.3E+0)•
	24	7.85E+3(3.3E+3)	6.31E+3(8.0E+2)§	1.16E+4(6.5E+2)•	6.13E+3(1.6E+3)§	**5.45E+3(7.2E+2)§**	5.98E+3(2.0E+3)§
	25	1.01E+3(1.3E+3)	1.58E+3(1.0E+3)•	5.32E+3(6.5E+3)•	1.54E+3(1.5E+3)§	4.28E+4(3.8E+4)•	**9.04E+2(9.7E+2)§**
	26	**1.46E−4(2.7E−4)**	1.99E+1(1.9E−1)•	2.12E+1(5.4E−2)•	5.34E+0(1.5E+0)•	5.31E+0(2.7E−1)•	1.92E+1(5.1E−1)•
	27	**3.00E+2(1.1E+2)**	6.93E+2(1.7E+2)•	3.28E+2(6.8E+1)§	4.04E+2(1.0E+2)§	5.14E+2(2.9E+1)•	6.06E+2(2.5E+2)•
	28	**3.27E−6(2.7E−6)**	2.11E−1(4.8E−2)•	1.05E−3(3.8E−4)•	9.01E−3(7.5E−3)•	1.04E+0(8.7E−3)•	8.13E−3(1.0E−2)•
	29	3.64E+1(3.3E+0)	5.60E+1(4.1E+0)•	3.77E+1(3.5E+0)§	**3.14E+1(3.0E+0)**∘	3.42E+1(3.7E+0)∘	4.56E+1(4.9E+0)•
	30	**4.82E+3(1.7E+3)**	6.47E+3(1.0E+3)•	1.18E+4(4.8E+2)•	6.35E+3(1.3E+3)•	5.54E+3(9.0E+2)•	5.96E+3(9.8E+2)•
	31	**5.84E+2(1.1E+3)**	2.03E+3(1.8E+3)•	1.14E+4(2.4E+4)•	1.09E+3(7.9E+2)•	3.74E+4(1.6E+4)•	9.41E+2(1.2E+3)§
	32	**5.31E−5(1.8E−4)**	2.00E+1(1.6E−1)•	1.72E+1(7.4E+0)•	6.02E+0(1.9E+0)•	5.34E+0(1.9E−1)•	1.68E+1(5.2E+0)•
	33	**2.67E+2(9.1E+1)**	6.58E+2(1.8E+2)•	2.97E+2(3.8E+1)§	4.57E+2(1.2E+2)•	5.16E+2(1.4E+1)•	6.75E+2(2.0E+2)•
	34	**3.99E−6(2.8E−6)**	2.20E−1(4.2E−2)•	9.86E−4(3.6E−4)•	2.48E−2(4.3E−2)•	1.04E+0(7.5E−3)•	7.50E−3(6.4E−3)•
	35	3.52E+1(5.2E+0)	5.48E+1(4.5E+0)•	3.81E+1(4.6E+0)§	**3.40E+1(3.3E+0)§**	3.43E+1(5.2E+0)§	4.46E+1(4.9E+0)•
	36	8.28E+3(3.1E+3)	6.72E+3(1.1E+3)§	1.18E+4(6.0E+2)•	7.01E+3(1.9E+3)§	**5.83E+3(9.0E+2)**∘	6.54E+3(1.3E+3)§
	37	2.11E+2(3.9E+2)	1.37E+3(1.8E+3)•	6.54E+3(1.6E+4)•	9.52E+2(2.0E+3)•	9.41E+4(9.1E+4)•	**8.33E+1(9.4E+1)§**
	38	**6.73E−1(2.7E+0)**	1.99E+1(2.3E−1)•	2.01E+1(4.2E+0)•	5.56E+0(2.0E+0)•	5.23E+0(2.8E−1)•	1.70E+1(5.3E+0)•
	39	3.52E+2(1.2E+1)	6.96E+2(1.6E+2)•	**3.35E+2(6.1E+1)**∘	4.33E+2(1.2E+2)§	5.06E+2(2.0E+1)•	7.09E+2(4.4E+2)•
	40	7.74E−4(3.1E−3)	2.41E−1(4.0E−2)•	**7.36E−4(2.8E−4)**∘	3.16E−2(8.3E−2)•	1.04E+0(7.4E−3)•	5.50E−3(5.4E−3)•
	41	3.56E+1(3.4E+0)	5.25E+1(4.3E+0)•	4.04E+1(3.7E+0)•	**3.02E+1(2.1E+0)**∘	3.31E+1(4.3E+0)∘	4.09E+1(3.6E+0)•
	42	6.56E+3(3.7E+3)	5.58E+3(1.1E+3)§	1.13E+4(6.4E+2)•	6.23E+3(2.6E+3)§	5.05E+3(6.5E+2)§	**4.88E+3(1.0E+3)§**
	43	2.61E+2(4.6E+2)	2.13E+3(4.2E+3)•	2.66E+4(3.6E+4)•	4.71E+2(4.6E+2)•	9.88E+4(2.1E+5)•	**8.95E+1(8.6E+1)§**
	44	**4.51E−1(1.2E+0)**	1.99E+1(3.7E−1)•	1.52E+1(9.2E+0)•	5.44E+0(1.8E+0)•	5.22E+0(2.2E−1)•	1.64E+1(6.3E+0)•
	45	3.49E+2(2.9E+1)	7.22E+2(1.5E+2)•	**3.39E+2(7.8E+1)§**	4.67E+2(1.1E+2)•	5.02E+2(2.0E+1)•	5.22E+2(1.4E+2)•
	46	**4.01E−6(3.2E−6)**	2.10E−1(5.2E−2)•	9.11E−4(2.9E−4)•	1.48E−2(1.6E−2)•	1.03E+0(1.1E−2)•	1.35E−2(8.7E−3)•
	47	3.71E+1(1.2E+0)	5.16E+1(5.3E+0)•	3.83E+1(3.5E+0)§	**2.95E+1(2.1E+0)**∘	3.36E+1(3.4E+0)∘	4.32E+1(3.8E+0)•
	48	**3.90E+3(2.7E+3)**	6.34E+3(9.5E+2)•	1.19E+4(6.9E+2)•	6.23E+3(1.8E+3)•	5.59E+3(7.9E+2)•	6.10E+3(1.0E+3)•
	49	**1.24E+2(2.7E+2)**	3.71E+3(5.5E+3)•	1.19E+4(2.4E+4)•	1.74E+3(4.4E+3)•	7.82E+4(9.3E+4)•	1.97E+2(1.7E+2)•
	50	**3.63E−2(1.4E−1)**	2.00E+1(1.5E−1)•	1.92E+1(5.6E+0)•	6.15E+0(1.8E+0)•	5.33E+0(2.5E−1)•	1.72E+1(5.6E+0)•
Best/All		29/50	0/50	4/50	9/50	3/50	5/50
Fried		1.88	5.06	3.96	2.92	3.66	3.52
R		1	6	5	2	4	3
$w/s/l$		–	45/5/0	37/11/2	32/11/7	37/6/7	38/11/1

Table 20
The comparison results of objective values for EMaTO-AMR, MFEA, EEMTA, AT-MFEA, SREMTO, MaTEA on WCCI2020-SOMATP10.

	k	EMaTO-AMR	AT-MFEA	EEMTA	MaTEA	MFEA	SREMTO
P10	1	**4.06E−4(1.4E−3)**	1.99E+1(3.2E−1)•	7.15E+0(7.2E+0)•	5.58E+0(1.7E+0)•	5.30E+0(2.5E−1)•	1.68E+1(5.4E+0)•
	2	**3.37E+2(7.0E+1)**	5.86E+2(1.3E+2)•	4.02E+2(1.5E+1)•	4.98E+2(1.3E+2)•	5.22E+2(2.0E+1)•	6.71E+2(1.5E+2)•
	3	**4.67E−4(1.8E−3)**	2.18E−1(4.4E−2)•	1.83E−1(4.3E−2)•	4.40E−2(1.3E−1)•	1.04E+0(9.0E−3)•	6.24E−3(8.0E−3)•
	4	3.55E+1(4.2E+0)	5.38E+1(5.9E+0)•	**4.66E+0(4.2E+0)**○	3.32E+1(3.0E+0)○	3.49E+1(2.6E+0)§	4.75E+1(5.4E+0)•
	5	7.24E+3(2.9E+3)	6.47E+3(1.3E+3)§	**5.54E+3(1.1E+3)§**	6.85E+3(1.7E+3)§	5.73E+3(5.8E+2)§	6.05E+3(1.6E+3)§
	6	**8.46E−3(3.3E−2)**	1.98E+1(4.1E−1)•	1.86E+1(7.0E+0)•	5.48E+0(1.5E+0)•	5.15E+0(2.9E−1)•	1.61E+1(5.8E+0)•
	7	**2.69E+2(1.1E+2)**	7.03E+2(1.4E+2)•	3.20E+2(5.1E+1)§	4.00E+2(9.7E+1)•	5.08E+2(3.0E+1)•	6.04E+2(1.7E+2)•
	8	**3.62E−6(4.9E−6)**	2.04E−1(4.3E−2)•	1.09E−3(4.5E−4)•	1.17E−2(1.3E−2)•	1.04E+0(9.3E−3)•	2.07E−3(4.5E−3)•
	9	3.69E+1(2.2E+0)	5.33E+1(3.6E+0)•	3.94E+1(3.9E+0)§	**3.07E+1(2.9E+0)**○	3.36E+1(4.4E+0)§	4.43E+1(4.6E+0)•
	10	**5.27E+3(2.6E+3)**	6.88E+3(1.0E+3)•	1.21E+4(6.1E+2)•	6.18E+3(2.3E+3)§	5.96E+3(7.6E+2)•	5.82E+3(1.2E+3)•
	11	**1.15E−4(3.4E−4)**	1.99E+1(1.7E−1)•	4.58E+0(4.2E+0)•	5.82E+0(2.2E+0)•	5.28E+0(1.8E−1)•	1.71E+1(5.4E+0)•
	12	**3.26E+2(7.5E+1)**	6.38E+2(1.6E+2)•	3.99E+2(1.8E+1)•	5.22E+2(1.7E+2)•	5.10E+2(2.7E+1)•	5.60E+2(1.6E+2)•
	13	**3.67E−6(2.7E−6)**	2.17E−1(5.4E−2)•	1.63E−1(3.7E−2)•	1.80E−2(1.6E−2)•	1.04E+0(1.0E−2)•	9.19E−3(6.8E−3)•
	14	3.64E+1(3.9E+0)	5.49E+1(4.0E+0)•	**4.27E+0(4.8E+0)**○	3.15E+1(2.8E+0)○	3.38E+1(3.6E+0)○	4.62E+1(6.0E+0)•
	15	8.10E+3(3.5E+3)	6.28E+3(1.2E+3)§	**5.04E+3(1.1E+3)**○	6.38E+3(2.0E+3)§	5.27E+3(7.5E+2)§	5.68E+3(1.3E+3)§
	16	**2.84E+0(7.2E+0)**	2.00E+1(4.4E−2)•	1.81E+1(5.8E+0)•	6.64E+0(2.0E+0)•	5.48E+0(2.4E−1)•	1.96E+1(1.7E−1)•
	17	**2.65E+2(1.2E+2)**	6.31E+2(1.1E+2)•	3.13E+2(6.3E+1)§	4.50E+2(1.6E+2)•	5.15E+2(2.9E+1)•	5.85E+2(2.6E+2)•
	18	**4.66E−4(1.9E−3)**	1.88E−1(4.4E−2)•	9.81E−4(3.2E−4)•	1.06E−2(1.2E−2)•	1.04E+0(9.4E−3)•	4.69E−3(7.6E−3)•
	19	3.73E+1(1.0E+0)	5.53E+1(4.5E+0)•	4.05E+1(3.8E+0)•	**3.27E+1(3.4E+0)**○	3.43E+1(4.4E+0)○	4.74E+1(6.7E+0)•
	20	6.27E+3(3.6E+3)	6.57E+3(9.1E+2)§	1.20E+4(4.4E+2)•	6.78E+3(1.7E+3)§	**5.46E+3(1.0E+3)§**	6.16E+3(8.3E+2)§
	21	**3.97E−5(1.0E−4)**	1.87E+1(4.3E+0)•	5.07E+0(4.3E+0)•	5.38E+0(2.0E+0)•	5.15E+0(2.7E−1)•	1.68E+1(5.4E+0)•
	22	**3.18E+2(8.4E+1)**	6.84E+2(1.3E+2)•	4.07E+2(1.3E+1)•	4.61E+2(1.4E+2)•	5.35E+2(1.7E+1)•	5.59E+2(1.4E+2)•
	23	**5.28E−6(7.7E−6)**	1.89E−1(3.5E−2)•	1.63E−1(3.2E−2)•	1.04E−2(9.3E−3)•	1.04E+0(9.3E−3)•	5.00E−3(6.2E−3)•
	24	3.61E+1(4.4E+0)	5.20E+1(5.9E+0)•	**1.92E+0(1.0E+0)**○	3.20E+1(2.5E+0)○	3.25E+1(3.8E+0)○	4.32E+1(3.3E+0)•
	25	6.87E+3(3.5E+3)	5.78E+3(9.7E+2)§	**4.58E+3(1.2E+3)§**	5.75E+3(1.6E+3)§	4.82E+3(8.3E+2)§	5.26E+3(1.5E+3)§
	26	**3.06E+0(7.2E+0)**	2.00E+1(3.9E−2)•	1.98E+1(5.3E+0)•	5.70E+0(1.4E+0)•	5.29E+0(3.4E−1)•	1.72E+1(5.6E+0)•
	27	**3.26E+2(7.3E+1)**	6.66E+2(1.1E+2)•	3.39E+2(5.2E+1)§	4.67E+2(1.4E+2)•	5.11E+2(1.8E+1)•	5.57E+2(1.9E+2)•
	28	**4.57E−6(3.8E−6)**	2.28E−1(3.0E−2)•	1.01E−3(3.7E−4)•	1.57E−2(1.5E−2)•	1.03E+0(1.6E−2)•	1.16E−2(7.4E−3)•
	29	3.71E+1(9.1E−1)	5.37E+1(5.5E+0)•	4.07E+1(5.0E+0)•	**3.21E+1(3.2E+0)**○	3.33E+1(3.9E+0)○	4.42E+1(5.2E+0)•
	30	**4.19E+3(2.9E+3)**	6.59E+3(1.3E+3)•	1.16E+4(5.6E+2)•	5.70E+3(1.7E+3)•	5.19E+3(5.9E+2)•	5.74E+3(1.6E+3)•
	31	**3.01E−1(1.2E+0)**	1.99E+1(2.3E−1)•	7.31E+0(6.9E+0)•	5.63E+0(1.4E+0)•	5.26E+0(1.8E−1)•	1.23E+1(7.8E+0)•
	32	**3.27E+2(6.8E+1)**	7.07E+2(1.5E+2)•	3.96E+2(1.7E+1)•	4.45E+2(1.8E+2)•	5.09E+2(2.6E+1)•	6.65E+2(2.0E+2)•
	33	**2.79E−6(2.1E−6)**	1.94E−1(3.8E−2)•	1.70E−1(2.3E−2)•	1.20E−2(1.3E−2)•	1.04E+0(8.5E−3)•	1.25E−2(1.3E−2)•
	34	3.58E+1(3.8E+0)	5.53E+1(3.8E+0)•	**2.29E+0(2.0E+0)**○	3.18E+1(3.1E+0)○	3.21E+1(3.7E+0)○	4.49E+1(4.1E+0)•
	35	9.42E+3(2.8E+3)	6.10E+3(1.1E+3)○	5.31E+3(1.1E+3)○	6.54E+3(2.5E+3)○	5.31E+3(7.0E+2)○	**5.19E+3(9.5E+2)**○
	36	**3.63E−4(1.4E−3)**	1.99E+1(2.8E−1)•	1.90E+1(4.8E+0)•	5.27E+0(1.4E+0)•	5.24E+0(2.4E−1)•	1.53E+1(7.2E+0)•
	37	3.18E+2(8.9E+1)	6.42E+2(1.1E+2)•	**3.09E+2(7.0E+1)§**	4.52E+2(1.6E+2)•	5.09E+2(2.4E+1)•	7.06E+2(3.0E+2)•
	38	**4.69E−6(3.4E−6)**	1.89E−1(3.5E−2)•	9.70E−4(3.6E−4)•	1.76E−2(2.2E−2)•	1.03E+0(1.3E−2)•	6.88E−3(8.9E−3)•
	39	3.65E+1(3.3E+0)	5.37E+1(4.4E+0)•	4.08E+1(3.5E+0)•	**2.94E+1(2.8E+0)**○	3.37E+1(3.9E+0)○	4.78E+1(4.5E+0)•
	40	7.34E+3(3.0E+3)	6.74E+3(9.1E+2)§	1.18E+4(4.7E+2)•	5.91E+3(1.6E+3)§	5.59E+3(1.1E+3)§	**5.52E+3(1.1E+3)§**
	41	**1.32E+0(5.3E+0)**	1.99E+1(3.3E−1)•	7.56E+0(7.7E+0)•	5.69E+0(1.5E+0)•	5.19E+0(2.8E−1)•	1.53E+1(7.1E+0)•
	42	**3.31E+2(7.8E+1)**	7.27E+2(1.4E+2)•	4.03E+2(1.5E+1)•	5.55E+2(1.5E+2)•	5.10E+2(1.5E+1)•	5.65E+2(1.7E+2)•
	43	**4.67E−4(1.8E−3)**	2.01E−1(4.7E−2)•	1.74E−1(2.9E−2)•	4.10E−2(1.0E−1)•	1.04E+0(1.0E−2)•	8.85E−3(8.0E−3)•
	44	3.58E+1(5.2E+0)	5.41E+1(3.4E+0)•	**3.45E+0(2.5E+0)**○	3.27E+1(4.0E+0)○	3.19E+1(4.3E+0)○	4.63E+1(6.3E+0)•
	45	1.07E+4(2.4E+3)	6.57E+3(7.5E+2)○	**5.07E+3(9.3E+2)**○	6.35E+3(1.9E+3)○	5.72E+3(9.2E+2)○	5.81E+3(1.3E+3)○
	46	**1.32E+0(5.3E+0)**	1.99E+1(5.8E−1)•	1.85E+1(7.2E+0)•	6.39E+0(2.5E+0)•	5.40E+0(2.4E−1)•	1.84E+1(3.9E+0)•
	47	**2.76E+2(1.1E+2)**	6.85E+2(1.5E+2)•	3.14E+2(6.1E+1)§	4.25E+2(1.5E+2)•	5.13E+2(2.3E+1)•	6.09E+2(1.7E+2)•
	48	**5.96E−6(5.8E−6)**	2.06E−1(4.7E−2)•	9.16E−4(3.4E−4)•	9.01E−3(9.2E−3)•	1.03E+0(8.2E−3)•	4.53E−3(5.9E−3)•
	49	3.65E+1(3.1E+0)	5.48E+1(3.9E+0)•	4.09E+1(3.7E+0)•	**3.25E+1(3.4E+0)**○	3.39E+1(4.7E+0)○	4.28E+1(4.8E+0)•
	50	9.05E+3(3.1E+3)	5.68E+3(9.0E+2)○	1.16E+4(6.7E+2)•	5.98E+3(2.3E+3)○	**5.02E+3(8.4E+2)**○	5.66E+3(1.3E+3)○
Best/All		31/50	0/50	10/50	5/50	2/50	2/50
Fried		2.22	5.42	3	3.12	3.28	3.96
R		1	6	2	3	4	5
$w/s/l$		–	42/5/3	34/8/8	31/6/13	32/7/11	42/5/3

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